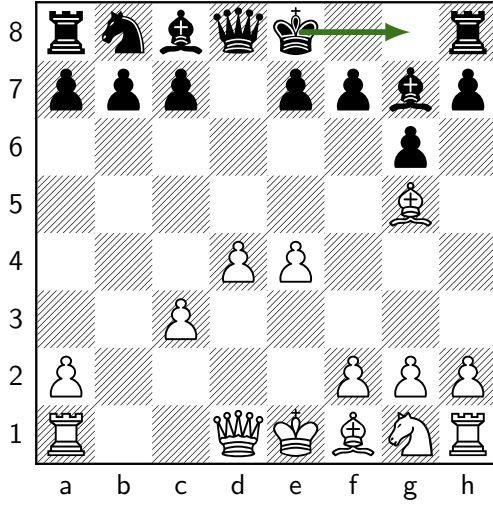


Melanie-BOTS (CCRL engines) — Grunfeld

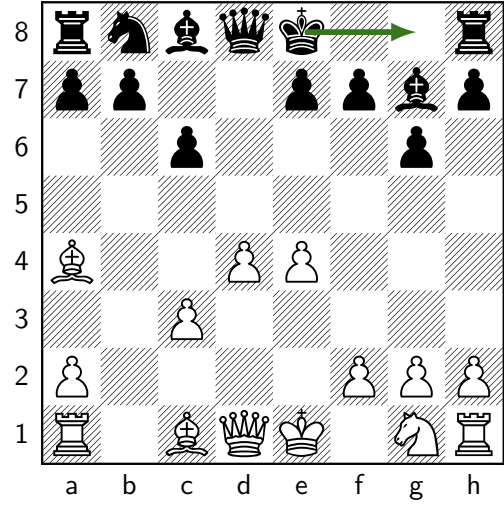
Melanie (Deep_Vision)

Abstract

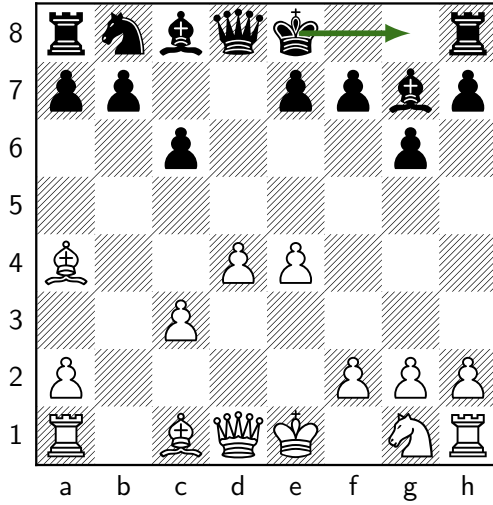
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie’s move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen’s role) are called out, as no material/eval engine explains a move that way.



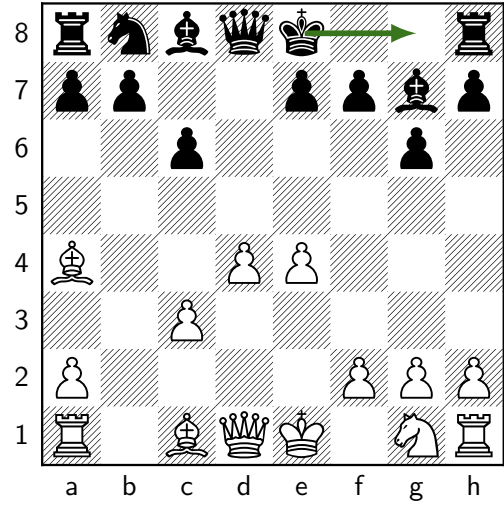
(a)



(b)

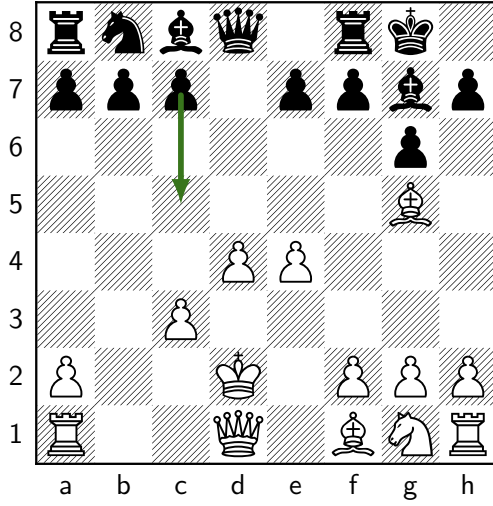


(c)

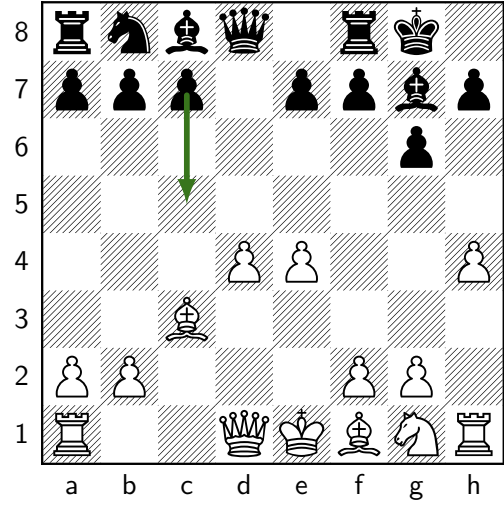


(d)

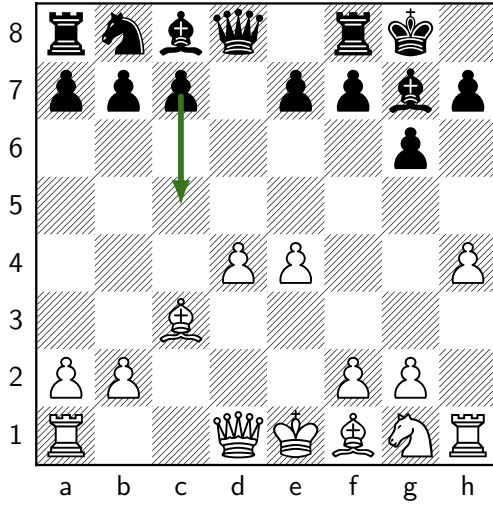
Figure 1: **Move 7, Black: 0-0.** (a) Our position; Melanie plays 0-0 (e8g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccl1_top_p0001#3093` (ply 15) [structural distance $d=35$, QAP obj=105]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Ph7* as their *Nb8*, our *Nb8* as their *Rh8*; transferred move e8g8 (**endorses this move**). (c) Retrieved from `ccl1_top_p0001#3094` (ply 15) [structural distance $d=35$, QAP obj=105]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Ph7* as their *Nb8*, our *Nb8* as their *Rh8*; transferred move e8g8 (**endorses this move**). (d) Retrieved from `ccl1_top_p0002#2321` (ply 15) [structural distance $d=35$, QAP obj=105]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Ph7* as their *Nb8*, our *Nb8* as their *Rh8*; transferred move e8g8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to 0-0 (summed QAP objective 418, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



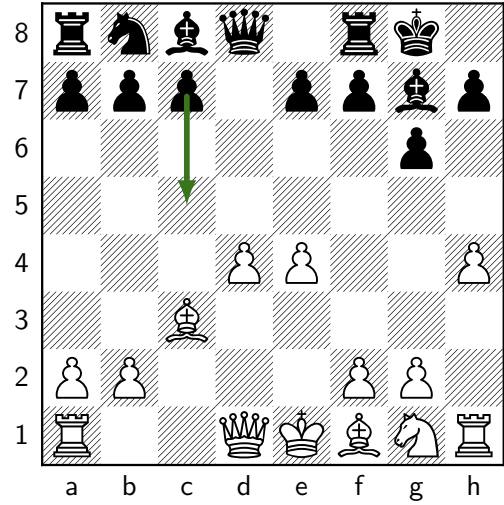
(a)



(b)

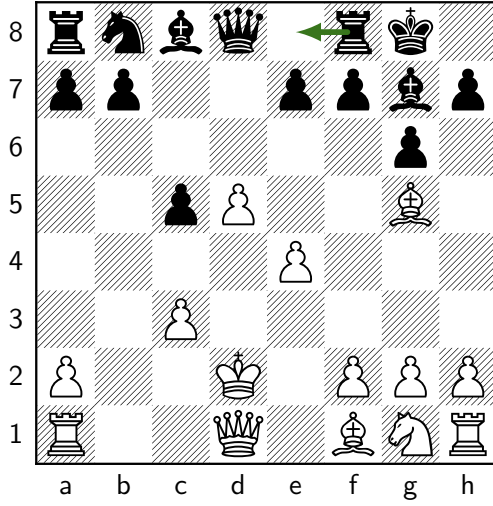


(c)

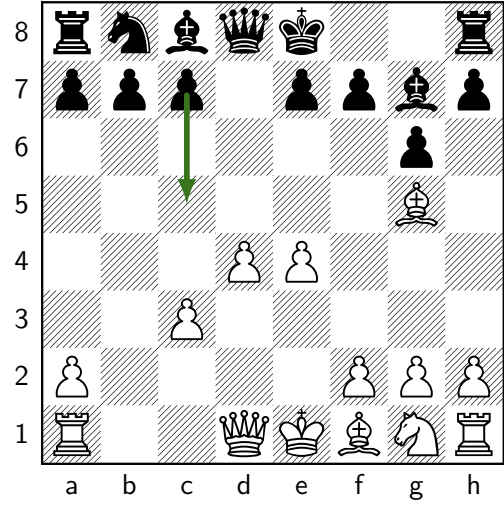


(d)

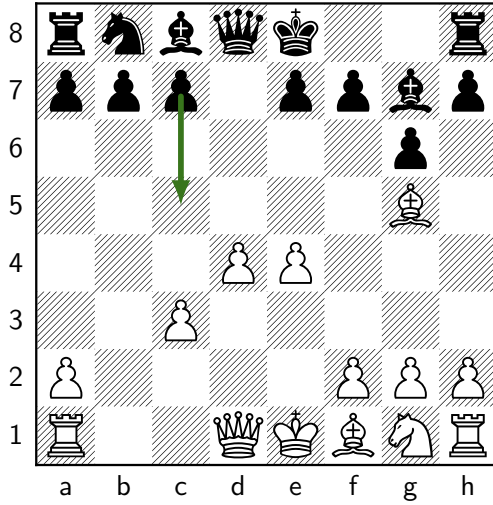
Figure 2: **Move 8, Black: c5**. (a) Our position; Melanie plays c5 (c7c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#240` (ply 15) [structural distance $d=35$, QAP obj=56]: the master played c5; under μ our *Pc2* plays the role of that game’s *Pc2*; cross-type roles: our *Bc8* as their *Ra8*, our *Ra8* as their *Pa7*; transferred move c7c5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#241` (ply 15) [structural distance $d=35$, QAP obj=56]: the master played c5; under μ our *Pc2* plays the role of that game’s *Pc2*; cross-type roles: our *Bc8* as their *Ra8*, our *Ra8* as their *Pa7*; transferred move c7c5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3201` (ply 15) [structural distance $d=35$, QAP obj=56]: the master played c5; under μ our *Pc2* plays the role of that game’s *Pc2*; cross-type roles: our *Bc8* as their *Ra8*, our *Ra8* as their *Pa7*; transferred move c7c5 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to c5 (summed QAP objective 224, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



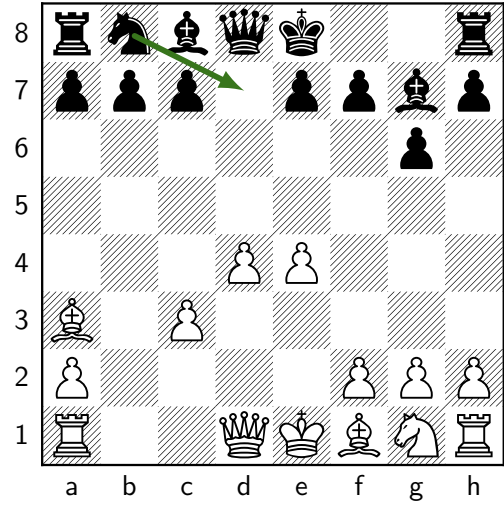
(a)



(b)

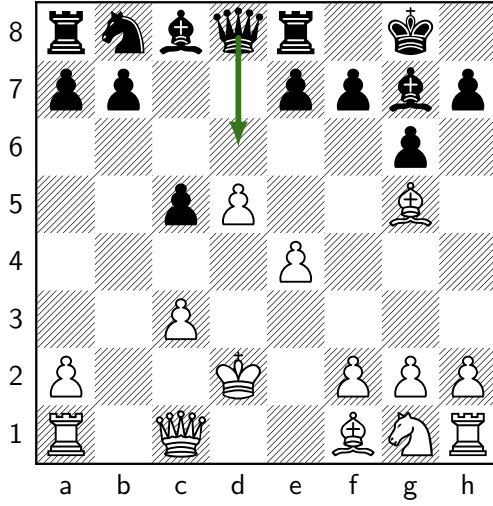


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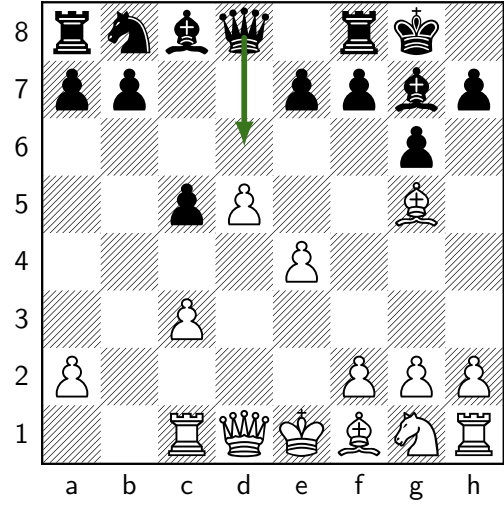


(d)

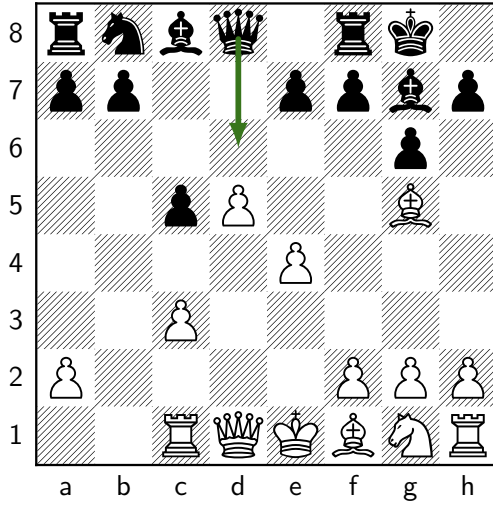
Figure 3: **Move 9, Black: Re8**. (a) Our position; Melanie plays Re8 (f8e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#296 (ply 13) [structural distance $d=52$, QAP obj=74]: the master played c5; under μ our *Rf1* plays the role of that game’s *Pc2* (a **cross-type** correspondence); cross-type roles: our *Pc5* as their *Ra8*, our *pg2* as their *Nb8*; transferred move f8e8 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#297 (ply 13) [structural distance $d=52$, QAP obj=74]: the master played c5; under μ our *Rf1* plays the role of that game’s *Pc2* (a **cross-type** correspondence); cross-type roles: our *Pc5* as their *Ra8*, our *pg2* as their *Nb8*; transferred move f8e8 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#3382 (ply 13) [structural distance $d=59$, QAP obj=43]: the master played Nd7; under μ our *Rf1* plays the role of that game’s *Nb1* (a **cross-type** correspondence); cross-type roles: our *pd5* as their *Ra8*, our *bg5* as their *Qd8*; transferred move f8e8 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Re8 (summed QAP objective 319, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



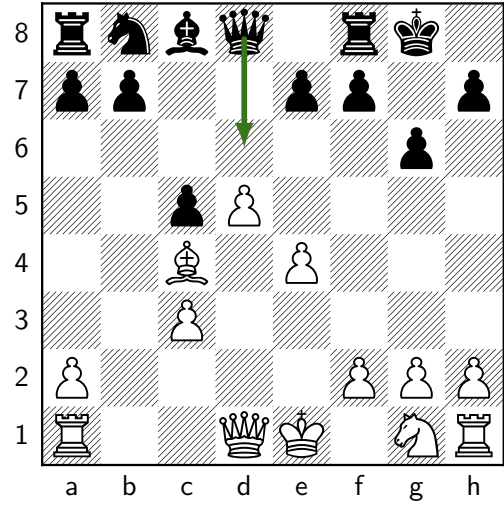
(a)



(b)

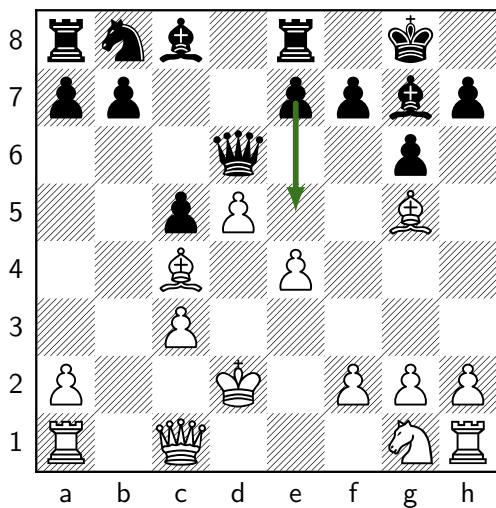


(c)

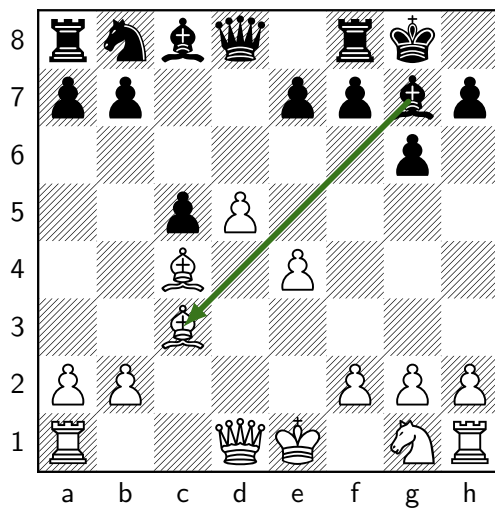


(d)

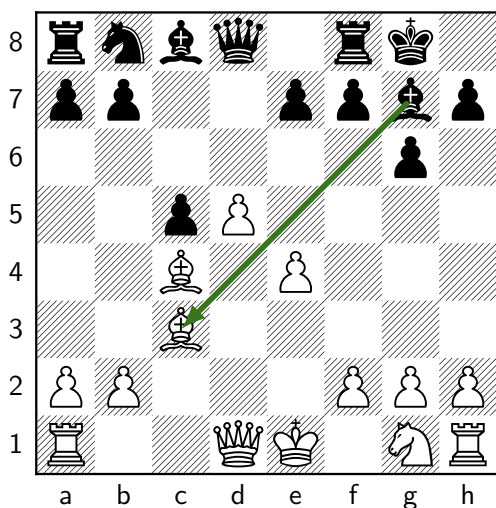
Figure 4: **Move 10, Black: Qd6.** (a) Our position; Melanie plays Qd6 (d8d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#296` (ply 17) [structural distance $d=27$, QAP obj=95]: the master played Qd6; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our pa2 as their Nb8, our qc1 as their Pc5; transferred move d8d6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#297` (ply 17) [structural distance $d=27$, QAP obj=95]: the master played Qd6; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our pa2 as their Nb8, our qc1 as their Pc5; transferred move d8d6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#65` (ply 19) [structural distance $d=42$, QAP obj=43]: the master played Qd6; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our Pf7 as their Nb8, our bg5 as their Pa7; transferred move d8d6 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Qd6 (summed QAP objective 234, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



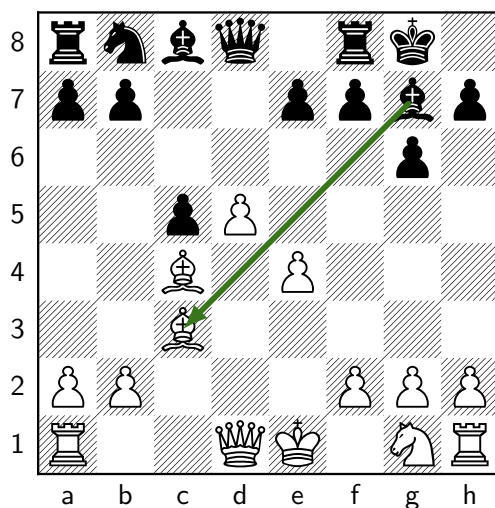
(a)



(b)

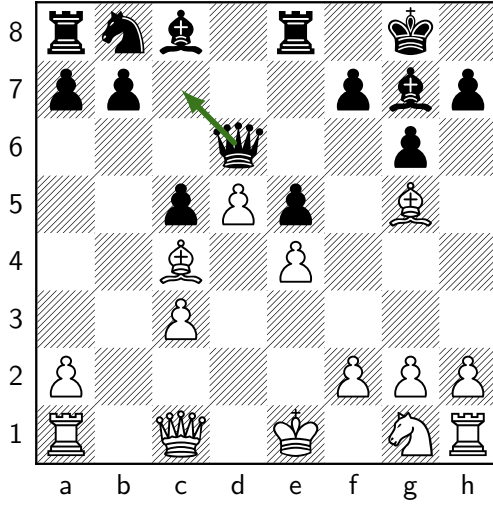


(c)

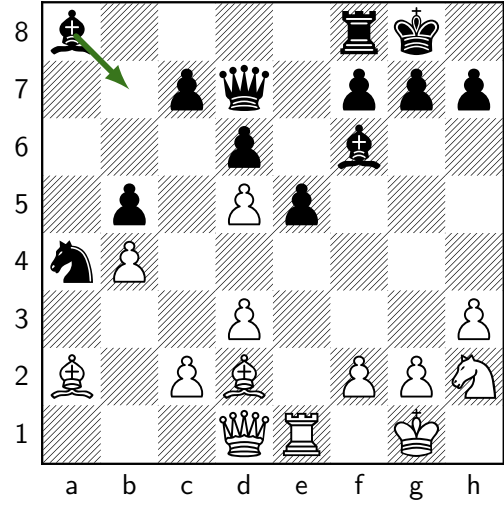


(d)

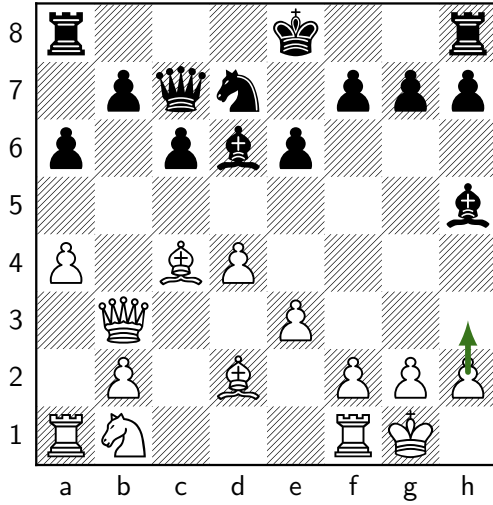
Figure 5: **Move 11, Black: e5.** (a) Our position; Melanie plays e5 (e7e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#153` (ply 17) [structural distance $d=82$, QAP obj=28]: the master played Bxc3+; under μ our *Pe2* plays the role of that game’s *Bg2* (a **cross-type** correspondence); cross-type roles: our *Pf7* as their *Nb8*, our *pc3* as their *Rf8*; transferred move e7e5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#154` (ply 17) [structural distance $d=82$, QAP obj=28]: the master played Bxc3+; under μ our *Pe2* plays the role of that game’s *Bg2* (a **cross-type** correspondence); cross-type roles: our *Pf7* as their *Nb8*, our *pc3* as their *Rf8*; transferred move e7e5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#3109` (ply 17) [structural distance $d=82$, QAP obj=28]: the master played Bxc3+; under μ our *Pe2* plays the role of that game’s *Bg2* (a **cross-type** correspondence); cross-type roles: our *Pf7* as their *Nb8*, our *pc3* as their *Rf8*; transferred move e7e5 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to e5 (summed QAP objective 165, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



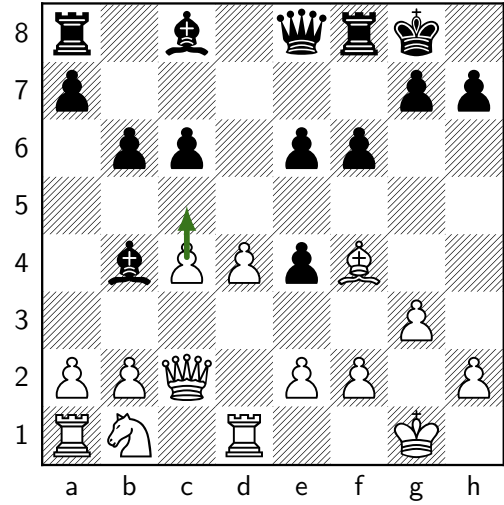
(a)



(b)

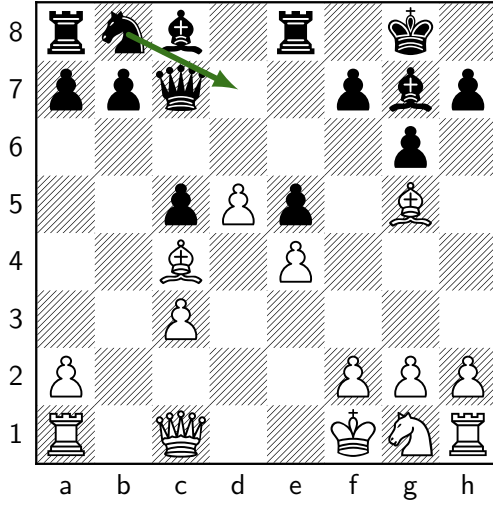


(c)

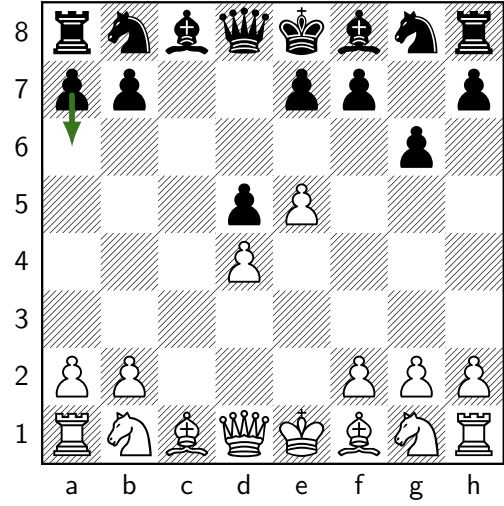


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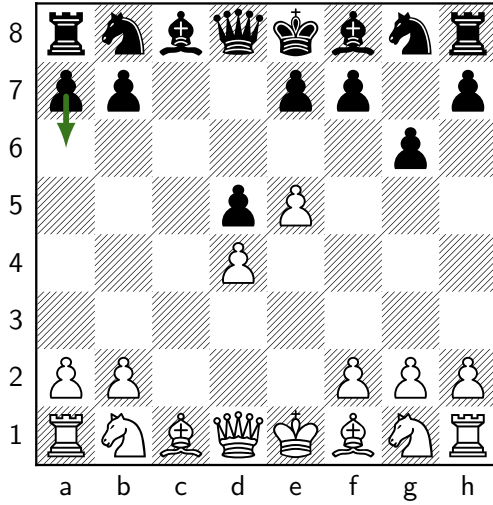
Figure 6: **Move 12, Black: Qc7.** (a) Our position; Melanie plays Qc7 (d6c7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1017` (ply 37) [structural distance $d=75$, QAP obj=11]: the master played Bb7; under μ our Qd3 plays the role of that game’s Ba1 (a **cross-type** correspondence); cross-type roles: our Ra8 as their Pc7, our Bc8 as their Qd7; transferred move d6c7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3672` (ply 24) [structural distance $d=63$, QAP obj=17]: the master played h3; under μ our Bg2 plays the role of that game’s Ph2 (a **cross-type** correspondence); cross-type roles: our pc3 as their Ra1, our Re8 as their Nb1; transferred move g7h6. (d) Retrieved from `ccr1_top_p0000#2011` (ply 30) [structural distance $d=64$, QAP obj=14]: the master played c5; under μ our Ph2 plays the role of that game’s Pc4; cross-type roles: our Nb8 as their Ra1, our Pg6 as their Nb1; transferred move h7h6. Synthesis: 1 retrieved analog transferred to Qc7 (summed QAP objective 11, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



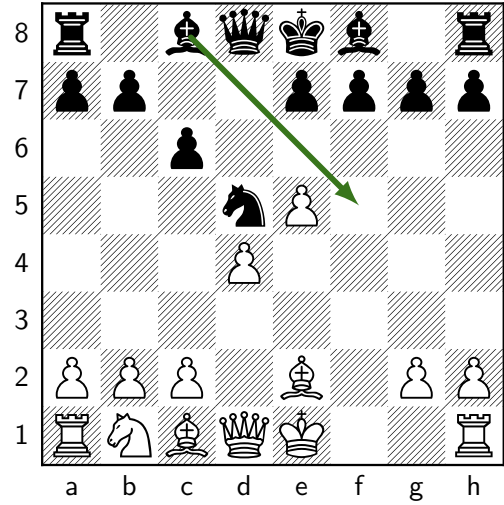
(a)



(b)

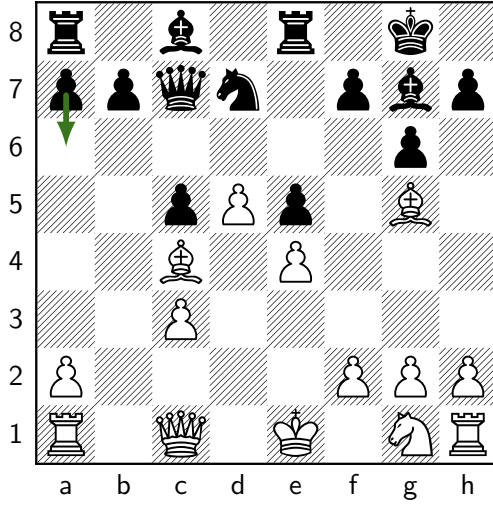


(c)

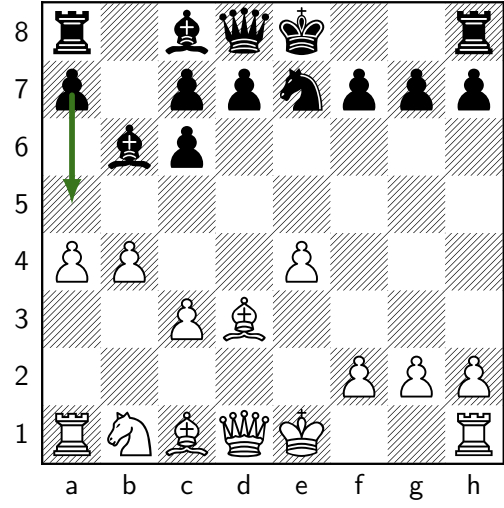


(d)

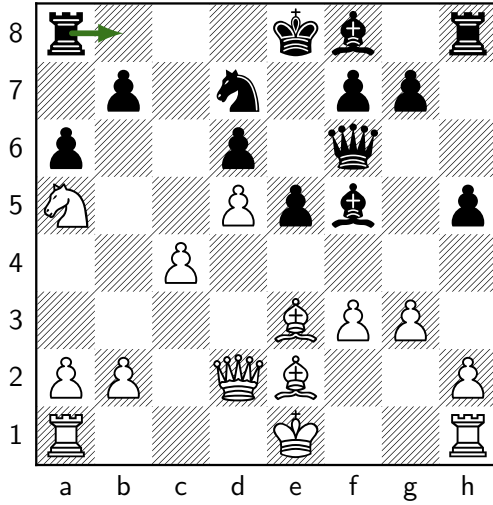
Figure 7: **Move 13, Black: Nd7.** (a) Our position; Melanie plays Nd7 (b8d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#715` (ply 9) [structural distance $d=64$, QAP obj=5]: the master played a6; under μ our Nb1 plays the role of that game’s Pa2 (a **cross-type** correspondence); cross-type roles: our bg5 as their Ra8, our pd5 as their Nb8; transferred move b8d7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#716` (ply 9) [structural distance $d=64$, QAP obj=5]: the master played a6; under μ our Nb1 plays the role of that game’s Pa2 (a **cross-type** correspondence); cross-type roles: our bg5 as their Ra8, our pd5 as their Nb8; transferred move b8d7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3135` (ply 15) [structural distance $d=75$, QAP obj=5]: the master played Bf5; under μ our Bc1 plays the role of that game’s Bc1; cross-type roles: our bc4 as their Ra8, our Re8 as their Qd8; transferred move c8f5. Synthesis: 2 retrieved analogs transferred to Nd7 (summed QAP objective 10, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



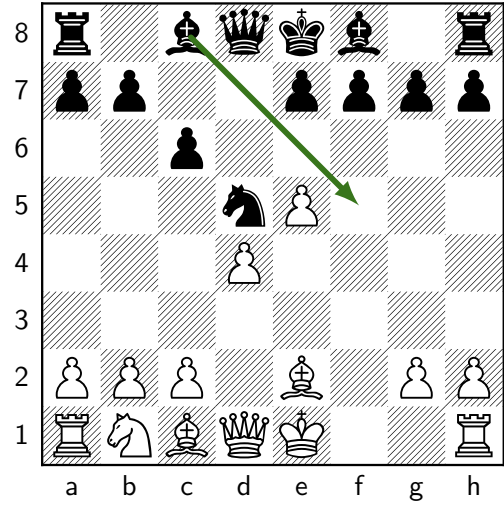
(a)



(b)

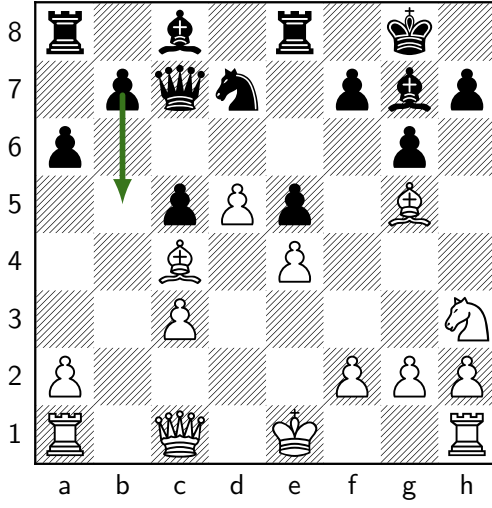


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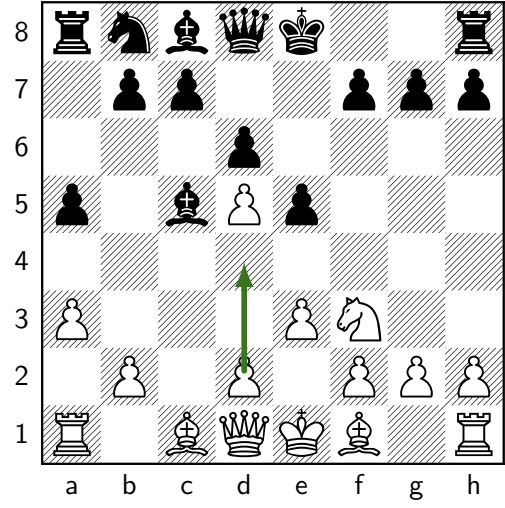


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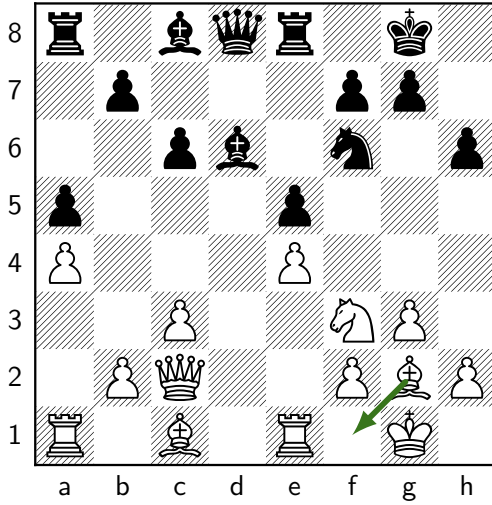
Figure 8: **Move 14, Black: a6.** (a) Our position; Melanie plays a6 (a7a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3859` (ply 17) [structural distance $d=58$, QAP obj=4]: the master played a5; under μ our *Pa2* plays the role of that game’s *Pa2*; cross-type roles: our *Pe5* as their *Ra8*, our *pd5* as their *Bc8*; transferred move a7a6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#2919` (ply 33) [structural distance $d=75$, QAP obj=6]: the master played Rb8; under μ our *Re1* plays the role of that game’s *Ra1*; cross-type roles: our *pe4* as their *Bf8*, our *pd5* as their *Rh8*; transferred move e8d8. (d) Retrieved from `ccrl_top_p0001#3135` (ply 15) [structural distance $d=57$, QAP obj=4]: the master played Bf5; under μ our *Pf2* plays the role of that game’s *Bc1* (a **cross-type** correspondence); cross-type roles: our *Nd7* as their *Ra8*, our *Re8* as their *Qd8*; transferred move f7f5. Synthesis: 1 retrieved analog transferred to a6 (summed QAP objective 4, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



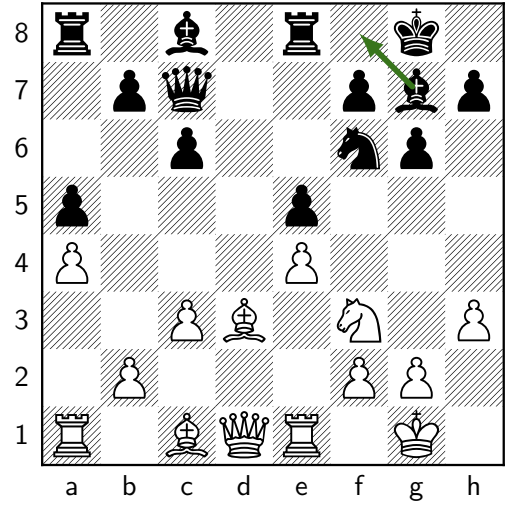
(a)



(b)

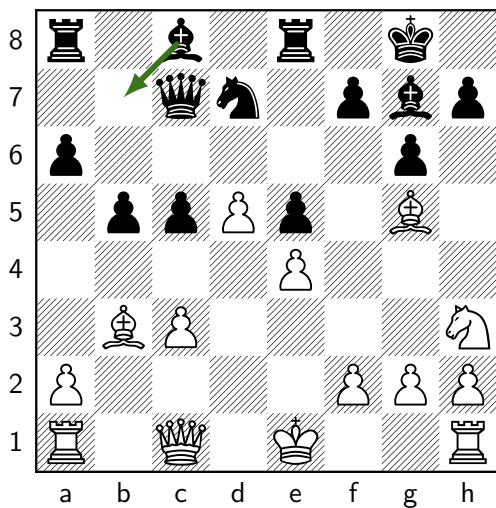


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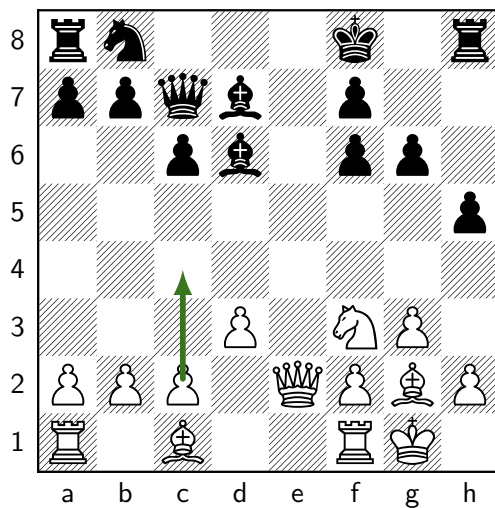


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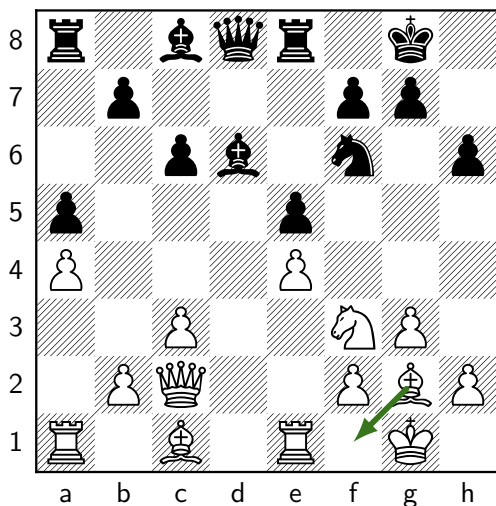
Figure 9: **Move 15, Black: b5.** (a) Our position; Melanie plays b5 (b7b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1428` (ply 14) [structural distance $d=57$, QAP obj=49]: the master played d4; under μ our *Pb2* plays the role of that game’s *Pd2*; cross-type roles: our *Pf7* as their *Ra1*, our *Re8* as their *Qd1*; transferred move b7b5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#909` (ply 26) [structural distance $d=54$, QAP obj=36]: the master played Bf1; under μ our *Pf2* plays the role of that game’s *Bg2* (a **cross-type** correspondence); cross-type roles: our *bg5* as their *Ra1*, our *pd5* as their *Bc1*; transferred move f7f6. (d) Retrieved from `ccr1_top_p0003#1420` (ply 27) [structural distance $d=54$, QAP obj=36]: the master played Bf8; under μ our *Pf2* plays the role of that game’s *Bg2* (a **cross-type** correspondence); cross-type roles: our *bg5* as their *Ra8*, our *pd5* as their *Bc8*; transferred move f7f6. Synthesis: 1 retrieved analog transferred to b5 (summed QAP objective 49, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



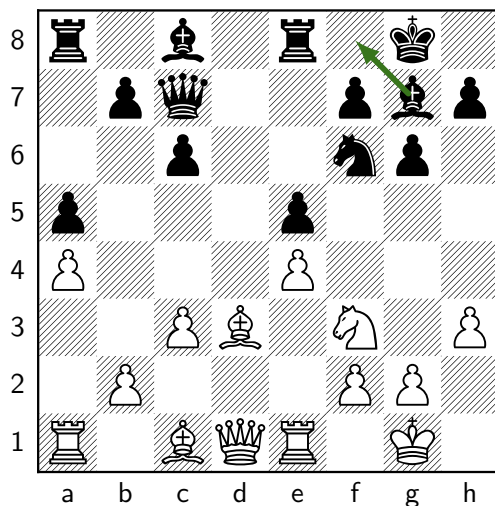
(a)



(b)

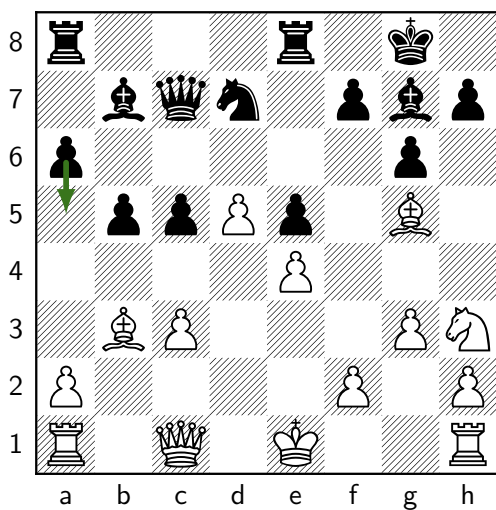


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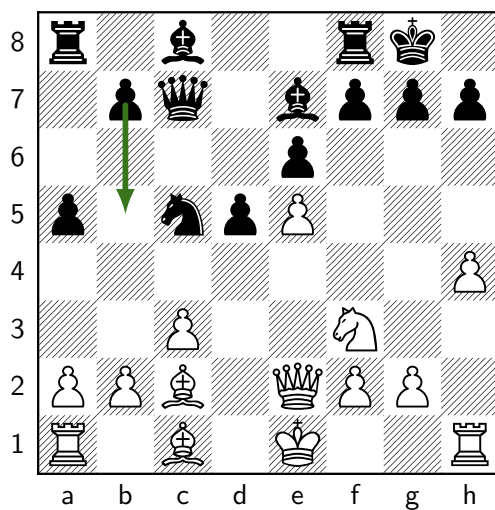


(d)

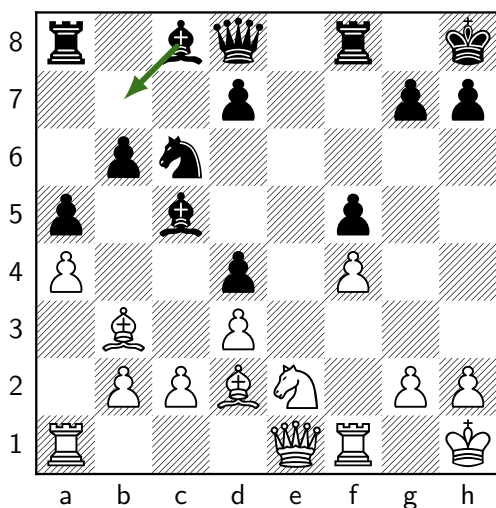
Figure 10: **Move 16, Black: Bb7.** (a) Our position; Melanie plays Bb7 (c8b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0002#372 (ply 24) [structural distance $d=71$, QAP obj=6]: the master played c4; under μ our Bc1 plays the role of that game’s Pc2 (a **cross-type** correspondence); cross-type roles: our Pe5 as their Bc1, our bg5 as their Pb2; transferred move c8b7 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#909 (ply 26) [structural distance $d=63$, QAP obj=8]: the master played Bf1; under μ our Pb4 plays the role of that game’s Bg2 (a **cross-type** correspondence); cross-type roles: our nh3 as their Ra1, our pd5 as their Bc1; transferred move b5b4. (d) Retrieved from ccr1_top_p0003#1420 (ply 27) [structural distance $d=63$, QAP obj=8]: the master played Bf8; under μ our Pb4 plays the role of that game’s Bg2 (a **cross-type** correspondence); cross-type roles: our nh3 as their Ra8, our pd5 as their Bc8; transferred move b5b4. Synthesis: 1 retrieved analog transferred to Bb7 (summed QAP objective 6, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



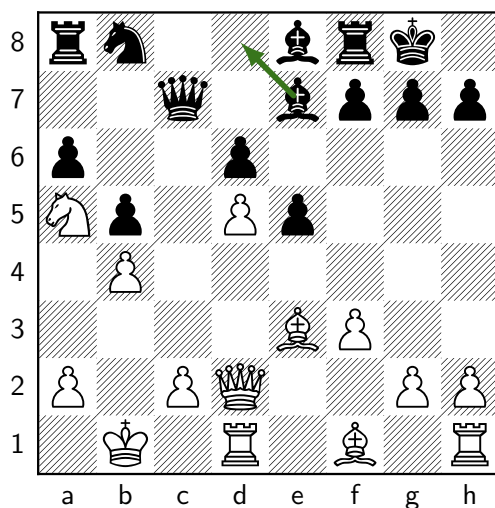
(a)



(b)

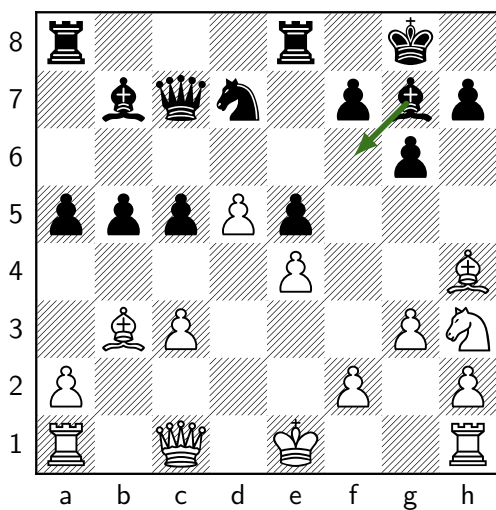


(c)

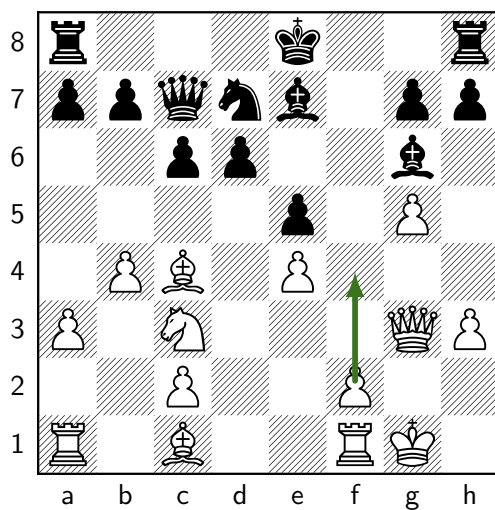


(d)

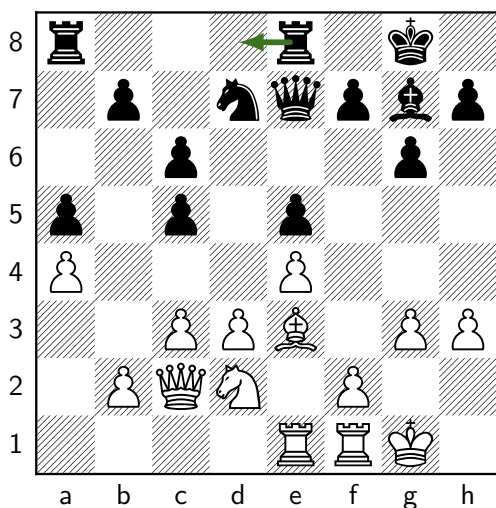
Figure 11: **Move 17, Black: a5**. (a) Our position; Melanie plays a5 (a6a5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1643` (ply 25) [structural distance $d=73$, QAP obj=31]: the master played b5; under μ our *Pa3* plays the role of that game’s *Pb2*; cross-type roles: our *ph2* as their *Ra8*, our *Ra8* as their *Bc8*; transferred move a6a5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#1352` (ply 31) [structural distance $d=60$, QAP obj=25]: the master played Bb7; under μ our *Pa3* plays the role of that game’s *Bc1* (a **cross-type** correspondence); cross-type roles: our *pc3* as their *Ra8*, our *bb3* as their *Qd8*; transferred move a6a5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#1951` (ply 29) [structural distance $d=73$, QAP obj=11]: the master played Bd8; under μ our *Pa3* plays the role of that game’s *Be2* (a **cross-type** correspondence); cross-type roles: our *Pf7* as their *Nb8*, our *Re8* as their *Be8*; transferred move a6a5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to a5 (summed QAP objective 66, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



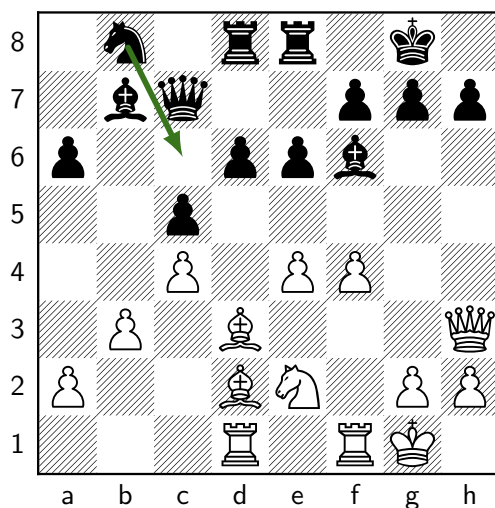
(a)



(b)

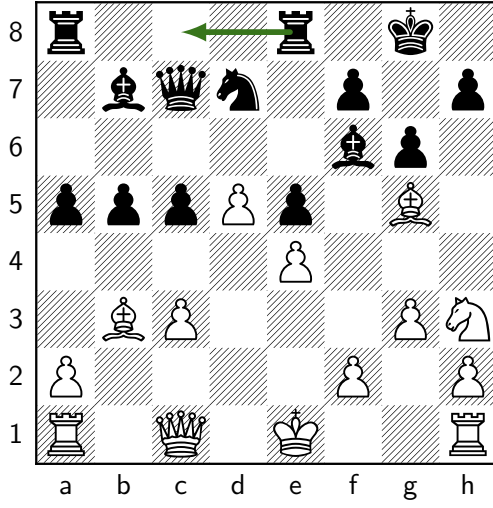


(c)

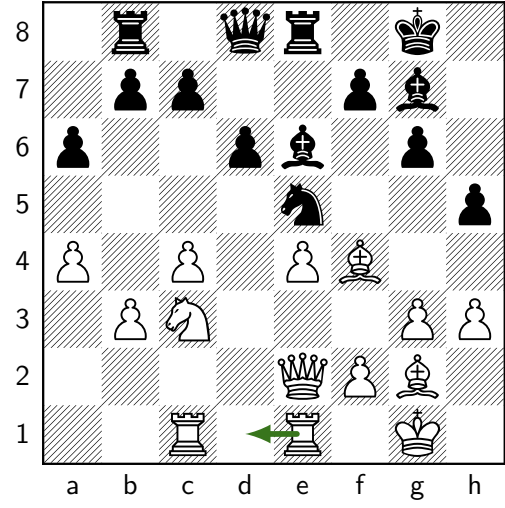


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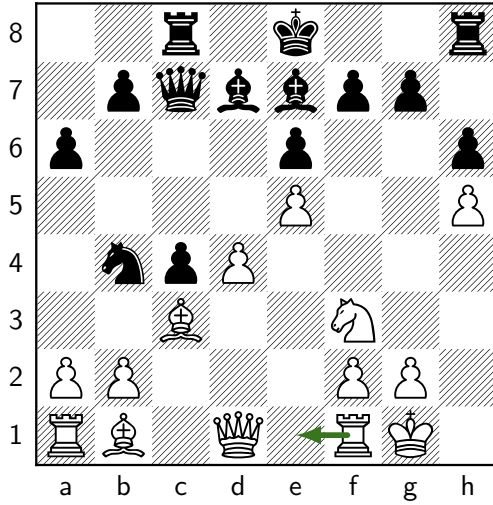
Figure 12: **Move 18, Black: Bf6**. (a) Our position; Melanie plays Bf6 (g7f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#576` (ply 30) [structural distance $d=72$, QAP obj=32]: the master played f4; under μ our *Bg2* plays the role of that game’s *Pf2* (a **cross-type** correspondence); cross-type roles: our *Pa5* as their *Ra1*, our *pe4* as their *Bc1*; transferred move g7f6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1084` (ply 35) [structural distance $d=76$, QAP obj=25]: the master played Red8; under μ our *Re1* plays the role of that game’s *Re1*; cross-type roles: our *qc1* as their *Ra8*, our *Qc7* as their *Pb7*; transferred move e8d8. (d) Retrieved from `ccrl_top_p0003#376` (ply 37) [structural distance $d=74$, QAP obj=25]: the master played Nc6; under μ our *Re1* plays the role of that game’s *Nb1* (a **cross-type** correspondence); cross-type roles: our *pf2* as their *Rd8*, our *bb3* as their *Re8*; transferred move e8c8. Synthesis: 1 retrieved analog transferred to Bf6 (summed QAP objective 32, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



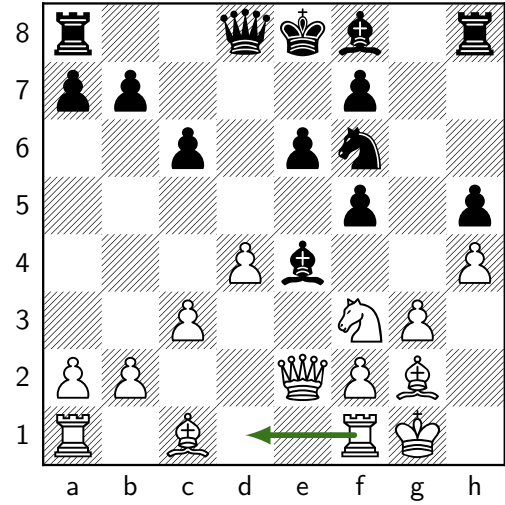
(a)



(b)

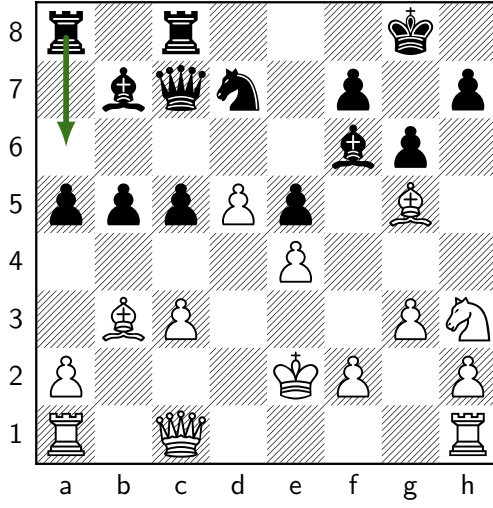


(c)

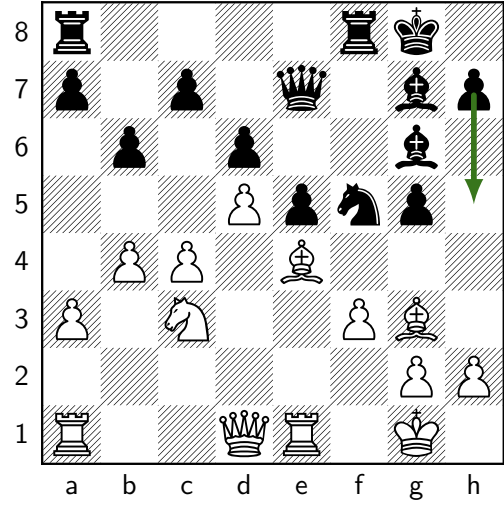


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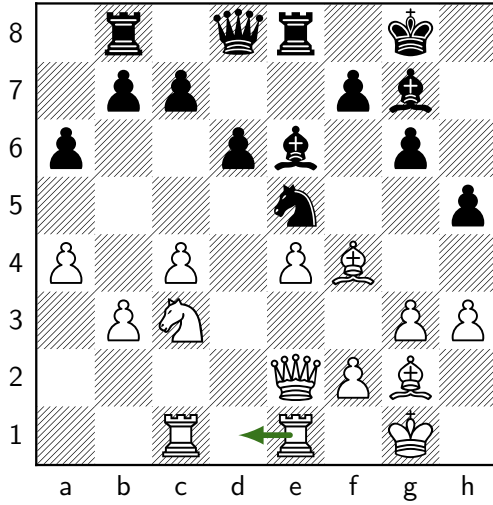
Figure 13: **Move 19, Black: Rec8.** (a) Our position; Melanie plays Rec8 (e8c8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#467 (ply 38) [structural distance $d=75$, QAP obj=27]: the master played Red1; under μ our Re1 plays the role of that game’s Re1; cross-type roles: our Pg6 as their Bg2, our pc3 as their Nc3; transferred move e8c8 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#3736 (ply 34) [structural distance $d=65$, QAP obj=26]: the master played Re1; under μ our Re1 plays the role of that game’s Rf1; cross-type roles: our Pf7 as their Ra1, our pe4 as their Bb1; transferred move e8c8 (**endorses this move**). (d) Retrieved from ccr1_top_p0001#968 (ply 24) [structural distance $d=67$, QAP obj=33]: the master played Rd1; under μ our Re1 plays the role of that game’s Rf1; cross-type roles: our pg3 as their Ra1, our Pa5 as their Bc1; transferred move e8e7. Synthesis: 2 retrieved analogs transferred to Rec8 (summed QAP objective 53, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



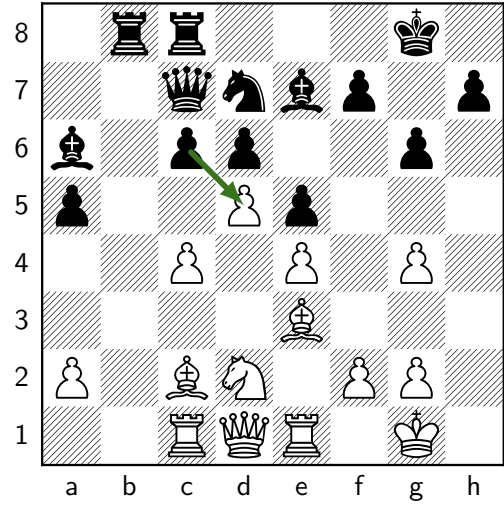
(a)



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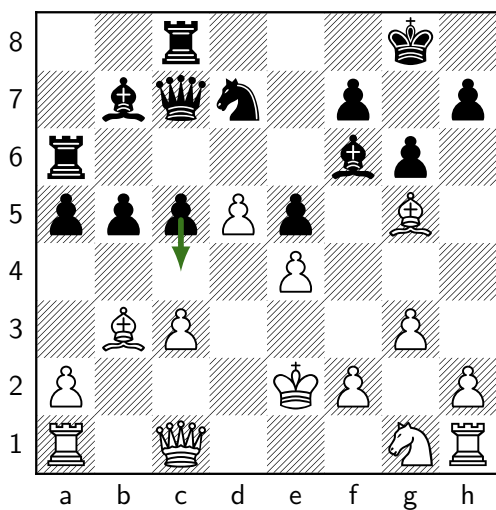


(c)

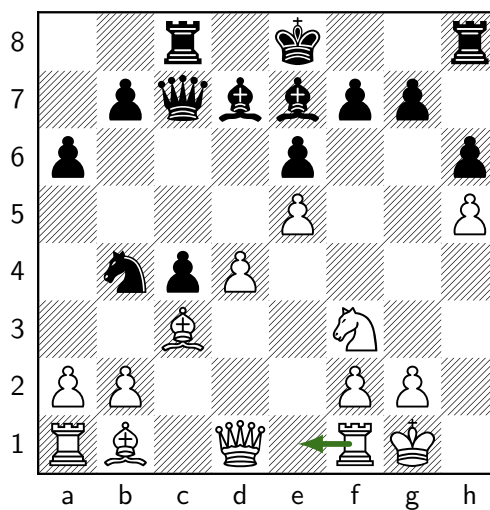


(d)

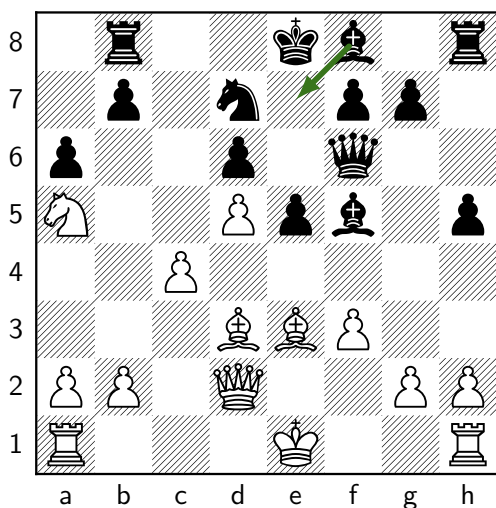
Figure 14: **Move 20, Black: Ra6.** (a) Our position; Melanie plays Ra6 (a8a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#961` (ply 47) [structural distance $d=65$, QAP obj=32]: the master played h5; under μ our *Ra1* plays the role of that game’s *Ph2* (a **cross-type** correspondence); cross-type roles: our *Pb5* as their *Ra8*, our *nh3* as their *Rf8*; transferred move a8a6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#467` (ply 38) [structural distance $d=67$, QAP obj=25]: the master played Red1; under μ our *Rc1* plays the role of that game’s *Re1*; cross-type roles: our *Pa5* as their *Rc1*, our *Ra8* as their *Qe2*; transferred move c8b8. (d) Retrieved from `ccr1_top_p0002#972` (ply 51) [structural distance $d=64$, QAP obj=19]: the master played cxd5; under μ our *Bf3* plays the role of that game’s *Pc3* (a **cross-type** correspondence); cross-type roles: our *bb3* as their *Rb8*, our *rh1* as their *Nd7*; transferred move f6g5. Synthesis: 1 retrieved analog transferred to Ra6 (summed QAP objective 32, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



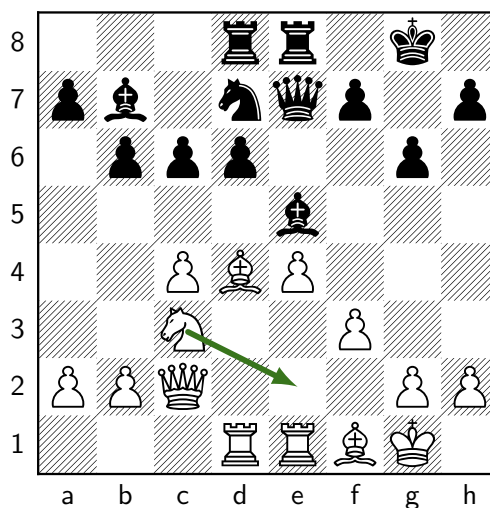
(a)



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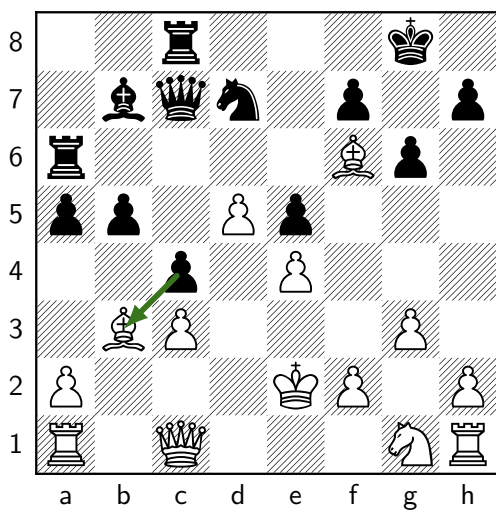


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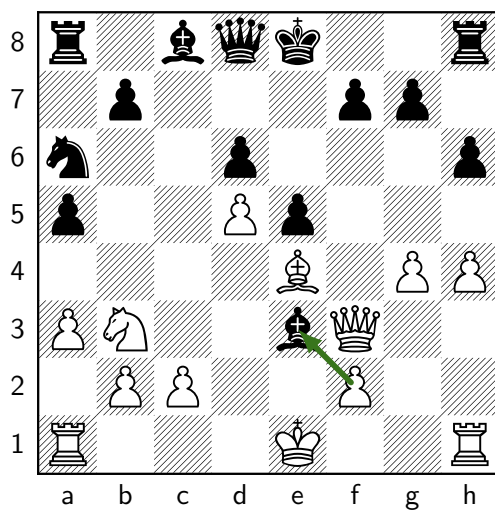


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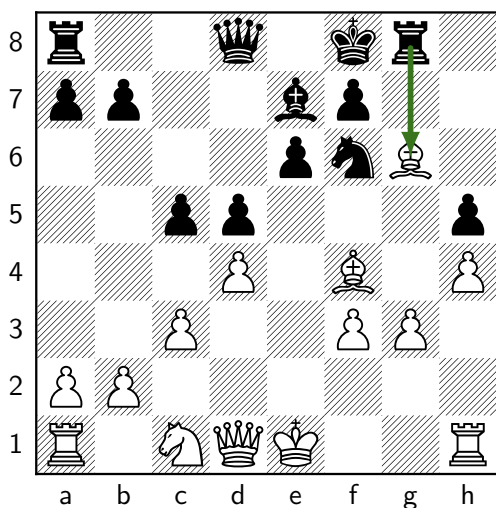
Figure 15: **Move 21, Black: c4**. (a) Our position; Melanie plays c4 (c5c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#3736` (ply 34) [structural distance $d=79$, QAP obj=24]: the master played Re1 ; under μ our Pc4 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our Nd7 as their Ra1 , our Pf7 as their Bb1 ; transferred move c5c4 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#1674` (ply 27) [structural distance $d=65$, QAP obj=24]: the master played Be7 ; under μ our Pc4 plays the role of that game’s Bf1 (a **cross-type** correspondence); cross-type roles: our bb3 as their Rb8 , our pa2 as their Rh8 ; transferred move c5c4 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#3423` (ply 38) [structural distance $d=80$, QAP obj=43]: the master played Ne2 ; under μ our Ra3 plays the role of that game’s Nc3 (a **cross-type** correspondence); cross-type roles: our qc1 as their Rd1 , our pa2 as their Re1 ; transferred move a6b6 . Synthesis: 2 retrieved analogs transferred to c4 (summed QAP objective 48, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



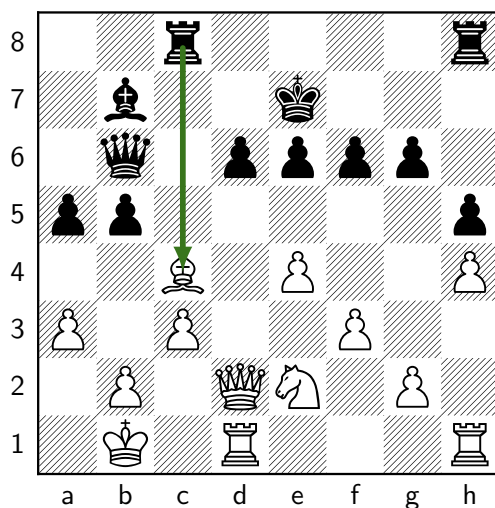
(a)



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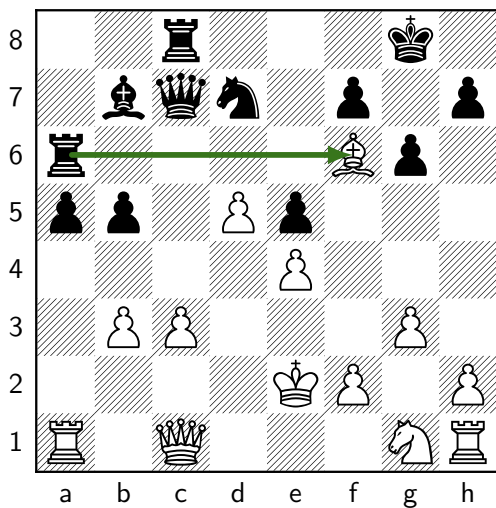


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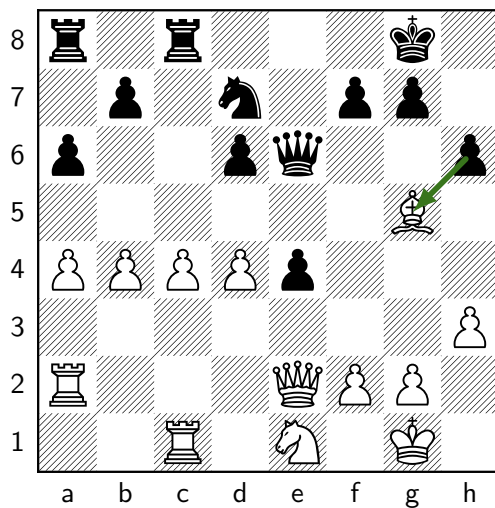


(d)

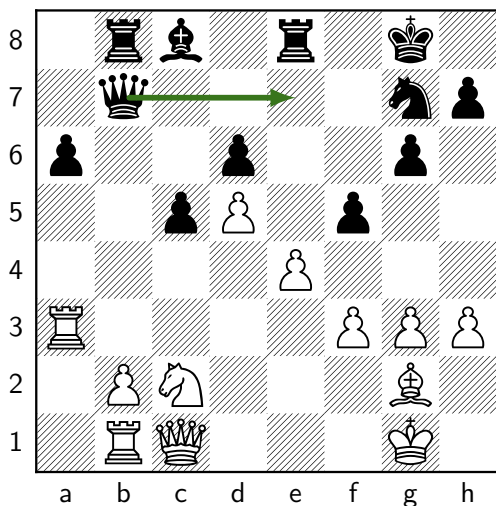
Figure 16: **Move 22, Black: cxb3**. (a) Our position; Melanie plays **cxb3** (**c4b3**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1069` (ply 32) [structural distance $d=77$, QAP obj=42]: the master played **fxe3**; under μ our *Pc5* plays the role of that game’s *Pf2*; cross-type roles: our *pe4* as their *Rh1*, our *Bb7* as their *Pc2*; transferred move **c4b3** (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#817` (ply 39) [structural distance $d=82$, QAP obj=41]: the master played **Rxc6**; under μ our *Pc5* plays the role of that game’s *Rg1* (a **cross-type** correspondence); cross-type roles: our *Bb7* as their *Ra8*, our *Pa5* as their *Qd8*; transferred move **c4b3** (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#1290` (ply 47) [structural distance $d=71$, QAP obj=33]: the master played **Rxc4**; under μ our *Pc5* plays the role of that game’s *Rc1* (a **cross-type** correspondence); cross-type roles: our *Pb5* as their *Bb7*, our *Nd7* as their *Pd6*; transferred move **c4b3** (**endorses this move**). Synthesis: 5 retrieved analogs transferred to **cxb3** (summed QAP objective 152, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



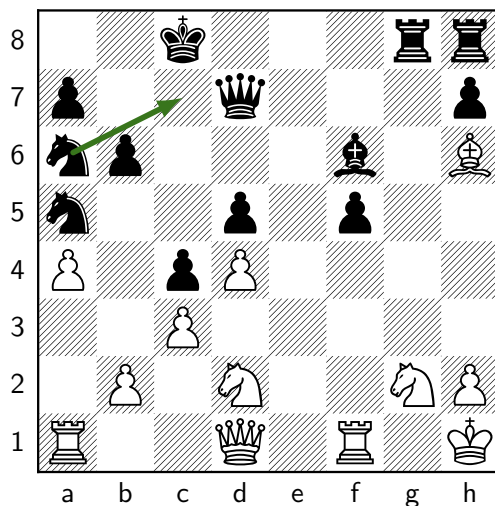
(a)



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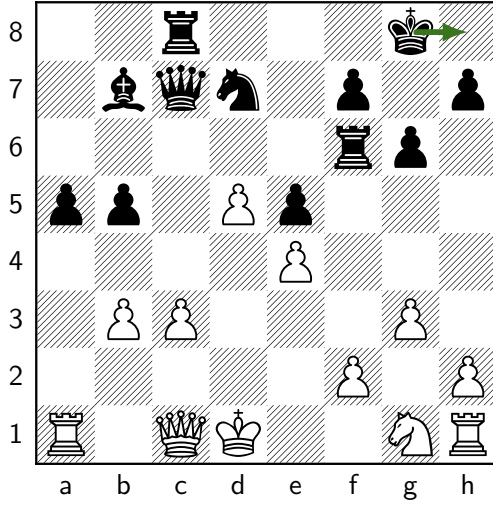


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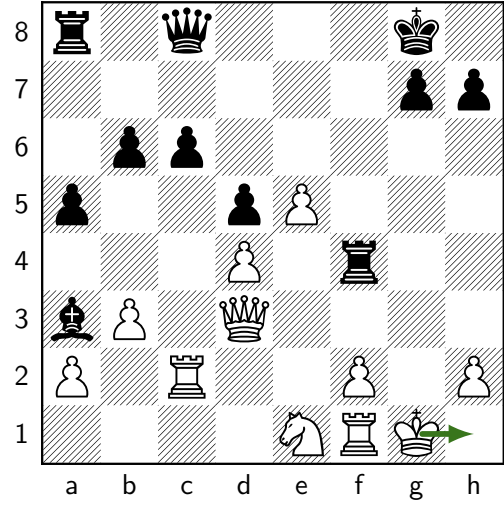


(d)

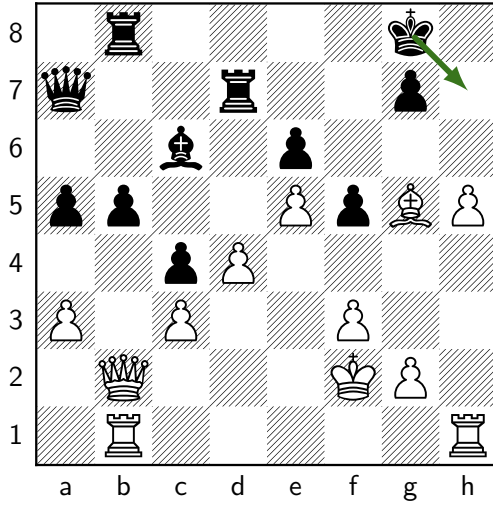
Figure 17: **Move 23, Black: Rxf6.** (a) Our position; Melanie plays **Rxf6** (**a6f6**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3257` (ply 45) [structural distance $d=78$, QAP obj=42]: the master played **hgx5**; under μ our *Nd2* plays the role of that game’s *Ph3* (a **cross-type** correspondence); cross-type roles: our *pe4* as their *Ra8*, our *Qc7* as their *Rc8*; transferred move **d7f6**. (c) Retrieved from `ccr1_top_p0001#523` (ply 61) [structural distance $d=74$, QAP obj=39]: the master played **Qe7**; under μ our *Qc2* plays the role of that game’s *Qb2*; cross-type roles: our *Ph7* as their *Bc8*, our *Nd7* as their *Re8*; transferred move **c7b8**. (d) Retrieved from `ccr1_top_p0000#3802` (ply 43) [structural distance $d=75$, QAP obj=37]: the master played **Nc7**; under μ our *Pa4* plays the role of that game’s *Na3* (a **cross-type** correspondence); cross-type roles: our *Ph7* as their *Rh8*, our *Rc8* as their *Pb6*; transferred move **a5a4**. Synthesis: no retrieved analog transferred to **Rxf6** – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



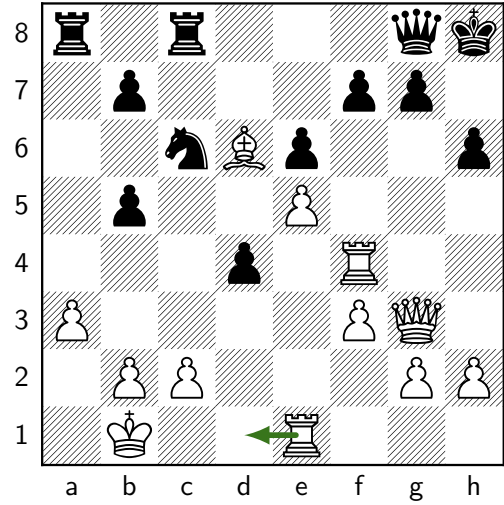
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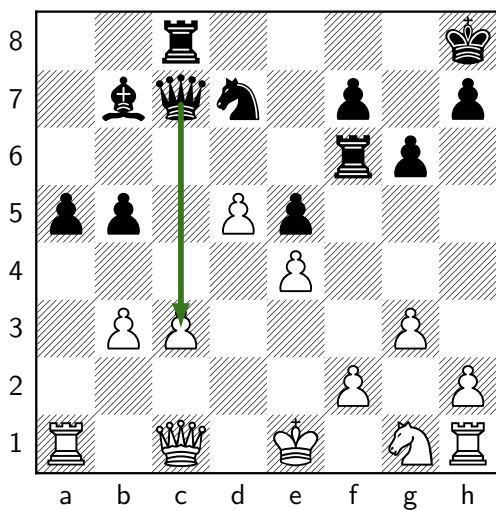


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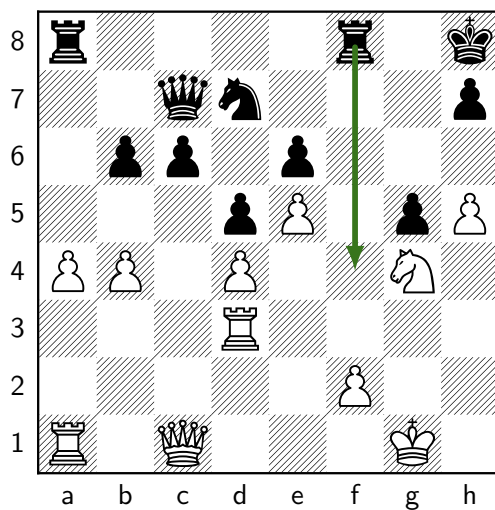


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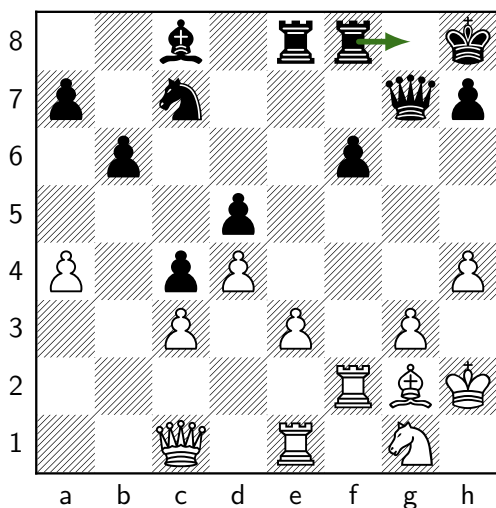
Figure 18: **Move 24, Black: Kh8**. (a) Our position; Melanie plays Kh8 (g8h8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0001#906 (ply 44) [structural distance $d=66$, QAP obj=67]: the master played Kh1; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our Pg6 as their Rf1, our Qc7 as their Rc2; transferred move g8h8 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#1320 (ply 61) [structural distance $d=80$, QAP obj=65]: the master played Kh7; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our qc1 as their Rb8, our Qc7 as their Rd7; transferred move g8h8 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#2291 (ply 52) [structural distance $d=80$, QAP obj=75]: the master played Rd1; under μ our Rc1 plays the role of that game’s Re1; cross-type roles: our Rf6 as their Pc2, our Bb7 as their Rf4; transferred move c8e8. Synthesis: 2 retrieved analogs transferred to Kh8 (summed QAP objective 131, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



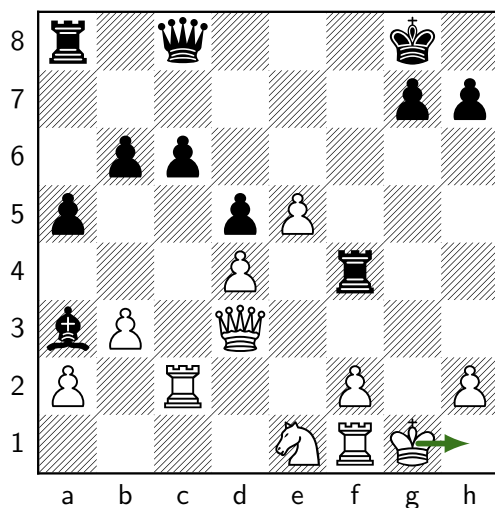
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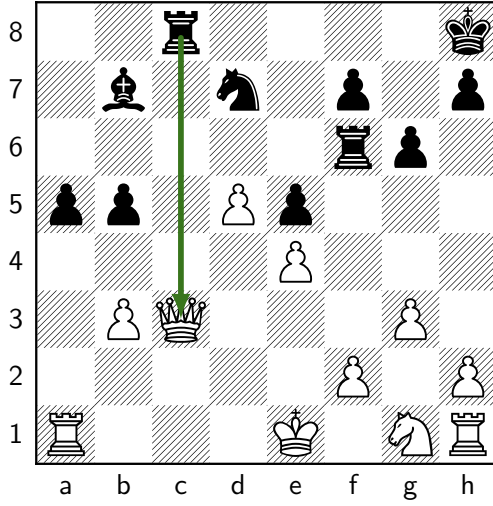


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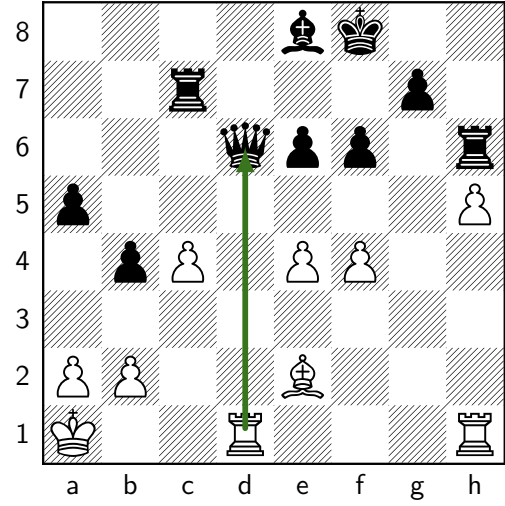


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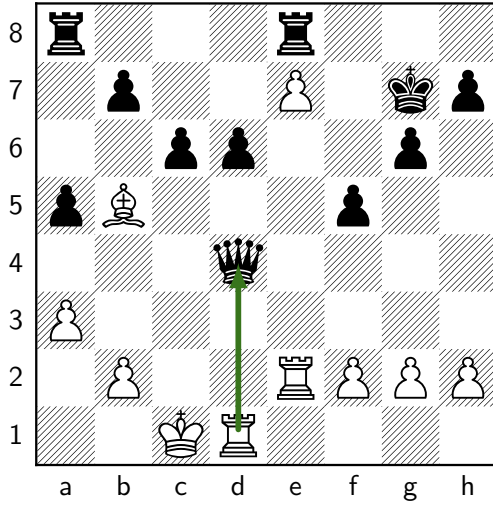
Figure 19: **Move 25, Black: Qxc3+.** (a) Our position; Melanie plays Qxc3+ (c7c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#290` (ply 53) [structural distance $d=81$, QAP obj=94]: the master played Rf4; under μ our Rf3 plays the role of that game’s Rf1; cross-type roles: our Qc7 as their Ra8, our Bb7 as their Qc7; transferred move f6f3. (c) Retrieved from `ccr1_top_p0002#3223` (ply 59) [structural distance $d=69$, QAP obj=75]: the master played Rg8; under μ our Rc1 plays the role of that game’s Rf1; cross-type roles: our Pa5 as their Bc8, our Qc7 as their Re8; transferred move c8g8. (d) Retrieved from `ccr1_top_p0001#906` (ply 44) [structural distance $d=69$, QAP obj=68]: the master played Kh1; under μ our Kh1 plays the role of that game’s Kg1; cross-type roles: our Pg6 as their Rf1, our Qc7 as their Rc2; transferred move h8g8. Synthesis: no retrieved analog transferred to Qxc3+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



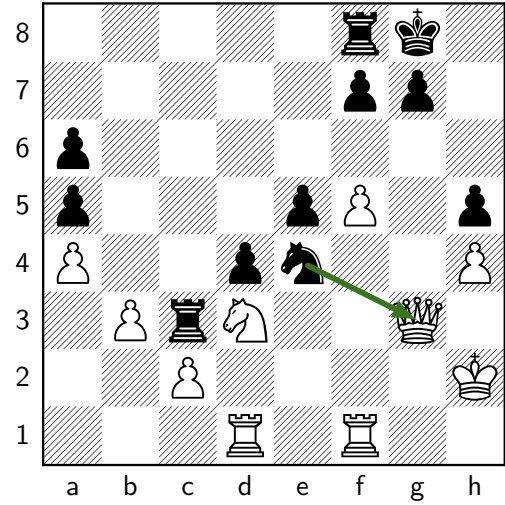
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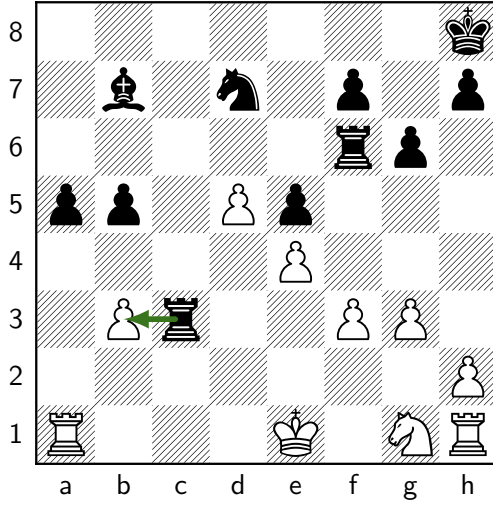


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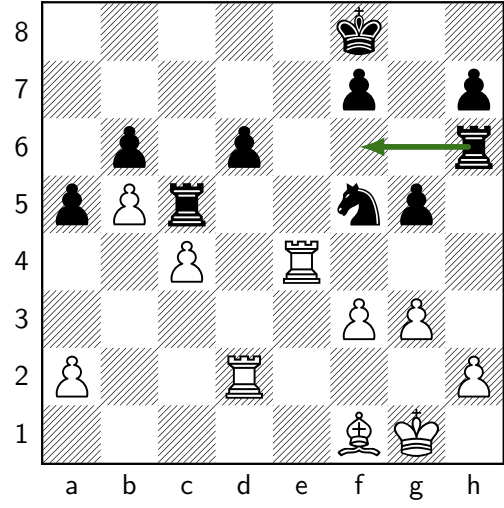


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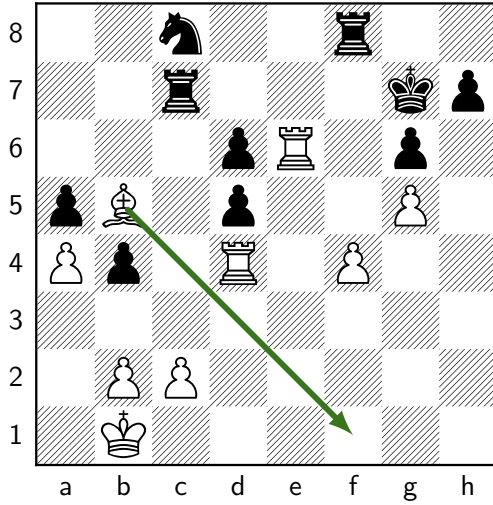
Figure 20: **Move 26, Black: Rxc3.** (a) Our position; Melanie plays **Rxc3** (**c8c3**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `cctl_top_p0000#2351` (ply 58) [structural distance $d=89$, QAP obj=66]: the master played **Rxd6**; under μ our *Rc1* plays the role of that game’s *Rd1*; cross-type roles: our *Bb7* as their *Rh1*, our *Nd7* as their *Be2*; transferred move **c8c3** (**endorses this move**). (c) Retrieved from `cctl_top_p0001#1427` (ply 48) [structural distance $d=85$, QAP obj=37]: the master played **Rxd4**; under μ our *Rc1* plays the role of that game’s *Rd1*; cross-type roles: our *ng1* as their *Re2*, our *Bb7* as their *Pf2*; transferred move **c8c3** (**endorses this move**). (d) Retrieved from `cctl_top_p0000#1588` (ply 61) [structural distance $d=86$, QAP obj=37]: the master played **Nxg3**; under μ our *Rc1* plays the role of that game’s *Ne5* (a **cross-type** correspondence); cross-type roles: our *Ph7* as their *Rf8*, our *Nd7* as their *Pa6*; transferred move **c8c3** (**endorses this move**). Synthesis: 4 retrieved analogs transferred to **Rxc3** (summed QAP objective 175, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



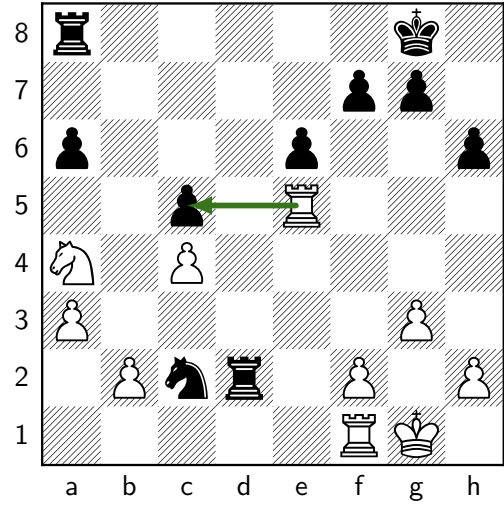
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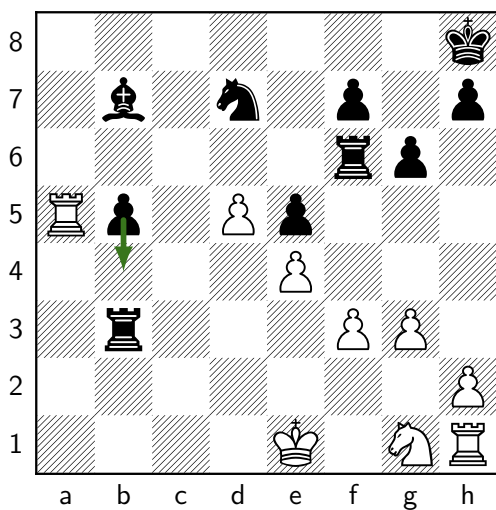


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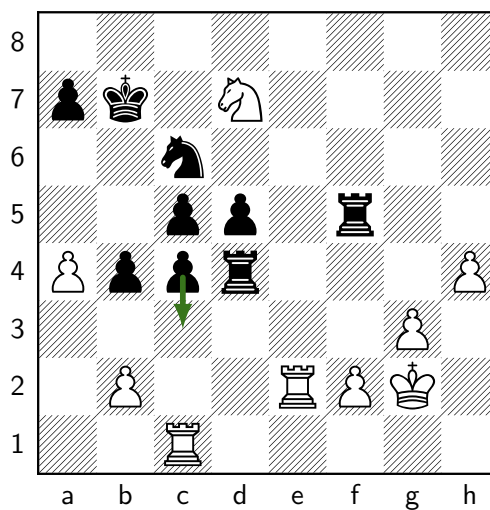


(d)

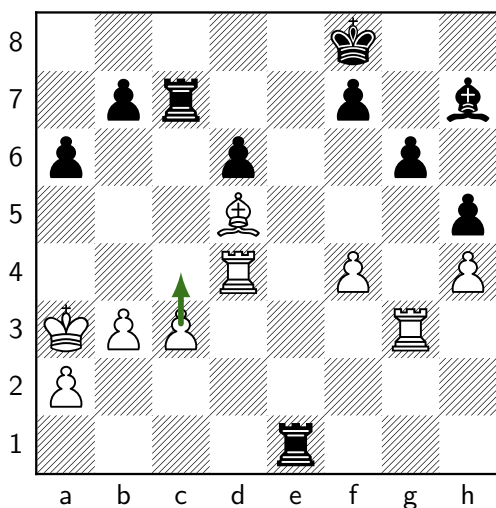
Figure 21: **Move 27, Black: Rxb3.** (a) Our position; Melanie plays **Rxb3** (**c3b3**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#709` (ply 69) [structural distance $d=63$, QAP obj=55]: the master played **Rf6**; under μ our **Rf3** plays the role of that game’s **Rh3**; cross-type roles: our **Nd7** as their **Pa5**, our **Bb7** as their **Nf5**; transferred move **f6d6**. (c) Retrieved from `ccrl_top_p0003#1608` (ply 62) [structural distance $d=60$, QAP obj=54]: the master played **Bf1**; under μ our **Pb4** plays the role of that game’s **Bb5** (a **cross-type** correspondence); cross-type roles: our **Rf6** as their **Pb2**, our **Bb7** as their **rf8**; transferred move **b5b4**. (d) Retrieved from `ccrl_top_p0000#385` (ply 50) [structural distance $d=52$, QAP obj=51]: the master played **Rxc5**; under μ our **Rc6** plays the role of that game’s **Re5**; cross-type roles: our **pe4** as their **Rf1**, our **ra1** as their **nc2**; transferred move **c3f3**. Synthesis: no retrieved analog transferred to **Rxb3** – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



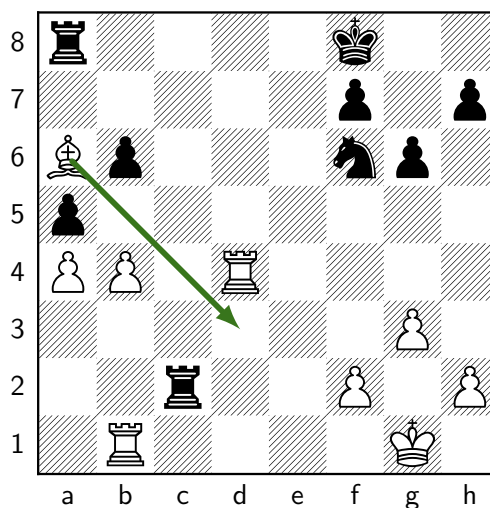
(a)



(b)

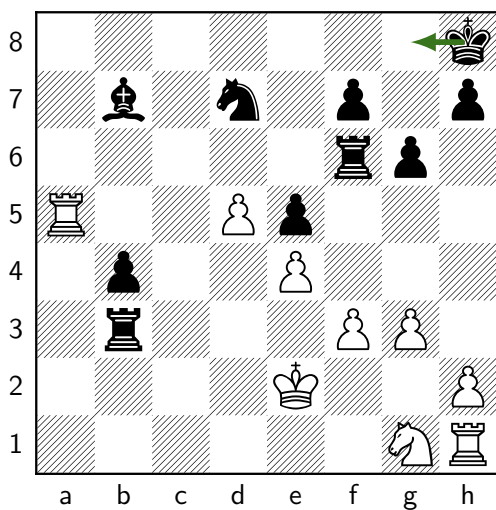


(c)

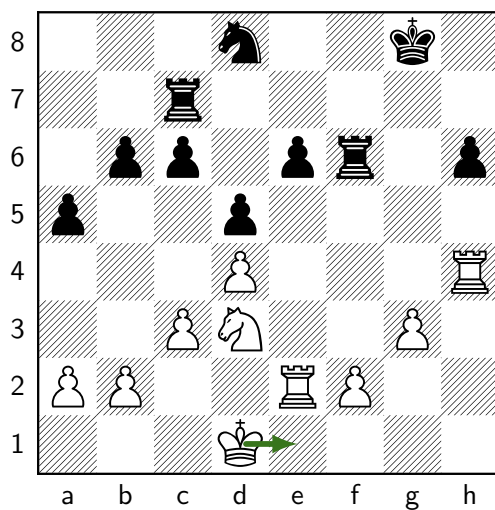


(d)

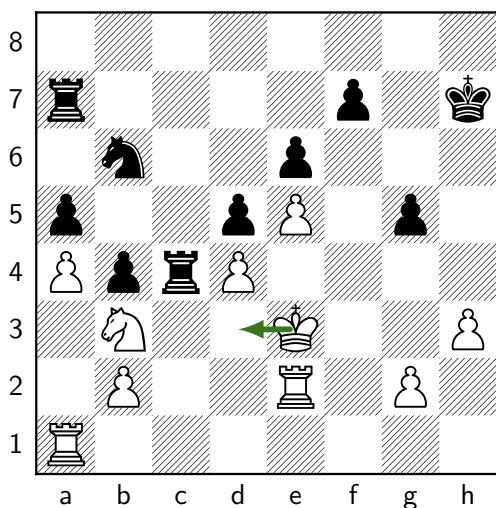
Figure 22: **Move 28, Black: b4**. (a) Our position; Melanie plays b4 (b5b4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#1545` (ply 65) [structural distance $d=48$, QAP obj=30]: the master played c3; under μ our Pb_4 plays the role of that game’s Pc_5 ; cross-type roles: our Bb_7 as their nd_7 , our ng_1 as their re_2 ; transferred move b5b4 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#2329` (ply 70) [structural distance $d=46$, QAP obj=29]: the master played c4; under μ our Pb_4 plays the role of that game’s Pc_3 ; cross-type roles: our Ph_7 as their re_1 , our Nd_7 as their Rd_4 ; transferred move b5b4 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#1067` (ply 52) [structural distance $d=52$, QAP obj=24]: the master played Bd3; under μ our Pb_4 plays the role of that game’s Ba_6 (a **cross-type** correspondence); cross-type roles: our Ph_7 as their Rd_4 , our Bb_7 as their pa_5 ; transferred move b5b4 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to b4 (summed QAP objective 83, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



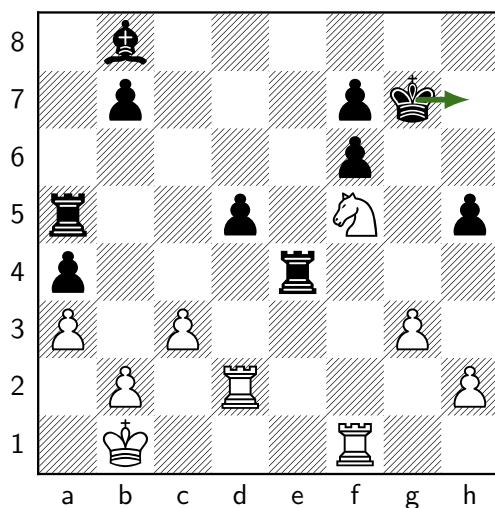
(a)



(b)



(c)



(d)

Figure 23: **Move 29, Black: Kg8**. (a) Our position; Melanie plays Kg8 (**h8g8**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#150` (ply 56) [structural distance $d=41$, QAP obj=35]: the master played Ke1; under μ our *Kh1* plays the role of that game’s *Kd1*; cross-type roles: our *ra5* as their *Pa2*, our *rh1* as their *Pg3*; transferred move **h8g8 (endorses this move)**. (c) Retrieved from `ccrl_top_p0002#2105` (ply 68) [structural distance $d=65$, QAP obj=18]: the master played Kd3; under μ our *Kh1* plays the role of that game’s *Ke3*; cross-type roles: our *Ph7* as their *Ra1*, our *Pf7* as their *Re2*; transferred move **h8g8 (endorses this move)**. (d) Retrieved from `ccrl_top_p0000#868` (ply 65) [structural distance $d=53$, QAP obj=3]: the master played Kh7; under μ our *Kh1* plays the role of that game’s *Kg2*; cross-type roles: our *Rf6* as their *Bb8*, our *Nd7* as their *Pb7*; transferred move **h8g8 (endorses this move)**. Synthesis: 4 retrieved analogs transferred to Kg8 (summed QAP objective 88, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

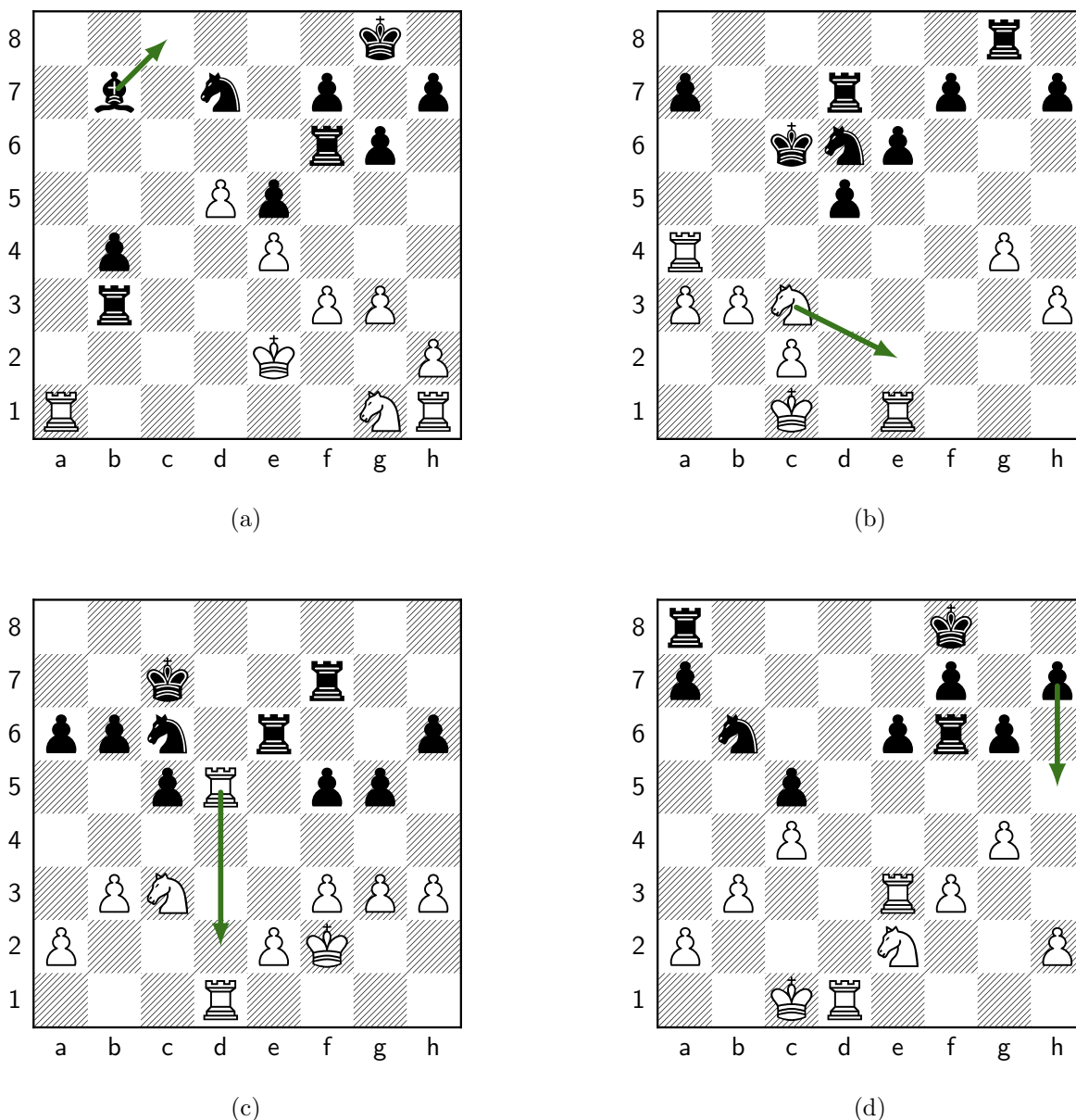
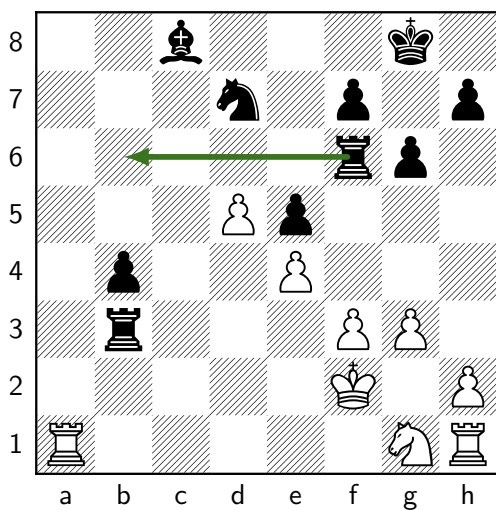
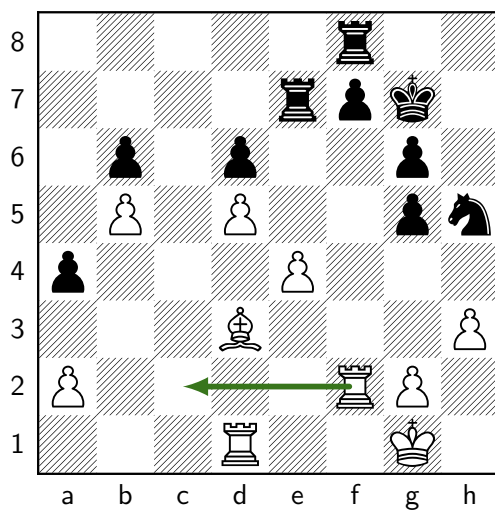


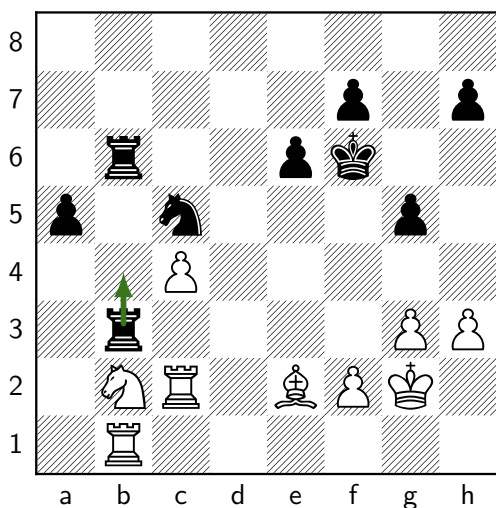
Figure 24: **Move 30, Black: Bc8.** (a) Our position; Melanie plays Bc8 (b7c8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1748` (ply 56) [structural distance $d=65$, QAP obj=39]: the master played Ne2; under μ our Bb2 plays the role of that game’s Nc3 (a **cross-type** correspondence); cross-type roles: our Ph7 as their Re1, our ra1 as their Pc2; transferred move b7c8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#199` (ply 66) [structural distance $d=69$, QAP obj=22]: the master played R5d2; under μ our Bb2 plays the role of that game’s Rd5 (a **cross-type** correspondence); cross-type roles: our Nd7 as their Pa2, our ra1 as their Nc3; transferred move b7c8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#2805` (ply 57) [structural distance $d=65$, QAP obj=40]: the master played h5; under μ our Ph2 plays the role of that game’s Ph2; cross-type roles: our Nd7 as their Ra8, our Bb7 as their Nb6; transferred move h7h5. Synthesis: 2 retrieved analogs transferred to Bc8 (summed QAP objective 61, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



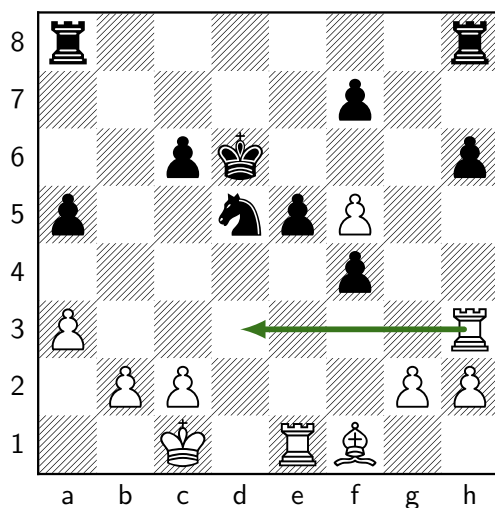
(a)



(b)

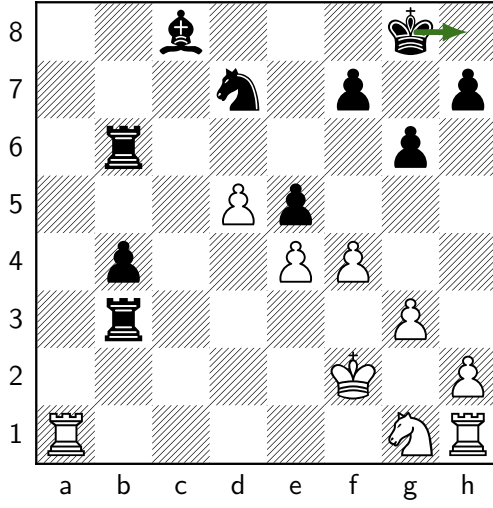


(c)

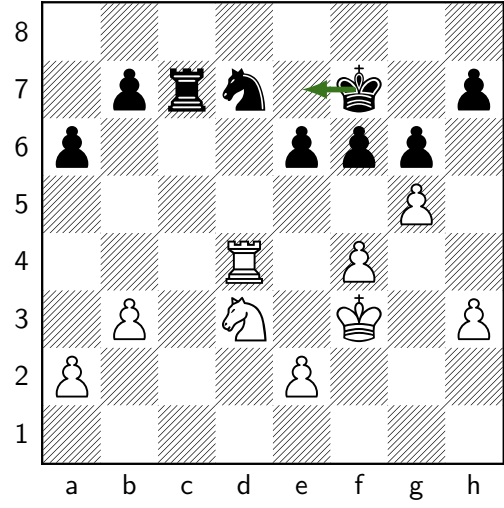


(d)

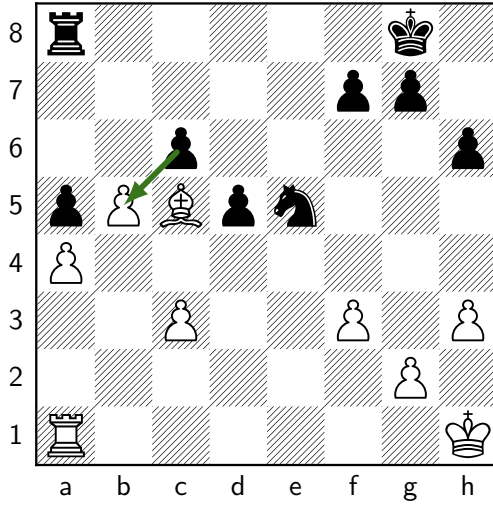
Figure 25: **Move 31, Black: Rb6**. (a) Our position; Melanie plays Rb6 (f6b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#691` (ply 66) [structural distance $d=67$, QAP obj=43]: the master played Rc2; under μ our Rf3 plays the role of that game’s Rf2; cross-type roles: our Nd7 as their Rd1, our ng1 as their Bd3; transferred move f6b6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#447` (ply 61) [structural distance $d=51$, QAP obj=6]: the master played R3b4; under μ our Rf3 plays the role of that game’s Rb6; cross-type roles: our Bc8 as their Ph7, our Pg6 as their Rb6; transferred move f6b6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1441` (ply 54) [structural distance $d=55$, QAP obj=58]: the master played Rd3; under μ our Rf3 plays the role of that game’s Rh3; cross-type roles: our Pg6 as their Bf1, our Nd7 as their Pg2; transferred move f6e6. Synthesis: 2 retrieved analogs transferred to Rb6 (summed QAP objective 49, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



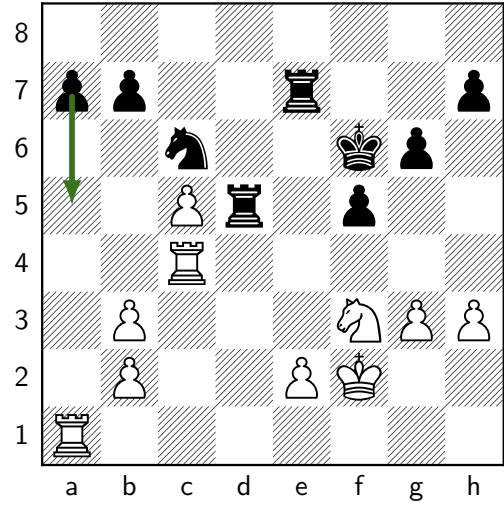
(a)



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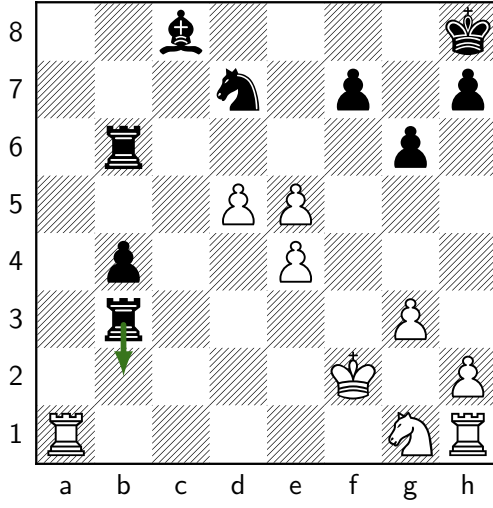


(c)

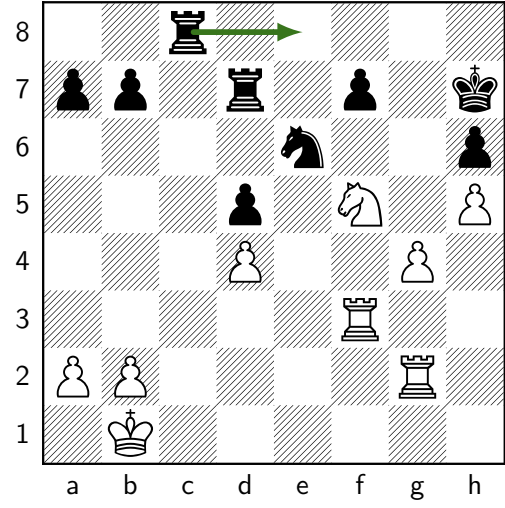


(d)

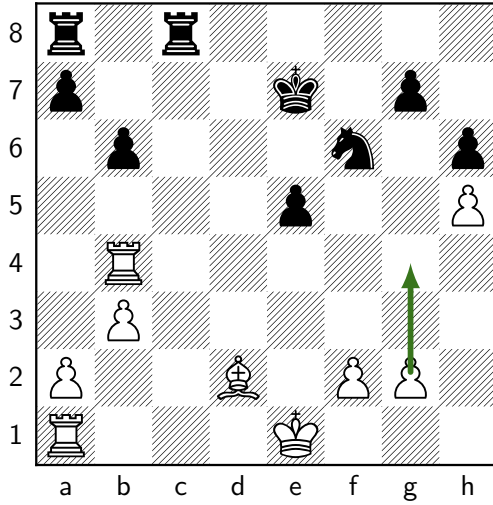
Figure 26: **Move 32, Black: Kh8**. (a) Our position; Melanie plays Kh8 (g8h8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#582` (ply 67) [structural distance $d=60$, QAP obj=49]: the master played Ke7; under μ our Kg1 plays the role of that game’s Kf2; cross-type roles: our Nd7 as their Rc7, our Pe5 as their Nd7; transferred move g8h8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#3858` (ply 65) [structural distance $d=61$, QAP obj=41]: the master played cxb5; under μ our Pe4 plays the role of that game’s Pc3; cross-type roles: our Pg6 as their Ra8, our Bc8 as their Pf7; transferred move e5f4. (d) Retrieved from `ccrl_top_p0003#131` (ply 61) [structural distance $d=55$, QAP obj=22]: the master played a5; under μ our Bc1 plays the role of that game’s Pa2 (a **cross-type** correspondence); cross-type roles: our rh1 as their Pb7, our pf4 as their Rd5; transferred move c8a6. Synthesis: 1 retrieved analog transferred to Kh8 (summed QAP objective 49, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



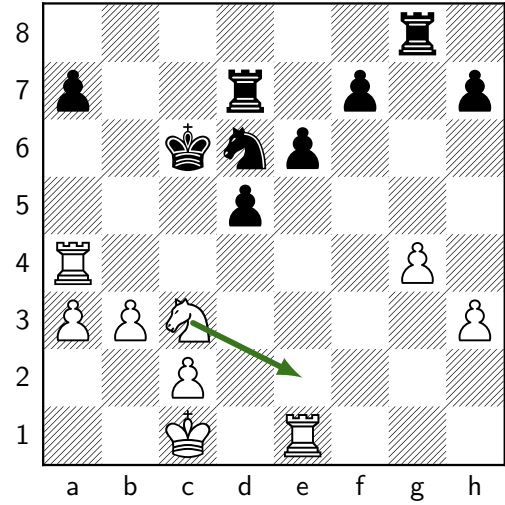
(a)



(b)

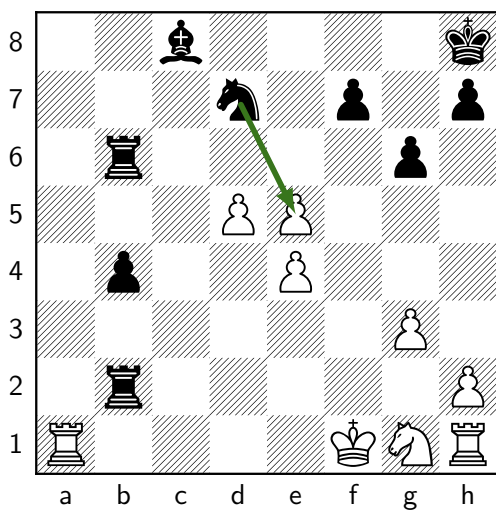


(c)

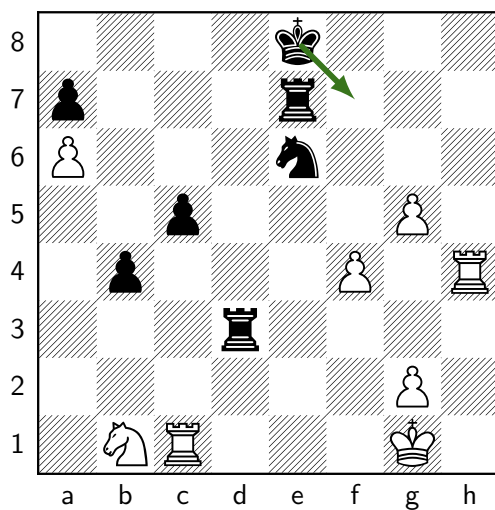


(d)

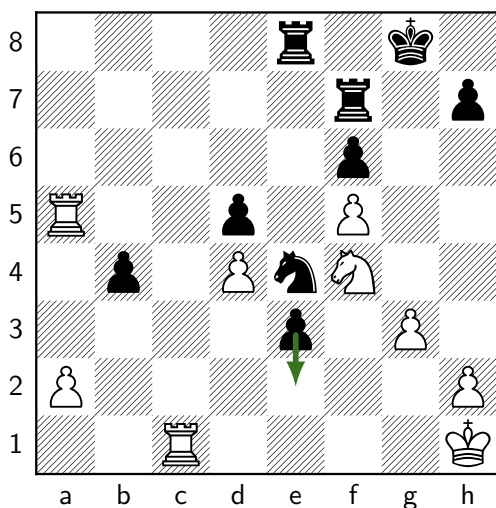
Figure 27: **Move 33, Black: Rb2+.** (a) Our position; Melanie plays Rb2+ (b3b2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#768` (ply 69) [structural distance $d=56$, QAP obj=29]: the master played Re8; under μ our Rb6 plays the role of that game’s Rc1; cross-type roles: our Rb6 as their Pa7, our Pf7 as their Rd7; transferred move b3b2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2738` (ply 54) [structural distance $d=55$, QAP obj=35]: the master played g4; under μ our Rb3 plays the role of that game’s Pg2 (a **cross-type** correspondence); cross-type roles: our Pf7 as their Ra1, our Rb3 as their Bd2; transferred move b6c6. (d) Retrieved from `ccr1_top_p0001#1748` (ply 56) [structural distance $d=41$, QAP obj=30]: the master played Ne2; under μ our Nd2 plays the role of that game’s Nc3; cross-type roles: our Bc8 as their Pa3, our ra1 as their Pb3; transferred move d7b8. Synthesis: 1 retrieved analog transferred to Rb2+ (summed QAP objective 29, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



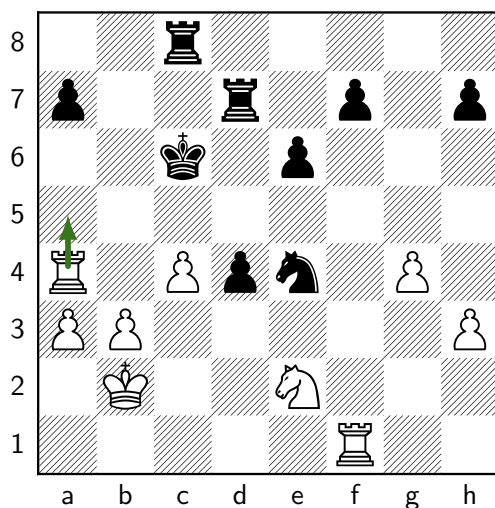
(a)



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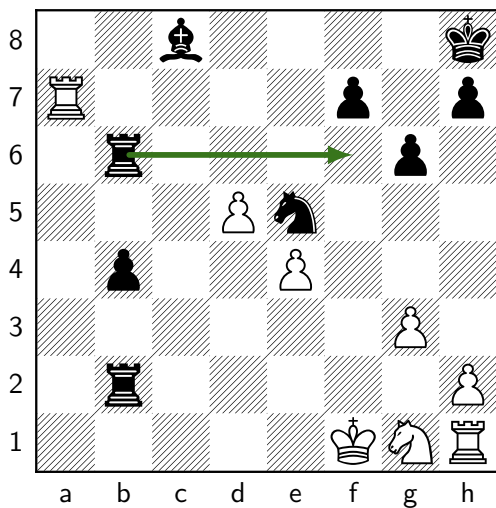


(c)

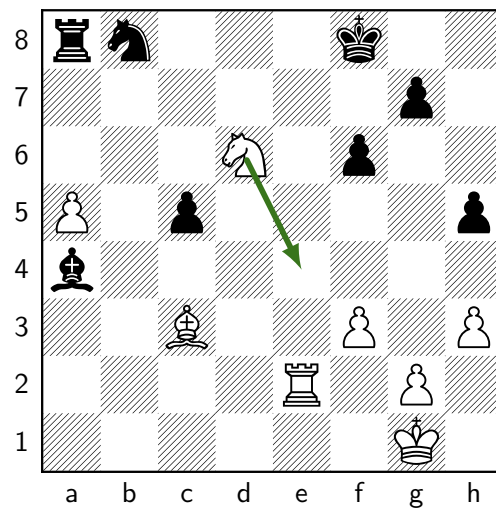


(d)

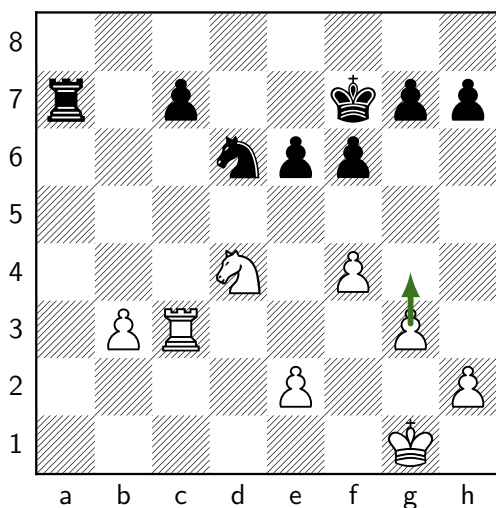
Figure 28: **Move 34, Black: Nxe5.** (a) Our position; Melanie plays **Nxe5** (d7e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2050` (ply 73) [structural distance $d=47$, QAP obj=42]: the master played **Kf7**; under μ our *Kh1* plays the role of that game’s *Ke1*; cross-type roles: our *Bc8* as their *Re7*, our *ng1* as their *Pb4*; transferred move **h8g8**. (c) Retrieved from `ccr1_top_p0001#469` (ply 79) [structural distance $d=44$, QAP obj=38]: the master played **e2**; under μ our *Ph2* plays the role of that game’s *Pe6*; cross-type roles: our *Bc8* as their *Re8*, our *Nd7* as their *ra5*; transferred move **h7h5**. (d) Retrieved from `ccr1_top_p0001#1748` (ply 66) [structural distance $d=47$, QAP obj=30]: the master played **Ra5**; under μ our *Nd2* plays the role of that game’s *Ra4* (a **cross-type** correspondence); cross-type roles: our *Rb6* as their *Pa3*, our *pe4* as their *ne4*; transferred move **d7c5**. Synthesis: no retrieved analog transferred to **Nxe5** – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



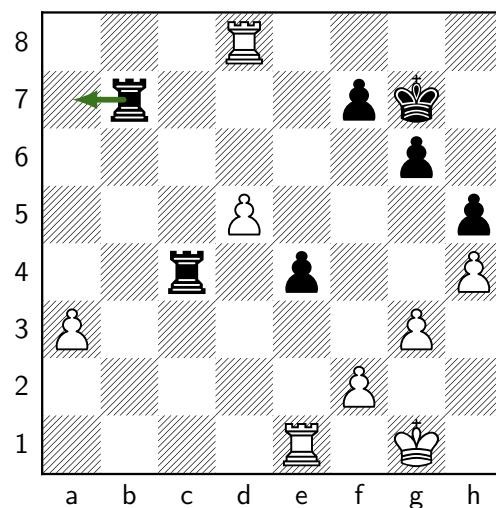
(a)



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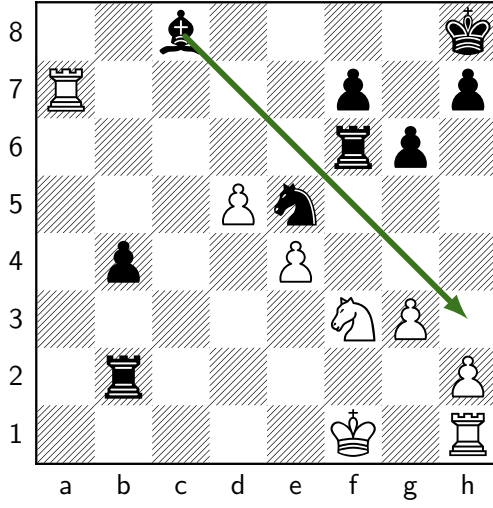


(c)

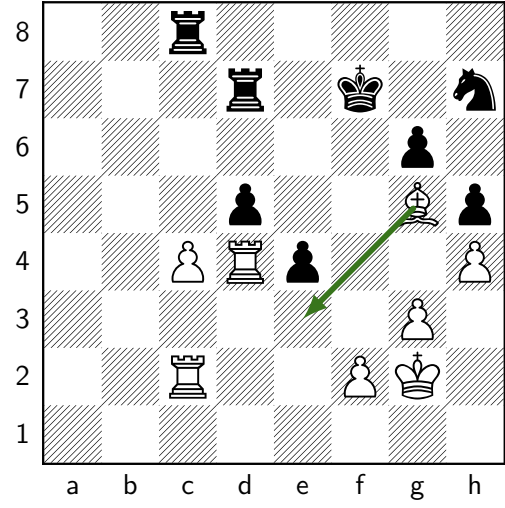


(d)

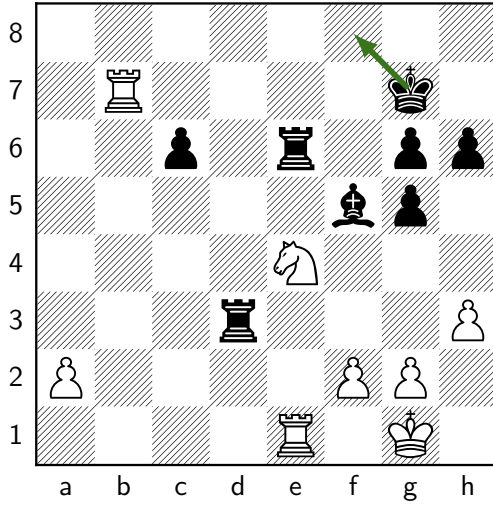
Figure 29: **Move 35, Black: Rf6+.** (a) Our position; Melanie plays Rf6+ (b6f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#2019` (ply 78) [structural distance $d=57$, QAP obj=23]: the master played Ne4; under μ our Bc1 plays the role of that game’s Nd6 (a **cross-type** correspondence); cross-type roles: our Ne5 as their Re2, our Rb2 as their Bc3; transferred move c8e6. (c) Retrieved from `ccrl_top_p0001#3376` (ply 56) [structural distance $d=56$, QAP obj=17]: the master played g4; under μ our Rb3 plays the role of that game’s Pg3 (a **cross-type** correspondence); cross-type roles: our Rb2 as their Nd4, our Bc8 as their Pf4; transferred move b6d6. (d) Retrieved from `ccrl_top_p0002#3238` (ply 75) [structural distance $d=57$, QAP obj=14]: the master played Ra7; under μ our Rb3 plays the role of that game’s Rb2; cross-type roles: our Ph7 as their rd8, our Rb2 as their Pf7; transferred move b6b8. Synthesis: no retrieved analog transferred to Rf6+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



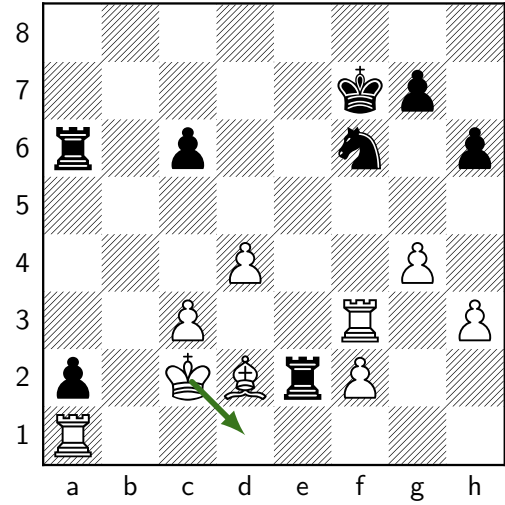
(a)



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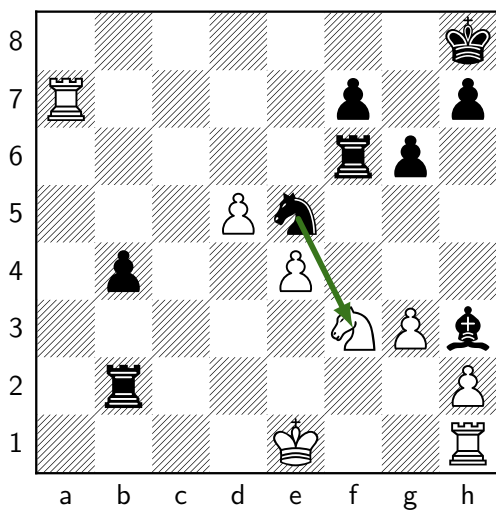


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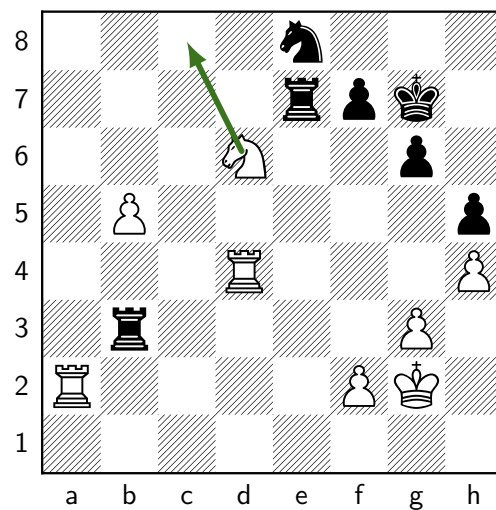


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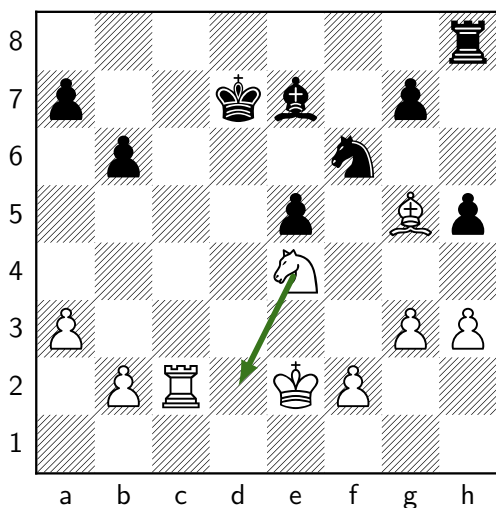
Figure 30: **Move 36, Black: Bh3+.** (a) Our position; Melanie plays Bh3+ (c8h3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#883` (ply 76) [structural distance $d=51$, QAP obj=104]: the master played Be3; under μ our Ne4 plays the role of that game’s Bg5 (a **cross-type** correspondence); cross-type roles: our Pf7 as their Rc2, our Bc8 as their Pg3; transferred move e5c6. (c) Retrieved from `ccrl_top_p0003#2233` (ply 63) [structural distance $d=58$, QAP obj=52]: the master played Kf8; under μ our Kh1 plays the role of that game’s Kg2; cross-type roles: our Ne5 as their Bf5, our Bc8 as their Pg5; transferred move h8g8. (d) Retrieved from `ccrl_top_p0000#3490` (ply 76) [structural distance $d=52$, QAP obj=40]: the master played Kd1; under μ our Kh1 plays the role of that game’s Kc2; cross-type roles: our nf3 as their pa2, our nf3 as their Bd2; transferred move h8g8. Synthesis: no retrieved analog transferred to Bh3+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



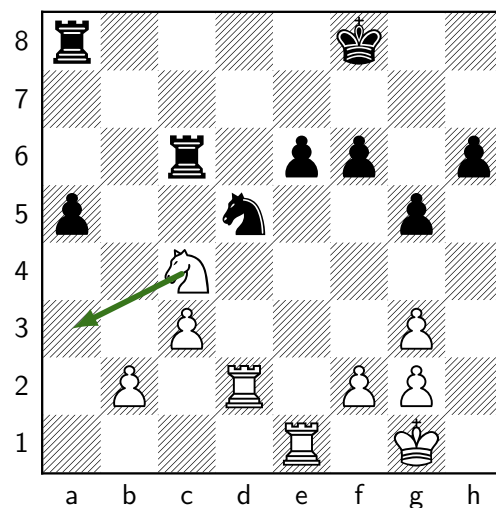
(a)



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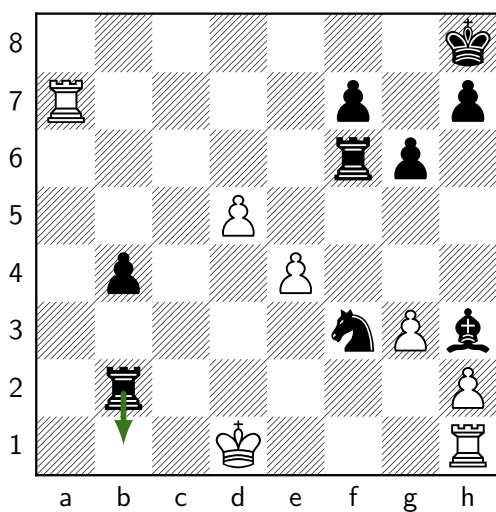


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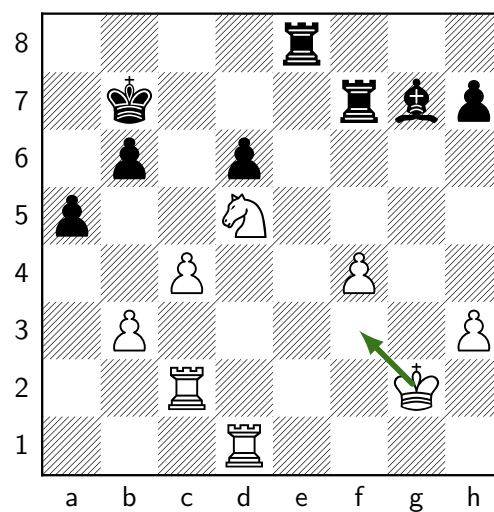


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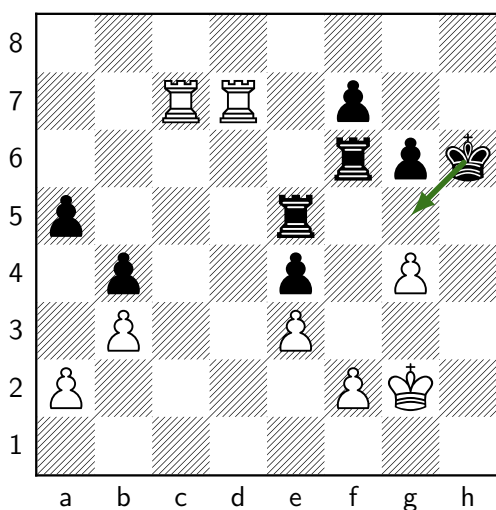
Figure 31: **Move 37, Black: Nxf3+.** (a) Our position; Melanie plays Nxf3+ (e5f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2630` (ply 82) [structural distance $d=54$, QAP obj=46]: the master played Nc8; under μ our Ne4 plays the role of that game’s Nd6; cross-type roles: our nf3 as their Pb5, our Bh3 as their ph5; transferred move e5c4. (c) Retrieved from `ccr1_top_p0001#3403` (ply 76) [structural distance $d=66$, QAP obj=45]: the master played Nd2; under μ our Ne4 plays the role of that game’s Ne4; cross-type roles: our Bh3 as their Ph3, our Rf6 as their Bg5; transferred move e5d7. (d) Retrieved from `ccr1_top_p0002#723` (ply 66) [structural distance $d=41$, QAP obj=39]: the master played Na3; under μ our Ne4 plays the role of that game’s Nc4; cross-type roles: our ra7 as their Pg2, our Ne5 as their Pc3; transferred move e5g4. Synthesis: no retrieved analog transferred to Nxf3+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



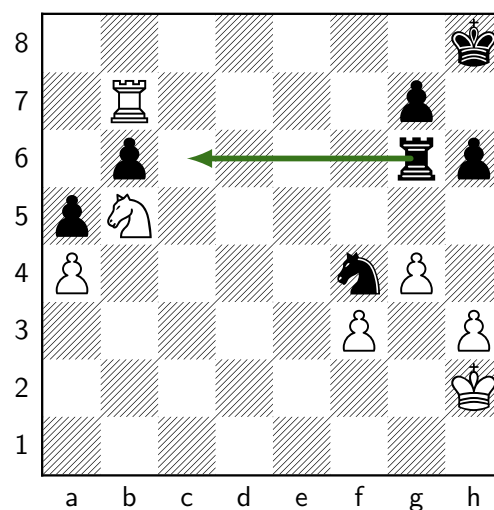
(a)



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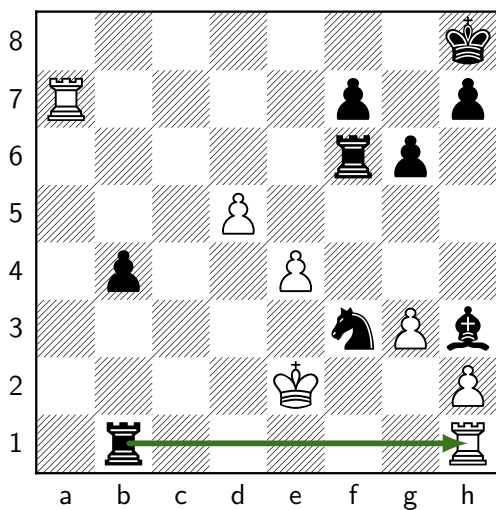


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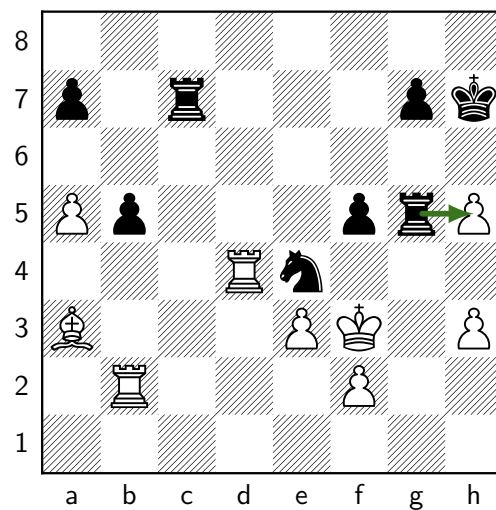


(d)

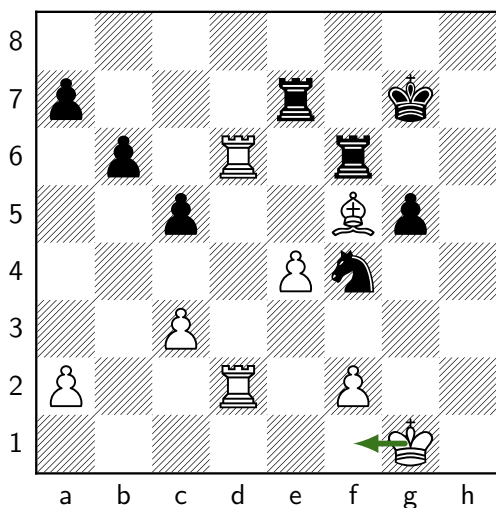
Figure 32: **Move 38, Black: Rb1+.** (a) Our position; Melanie plays Rb1+ (b2b1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#793` (ply 70) [structural distance $d=41$, QAP obj=28]: the master played Kf3; under μ our Kh1 plays the role of that game’s Kg2; cross-type roles: our Bh3 as their Pc4, our rh1 as their pd6; transferred move h8g8. (c) Retrieved from `ccrl_top_p0001#1312` (ply 83) [structural distance $d=44$, QAP obj=28]: the master played Kg5; under μ our Kh1 plays the role of that game’s Kh3; cross-type roles: our Nf3 as their rd7, our Bh3 as their Pg6; transferred move h8g7. (d) Retrieved from `ccrl_top_p0001#1203` (ply 87) [structural distance $d=44$, QAP obj=14]: the master played Rc6; under μ our Rf3 plays the role of that game’s Rg3; cross-type roles: our Bh3 as their nb5, our rh1 as their pf3; transferred move f6b6. Synthesis: no retrieved analog transferred to Rb1+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



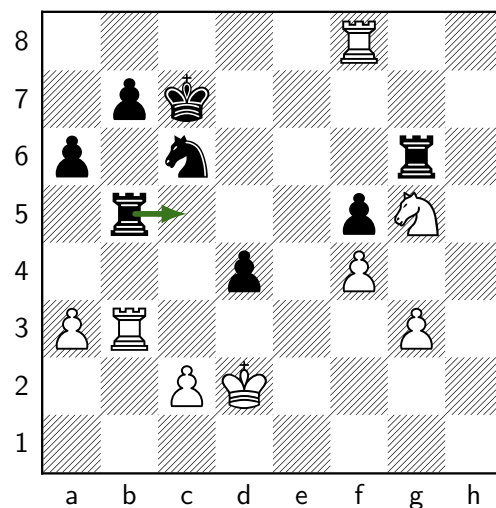
(a)



(b)

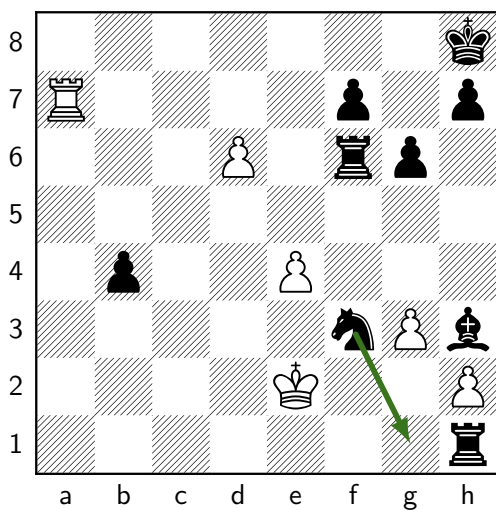


(c)

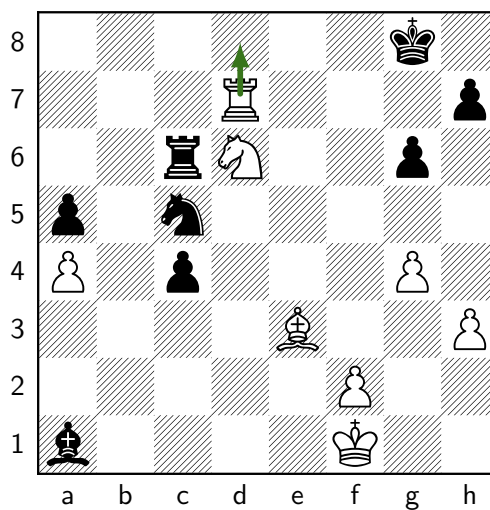


(d)

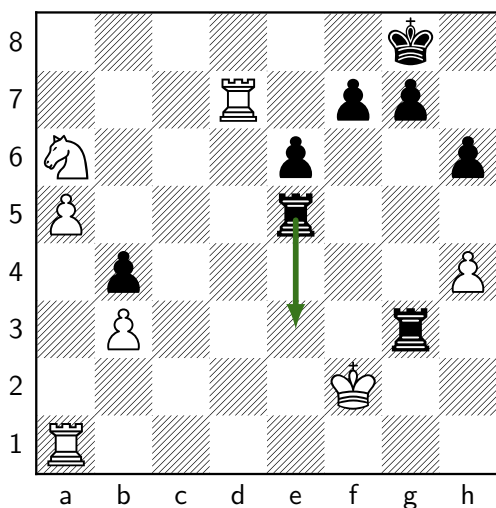
Figure 33: **Move 39, Black: Rxh1.** (a) Our position; Melanie plays Rxh1 (b1h1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#484` (ply 65) [structural distance $d=50$, QAP obj=40]: the master played Rxh5; under μ our *Rb8* plays the role of that game’s *Rg4*; cross-type roles: our *ra7* as their *Pa7*, our *Bh3* as their *Rc7*; transferred move b1b2. (c) Retrieved from `ccr1_top_p0001#1727` (ply 80) [structural distance $d=46$, QAP obj=27]: the master played Kf1; under μ our *Kh1* plays the role of that game’s *Kg1*; cross-type roles: our *Rb1* as their *Pe4*, our *pe4* as their *nf4*; transferred move h8g8. (d) Retrieved from `ccr1_top_p0000#524` (ply 85) [structural distance $d=51$, QAP obj=26]: the master played Rc5; under μ our *Rb8* plays the role of that game’s *Rb4*; cross-type roles: our *Ph7* as their *rf8*, our *ra7* as their *Pa6*; transferred move b1b2. Synthesis: no retrieved analog transferred to Rxh1 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



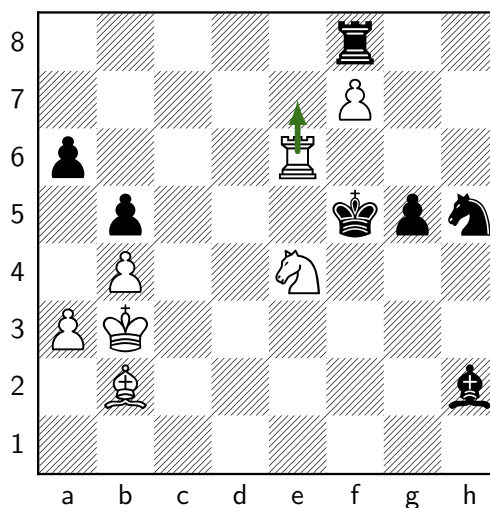
(a)



(b)

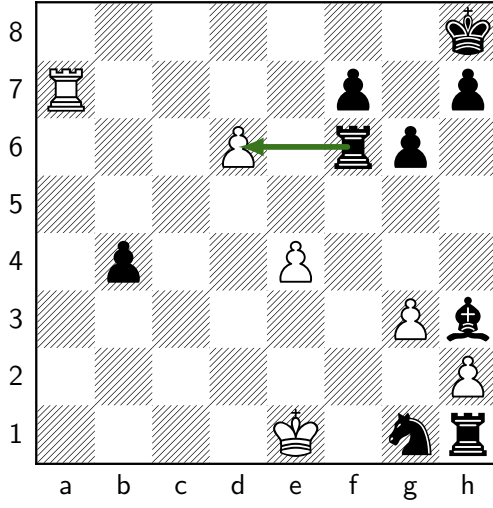


(c)

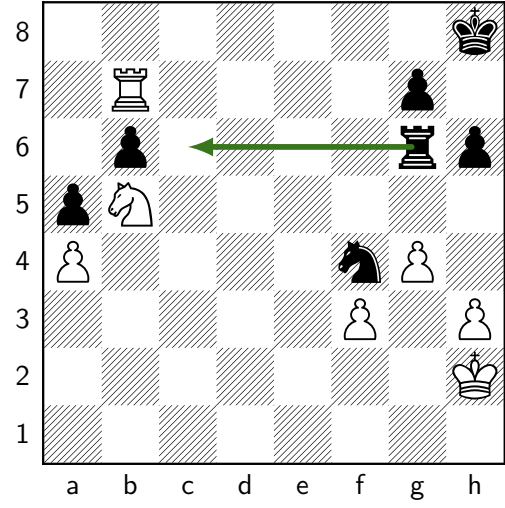


(d)

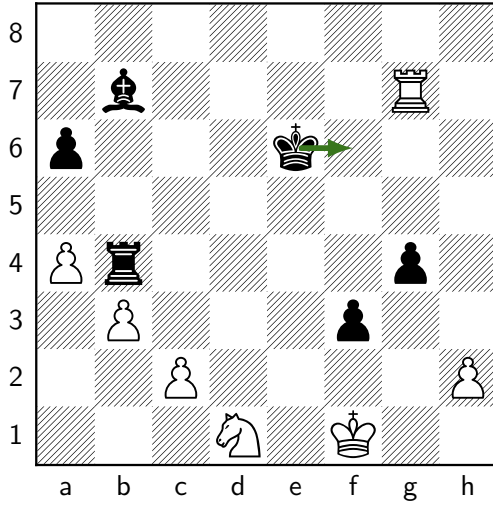
Figure 34: **Move 40, Black: Ng1+.** (a) Our position; Melanie plays Ng1+ (f3g1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1172` (ply 82) [structural distance $d=42$, QAP obj=27]: the master played Rd8+; under μ our Nf6 plays the role of that game’s Rd7 (a **cross-type** correspondence); cross-type roles: our pg3 as their ba1, our Rh1 as their Be3; transferred move f3g1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#921` (ply 81) [structural distance $d=36$, QAP obj=28]: the master played Ree3; under μ our Rf3 plays the role of that game’s Re4; cross-type roles: our Bh3 as their rd7, our Rh1 as their Pf7; transferred move f6f5. (d) Retrieved from `ccrl_top_p0003#1118` (ply 92) [structural distance $d=43$, QAP obj=26]: the master played Re7; under μ our Nf6 plays the role of that game’s Re6 (a **cross-type** correspondence); cross-type roles: our Rf6 as their Bb2, our pe4 as their bh2; transferred move f3h2. Synthesis: 1 retrieved analog transferred to Ng1+ (summed QAP objective 27, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



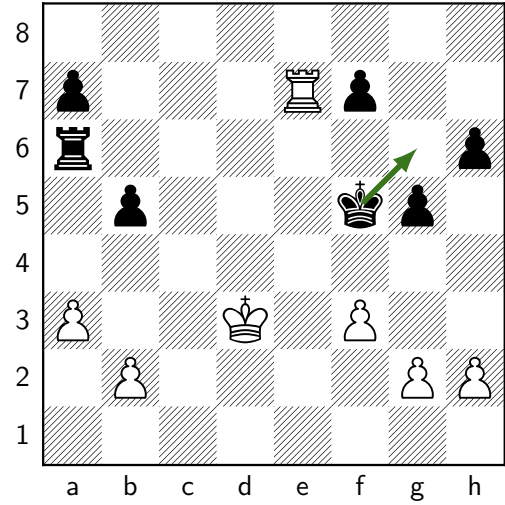
(a)



(b)

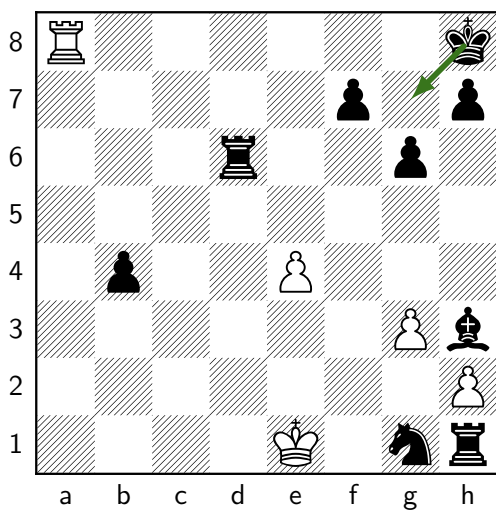


(c)

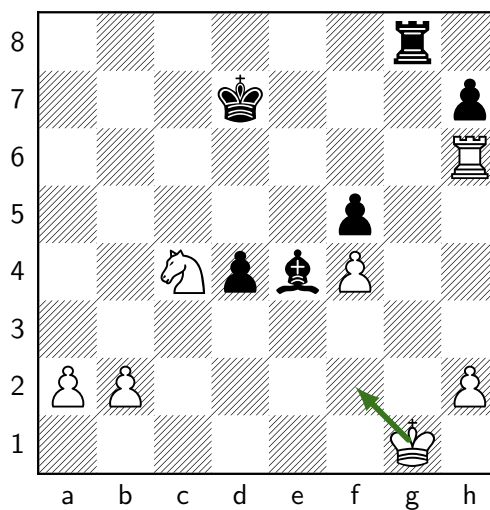


(d)

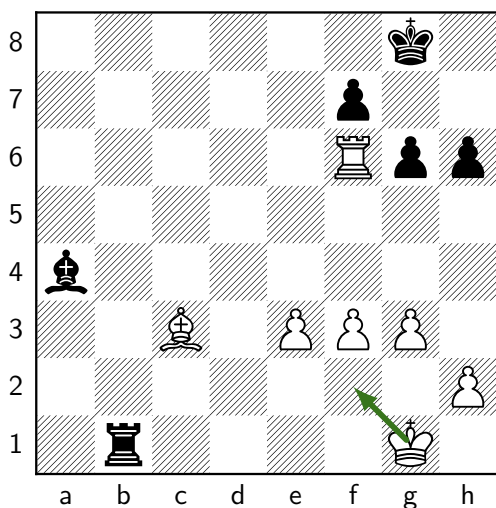
Figure 35: **Move 41, Black: Rxd6.** (a) Our position; Melanie plays Rxd6 (f6d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1203` (ply 87) [structural distance $d=38$, QAP obj=35]: the master played Rc6; under μ our Rf3 plays the role of that game’s Rg3; cross-type roles: our Bh3 as their nb5, our Ng1 as their pa4; transferred move f6d6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#3129` (ply 79) [structural distance $d=40$, QAP obj=42]: the master played Kf6; under μ our Kh1 plays the role of that game’s Ke3; cross-type roles: our pd6 as their Bb7, our Ng1 as their Pf3; transferred move h8g8. (d) Retrieved from `ccrl_top_p0001#339` (ply 75) [structural distance $d=33$, QAP obj=26]: the master played Kg6; under μ our Kh1 plays the role of that game’s Kf4; cross-type roles: our Ng1 as their Ph6, our Bh3 as their Pg5; transferred move h8g8. Synthesis: 1 retrieved analog transferred to Rxd6 (summed QAP objective 35, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



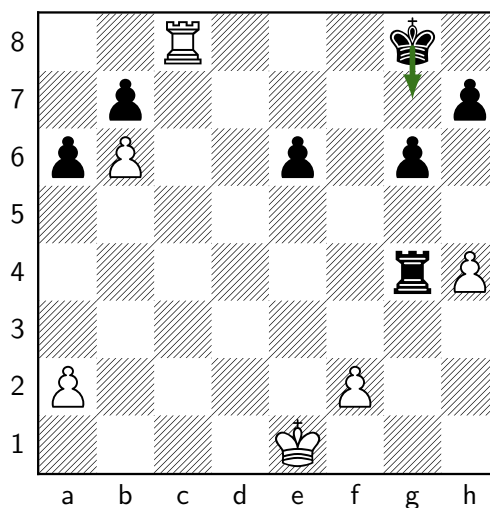
(a)



(b)

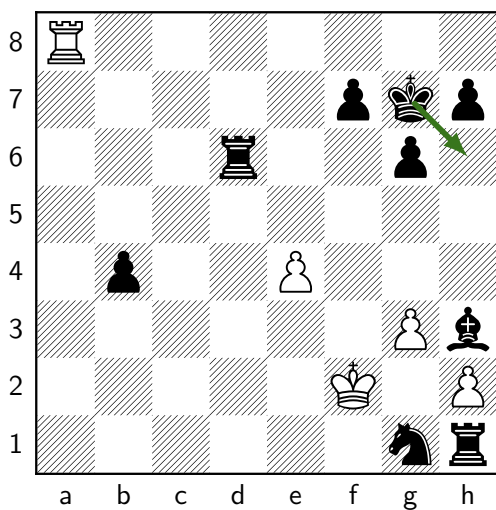


(c)

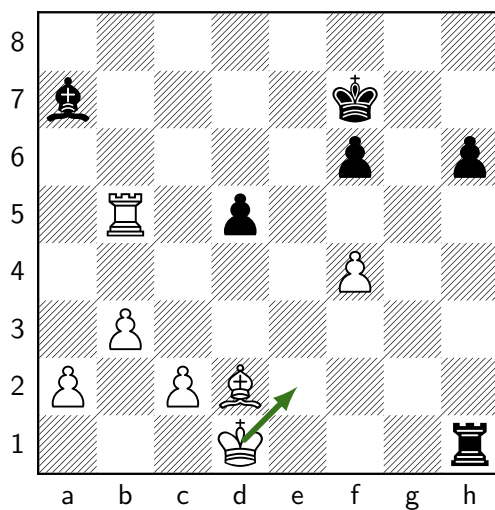


(d)

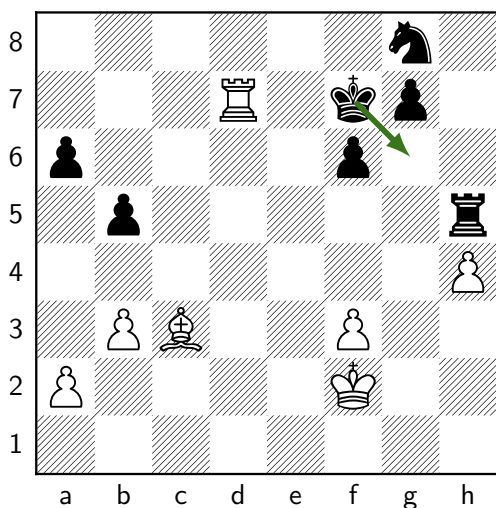
Figure 36: **Move 42, Black: Kg7**. (a) Our position; Melanie plays Kg7 (**h8g7**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#2999` (ply 76) [structural distance $d=27$, QAP obj=69]: the master played Kf2; under μ our *Kh1* plays the role of that game’s *Kg1*; cross-type roles: our *Rd6* as their *Nc4*, our *Ng1* as their *be4*; transferred move **h8g7 (endorses this move)**. (c) Retrieved from `ccrl_top_p0003#2217` (ply 74) [structural distance $d=30$, QAP obj=51]: the master played Kf2; under μ our *Kh1* plays the role of that game’s *Kg1*; cross-type roles: our *Rh1* as their *Bc3*, our *Bh3* as their *Pf3*; transferred move **h8g7 (endorses this move)**. (d) Retrieved from `ccrl_top_p0001#858` (ply 75) [structural distance $d=26$, QAP obj=44]: the master played Kg7; under μ our *Kh1* plays the role of that game’s *Kg1*; cross-type roles: our *Ng1* as their *pb6*, our *Bh3* as their *pf2*; transferred move **h8g7 (endorses this move)**. Synthesis: 8 retrieved analogs transferred to Kg7 (summed QAP objective 283, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



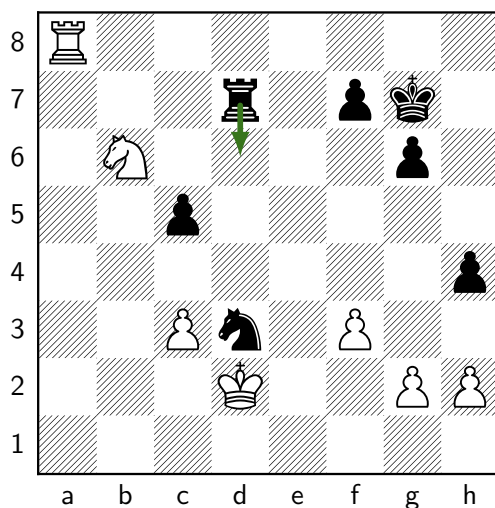
(a)



(b)

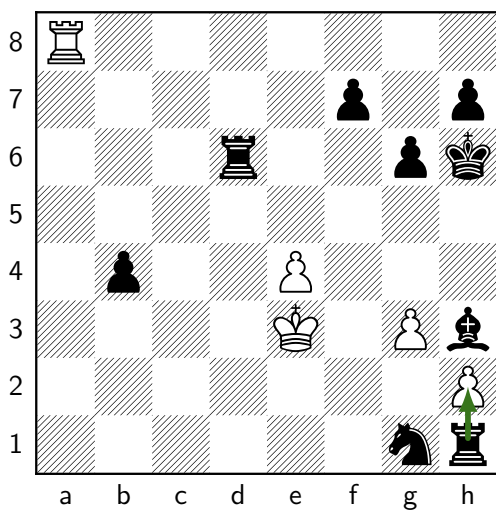


(c)

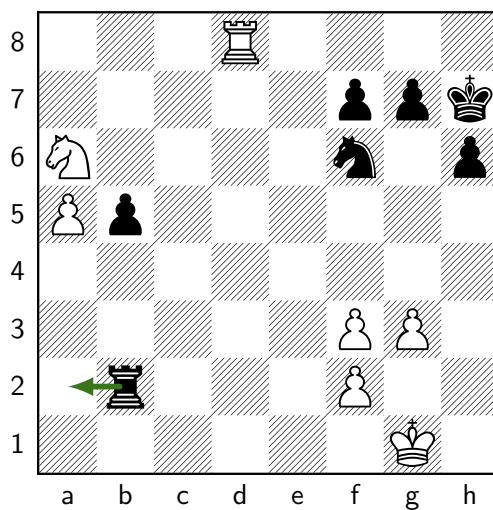


(d)

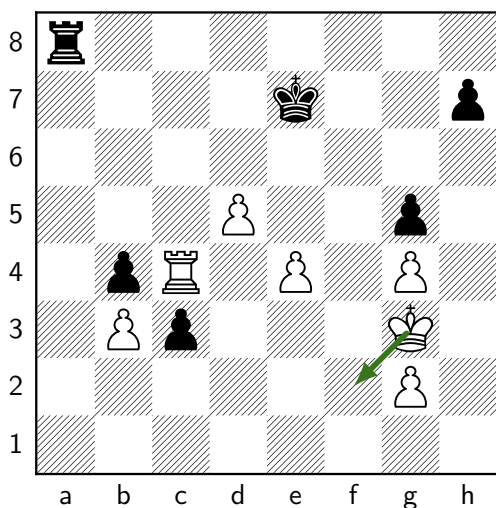
Figure 37: **Move 43, Black: Kh6.** (a) Our position; Melanie plays Kh6 (g7h6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1102` (ply 90) [structural distance $d=35$, QAP obj=3]: the master played Ke2; under μ our Kg2 plays the role of that game’s Kd1; cross-type roles: our Ng1 as their Pa2, our Pg6 as their Bd2; transferred move g7h6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#2742` (ply 73) [structural distance $d=35$, QAP obj=3]: the master played Kg6; under μ our Kg2 plays the role of that game’s Kf2; cross-type roles: our Ph7 as their Ng8, our Bh3 as their pb3; transferred move g7h6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#2181` (ply 81) [structural distance $d=26$, QAP obj=30]: the master played Rd6; under μ our Ng8 plays the role of that game’s Rd2 (a **cross-type** correspondence); cross-type roles: our pg3 as their ra8, our kf2 as their nb6; transferred move g1f3. Synthesis: 3 retrieved analogs transferred to Kh6 (summed QAP objective 10, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



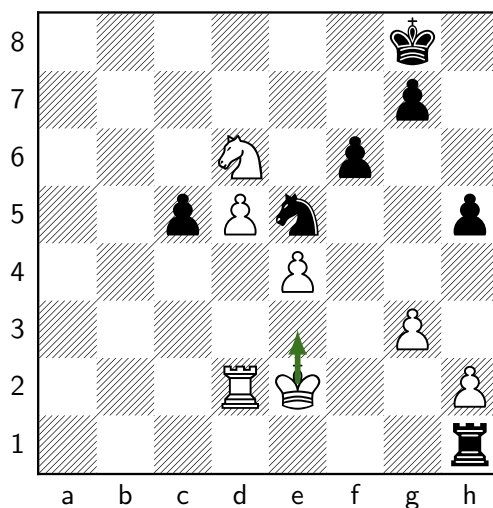
(a)



(b)

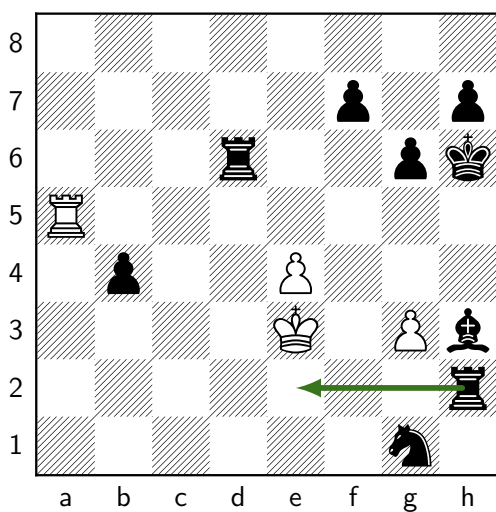


(c)

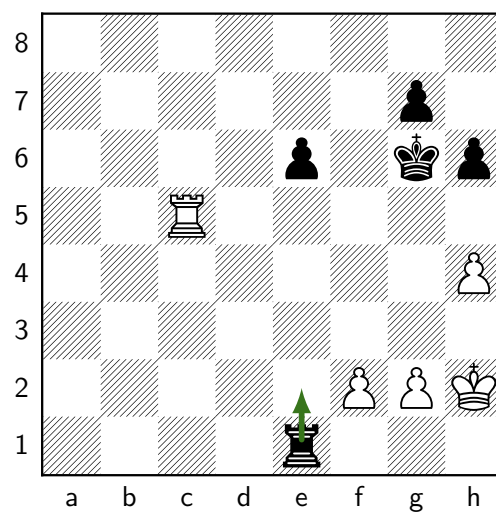


(d)

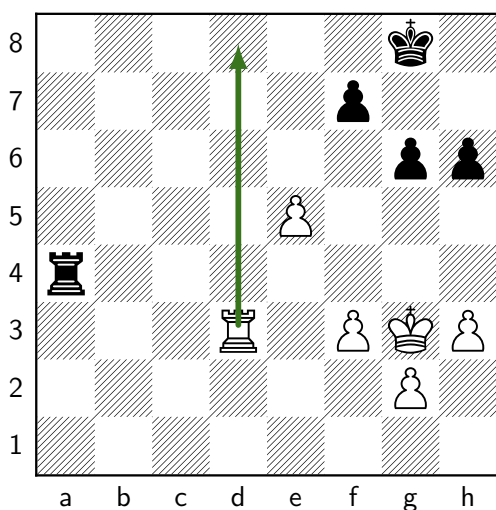
Figure 38: **Move 44, Black: Rxxh2.** (a) Our position; Melanie plays Rxxh2 (h1h2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#2047` (ply 71) [structural distance $d=24$, QAP obj=7]: the master played Ra2; under μ our Rh8 plays the role of that game’s Rb7; cross-type roles: our Pg6 as their rd8, our Bh3 as their na6; transferred move h1h2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#956` (ply 96) [structural distance $d=30$, QAP obj=34]: the master played Kf2; under μ our Kh3 plays the role of that game’s Kg3; cross-type roles: our Ng1 as their Pe4, our Rd6 as their Pd5; transferred move h6g7. (d) Retrieved from `ccrl_top_p0003#888` (ply 82) [structural distance $d=30$, QAP obj=22]: the master played Ke3; under μ our Kh3 plays the role of that game’s Ke2; cross-type roles: our Ph7 as their Rd2, our ra8 as their Pd5; transferred move h6g7. Synthesis: 1 retrieved analog transferred to Rxxh2 (summed QAP objective 7, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



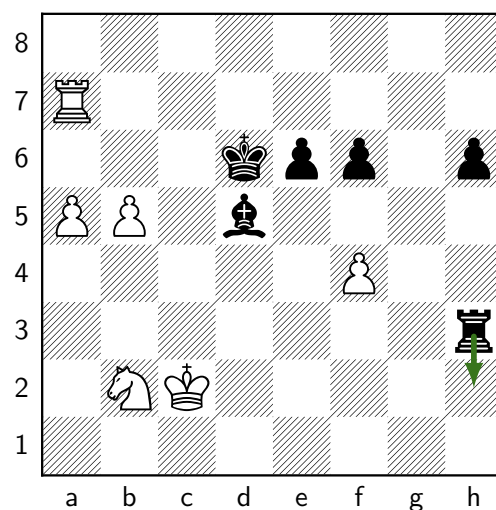
(a)



(b)

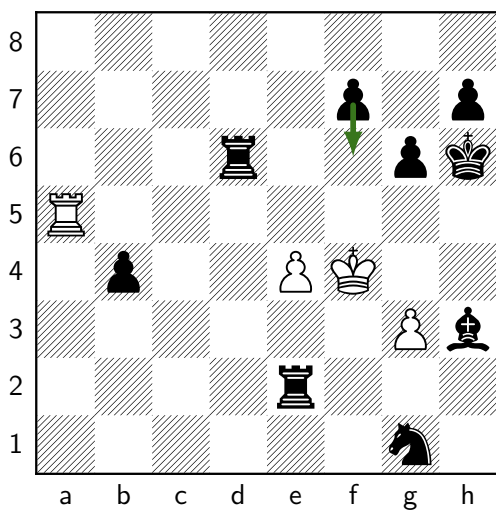


(c)

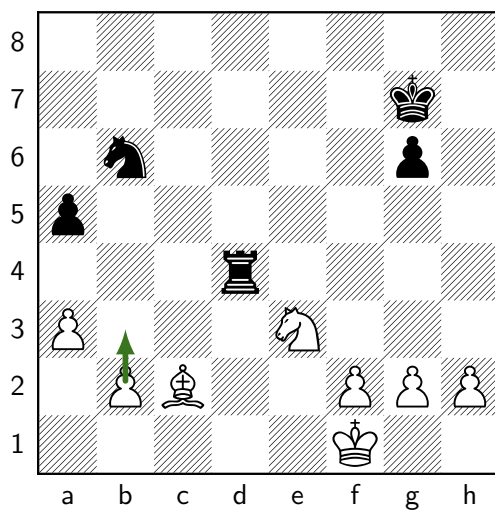


(d)

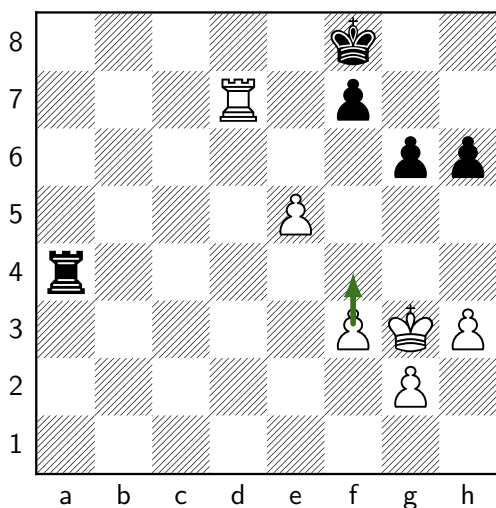
Figure 39: **Move 45, Black: Re2+.** (a) Our position; Melanie plays Re2+ (h2e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2677` (ply 77) [structural distance $d=34$, QAP obj=4]: the master played Re2; under μ our Rh7 plays the role of that game’s Re8; cross-type roles: our Rd6 as their pf2; transferred move h2c2. (c) Retrieved from `ccr1_top_p0002#2875` (ply 100) [structural distance $d=33$, QAP obj=4]: the master played Rd8+; under μ our Rd3 plays the role of that game’s Rd3; cross-type roles: our Bh3 as their Pe5; transferred move d6d1. (d) Retrieved from `ccr1_top_p0002#94` (ply 99) [structural distance $d=33$, QAP obj=2]: the master played Rh2+; under μ our Rd3 plays the role of that game’s Rh6; cross-type roles: our pg3 as their ra7, our Rh2 as their Pe6; transferred move d6d1. Synthesis: no retrieved analog transferred to Re2+ – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



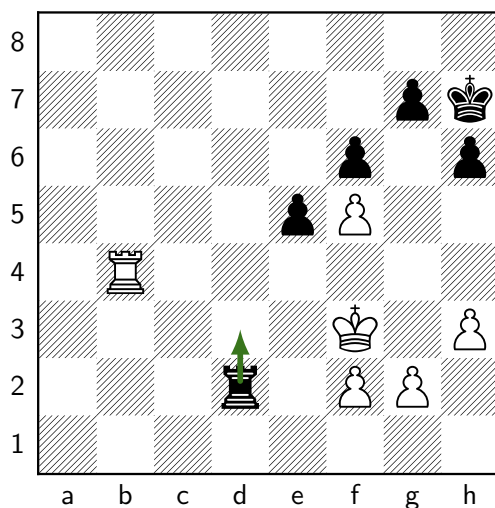
(a)



(b)

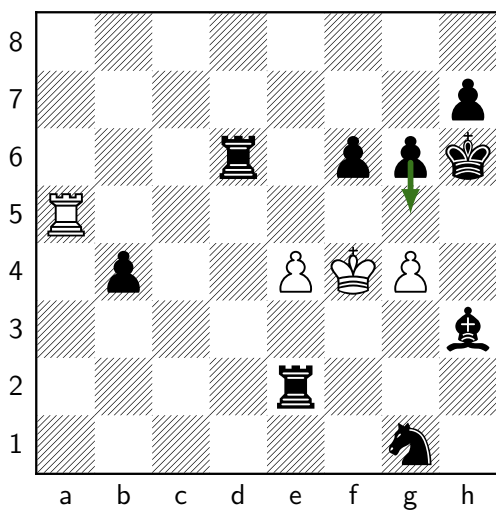


(c)

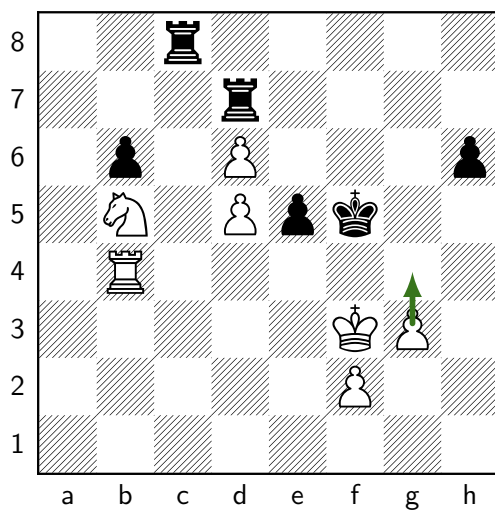


(d)

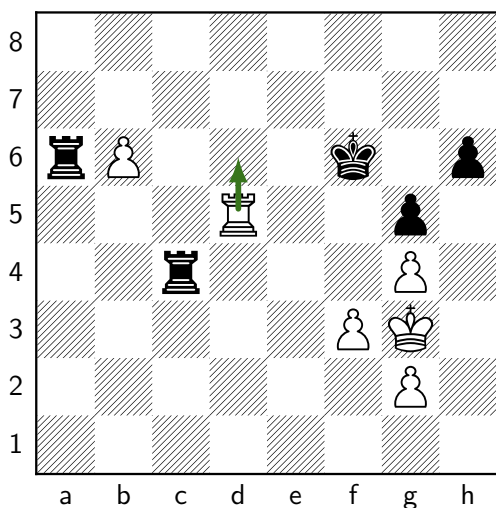
Figure 40: **Move 46, Black: f6 (f7f6)**. (a) Our position; Melanie plays f6 (f7f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#2295` (ply 82) [structural distance $d=35$, QAP obj=12]: the master played b3; under μ our *Pf2* plays the role of that game’s *Pb2*; cross-type roles: our *Re2* as their *Bc2*, our *Rd6* as their *Pf2*; transferred move **f7f6 (endorses this move)**. (c) Retrieved from `ccrl_top_p0002#2875` (ply 104) [structural distance $d=37$, QAP obj=23]: the master played f4; under μ our *Pg3* plays the role of that game’s *Pf3*; cross-type roles: our *Bh3* as their *Pe5*, our *Rd6* as their *pg6*; transferred move **g6g5**. (d) Retrieved from `ccrl_top_p0001#160` (ply 79) [structural distance $d=37$, QAP obj=22]: the master played **Rd3+**; under μ our *Re7* plays the role of that game’s *Rd7*; cross-type roles: our *Bh3* as their *Pg7*, our *Ng1* as their *pf5*; transferred move **e2f2**. Synthesis: 1 retrieved analog transferred to **f6** (summed QAP objective 12, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



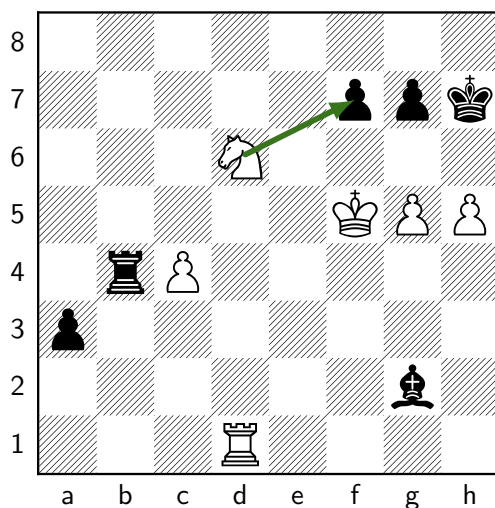
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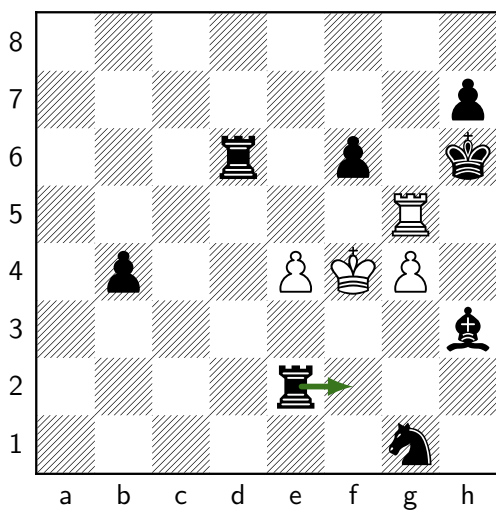


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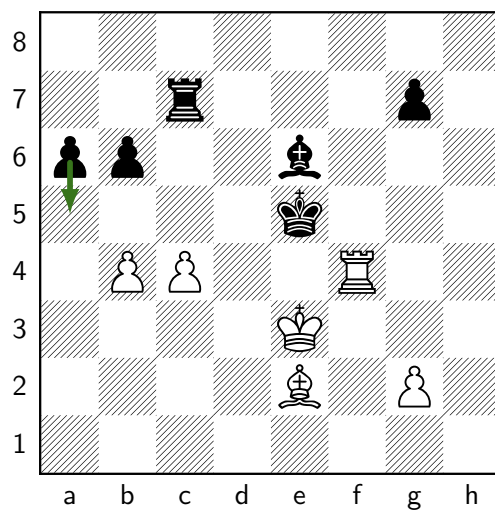


(d)

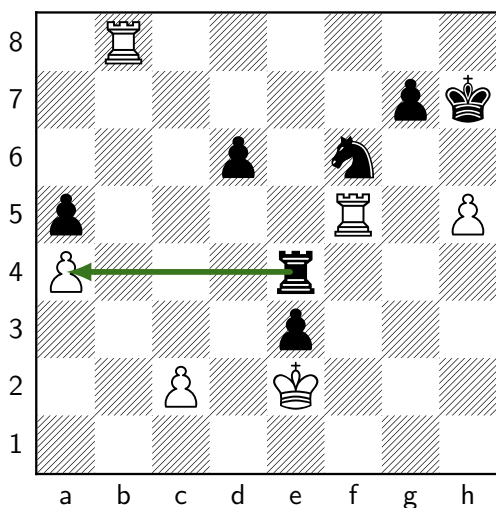
Figure 41: **Move 47, Black: g5+.** (a) Our position; Melanie plays g5+ (g6g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#3049` (ply 96) [structural distance $d=42$, QAP obj=22]: the master played g4+; under μ our *Pg3* plays the role of that game’s *Pg3*; cross-type roles: our *Pf6* as their *Rb4*, our *Pb4* as their *Nb5*; transferred move g6g5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#3765` (ply 86) [structural distance $d=52$, QAP obj=30]: the master played Rd6+; under μ our *Re7* plays the role of that game’s *Rd5*; cross-type roles: our *Bh3* as their *ra6*, our *Ng1* as their *ph6*; transferred move e2f2. (d) Retrieved from `ccrl_top_p0003#1303` (ply 90) [structural distance $d=50$, QAP obj=29]: the master played Nxf7; under μ our *Re7* plays the role of that game’s *Nd6* (a **cross-type** correspondence); cross-type roles: our *Ng1* as their *bg2*, our *Bh3* as their *rb4*; transferred move e2c2. Synthesis: 1 retrieved analog transferred to g5+ (summed QAP objective 22, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



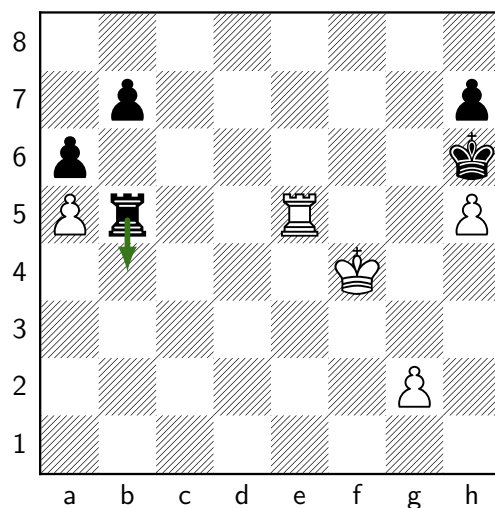
(a)



(b)

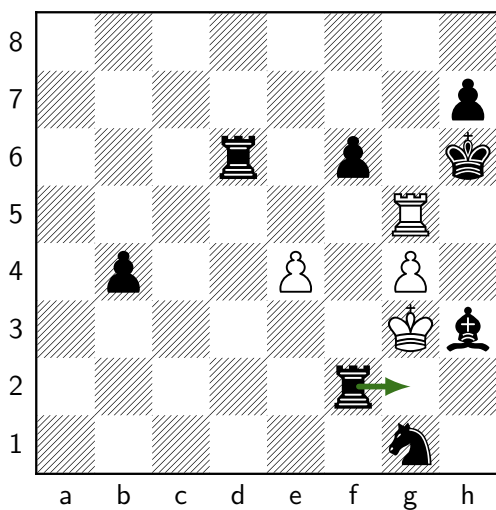


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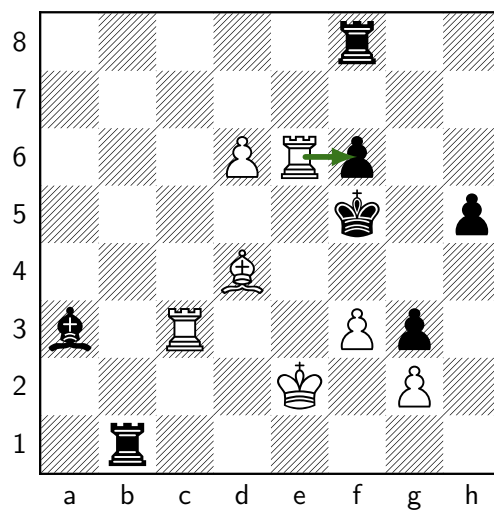


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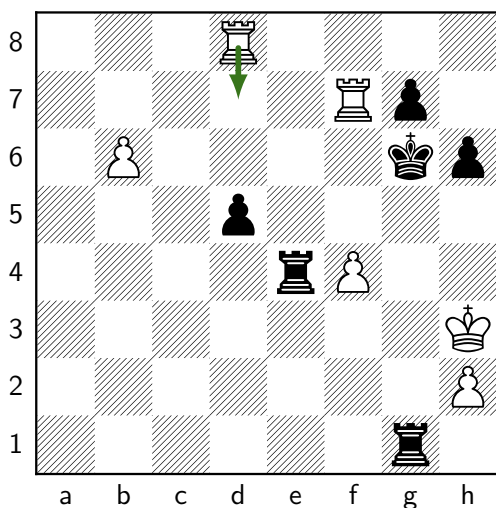
Figure 42: **Move 48, Black: Rf2+.** (a) Our position; Melanie plays Rf2+ (e2f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2903` (ply 107) [structural distance $d=45$, QAP obj=23]: the master played a5; under μ our *Re7* plays the role of that game’s *Pa3* (a **cross-type** correspondence); cross-type roles: our *Bh3* as their *Rc7*, our *Rd6* as their *Pg7*; transferred move e2f2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#739` (ply 99) [structural distance $d=52$, QAP obj=12]: the master played Rxa4; under μ our *Re7* plays the role of that game’s *Re5*; cross-type roles: our *Ng1* as their *Pg7*, our *rg5* as their *Nf6*; transferred move e2c2. (d) Retrieved from `ccr1_top_p0003#1438` (ply 99) [structural distance $d=51$, QAP obj=11]: the master played Rb4+; under μ our *Pf3* plays the role of that game’s *Rb4* (a **cross-type** correspondence); cross-type roles: our *Bh3* as their *Ph7*, our *Rd6* as their *Pa6*; transferred move f6g5. Synthesis: 1 retrieved analog transferred to Rf2+ (summed QAP objective 23, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



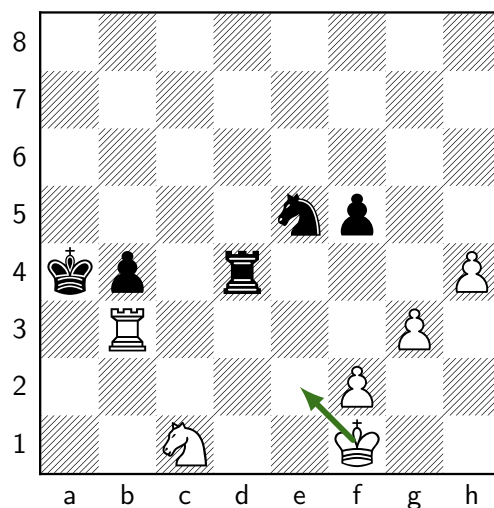
(a)



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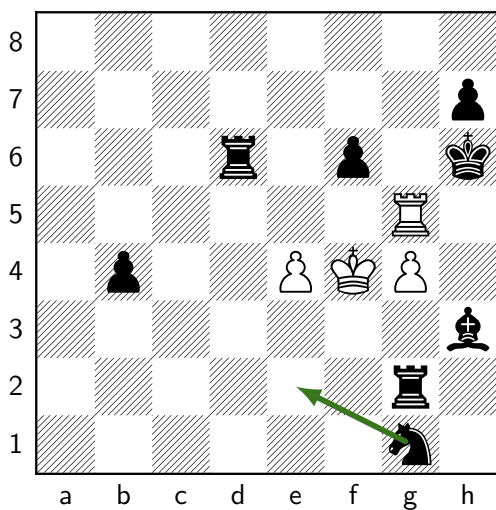


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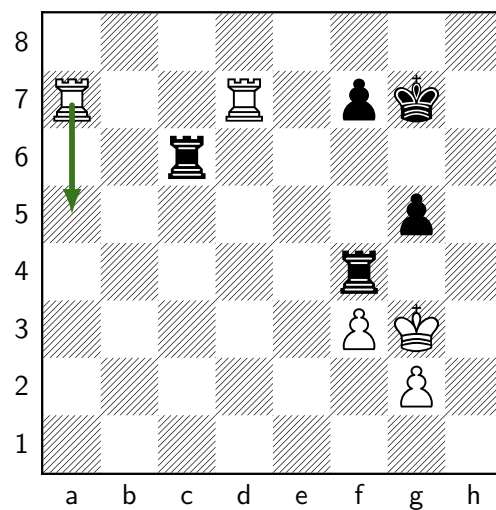


(d)

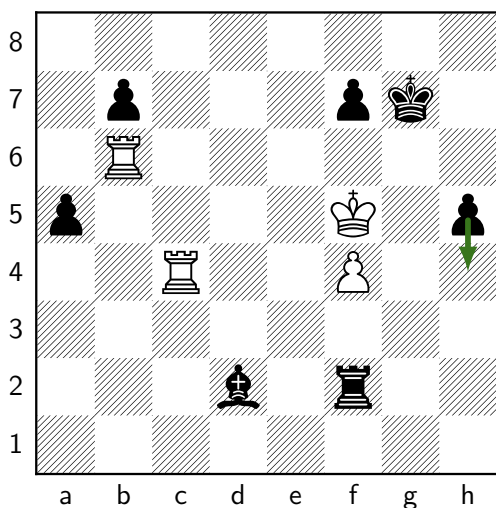
Figure 43: **Move 49, Black: Rg2+.** (a) Our position; Melanie plays Rg2+ (f2g2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1629` (ply 92) [structural distance $d=49$, QAP obj=42]: the master played Rxf6+; under μ our Rf7 plays the role of that game’s Re6; cross-type roles: our Ng1 as their Pg2, our rg5 as their ba3; transferred move f2f3. (c) Retrieved from `ccrl_top_p0001#1222` (ply 90) [structural distance $d=48$, QAP obj=38]: the master played Rdd7; under μ our Pb5 plays the role of that game’s Rd8 (a **cross-type** correspondence); cross-type roles: our Bh3 as their re4, our Ng1 as their pd5; transferred move b4b3. (d) Retrieved from `ccrl_top_p0003#2137` (ply 82) [structural distance $d=48$, QAP obj=38]: the master played Ke2; under μ our Kh3 plays the role of that game’s Kf1; cross-type roles: our Rd6 as their Ph4, our Bh3 as their ne5; transferred move h6g5. Synthesis: no retrieved analog transferred to Rg2+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



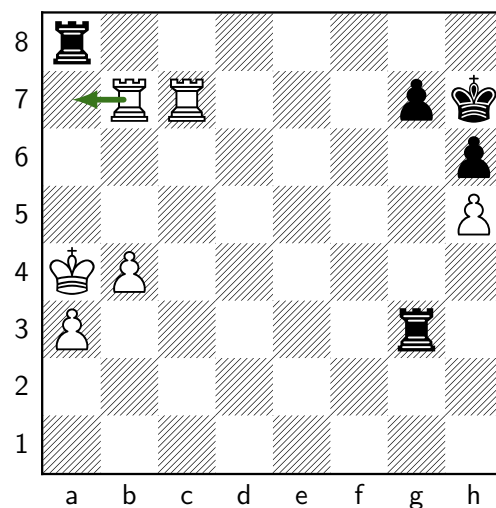
(a)



(b)

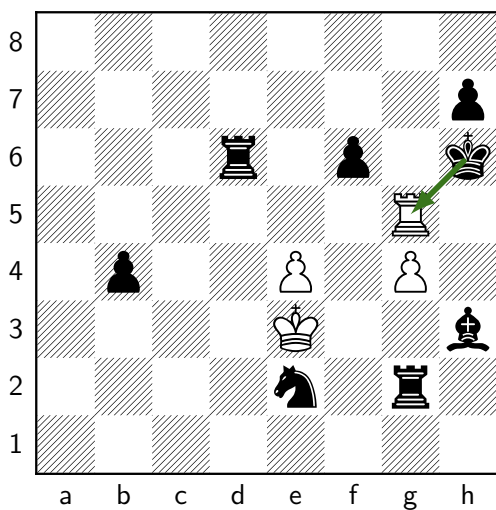


(c)

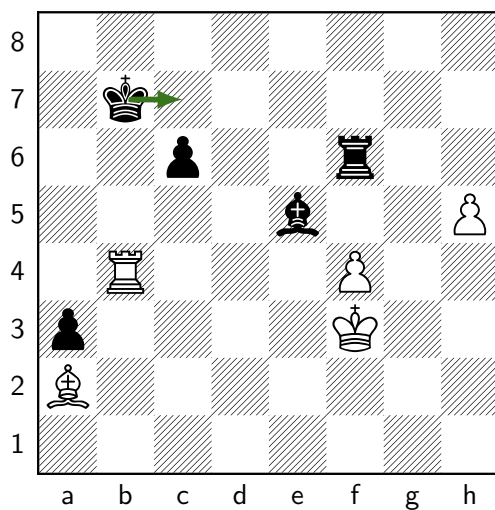


(d)

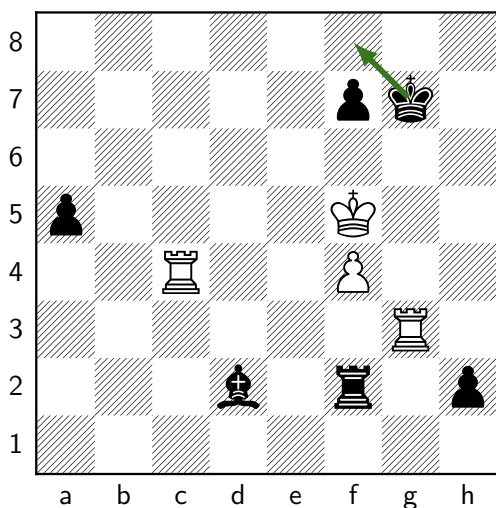
Figure 44: **Move 50, Black: Ne2+ (g1e2)**. (a) Our position; Melanie plays Ne2+ (g1e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#272 (ply 98) [structural distance $d=48$, QAP obj=47]: the master played Ra5; under μ our Ng8 plays the role of that game’s Ra7 (a **cross-type** correspondence); transferred move g1e2 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#2591 (ply 87) [structural distance $d=48$, QAP obj=32]: the master played h4; under μ our Ng8 plays the role of that game’s Ph4 (a **cross-type** correspondence); cross-type roles: our pe4 as their rb6, our Rd6 as their Pa5; transferred move g1e2 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#2650 (ply 90) [structural distance $d=50$, QAP obj=18]: the master played Ra7; under μ our Ng8 plays the role of that game’s Rb7 (a **cross-type** correspondence); cross-type roles: our Rd6 as their Pb4, our Pb4 as their ra8; transferred move g1e2 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Ne2+ (summed QAP objective 112, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



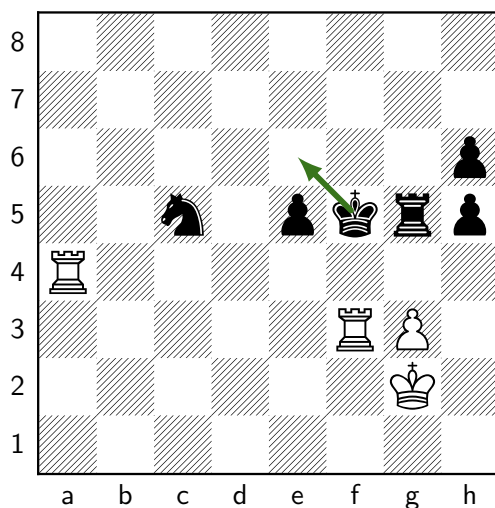
(a)



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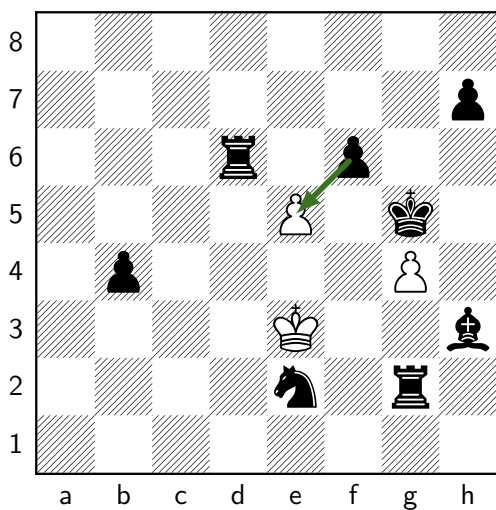


(c)

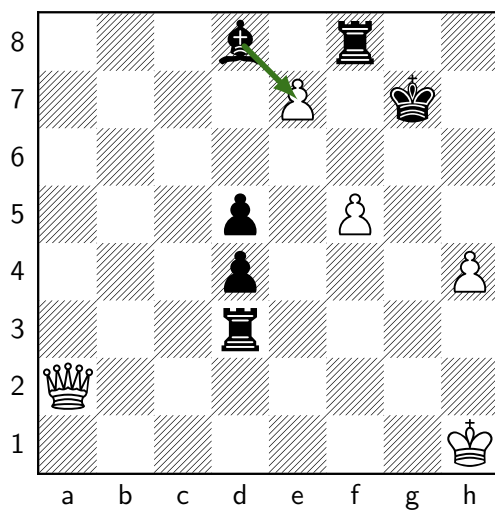


(d)

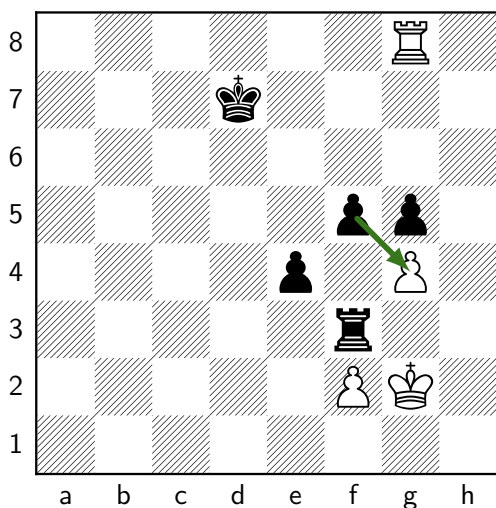
Figure 45: **Move 51, Black: Kxg5**. (a) Our position; Melanie plays Kxg5 (h6g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#661` (ply 109) [structural distance $d=44$, QAP obj=42]: the master played Kc7; under μ our Kh3 plays the role of that game’s Kb2; cross-type roles: our Ne2 as their Pc6, our pe4 as their ba2; transferred move h6g5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2591` (ply 93) [structural distance $d=46$, QAP obj=32]: the master played Kf8; under μ our Kh3 plays the role of that game’s Kg2; cross-type roles: our Rd6 as their Pf7, our Ph7 as their rg3; transferred move h6g5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1082` (ply 101) [structural distance $d=41$, QAP obj=25]: the master played Ke6; under μ our Kh3 plays the role of that game’s Kf4; cross-type roles: our Bh3 as their Ph6, our Ph7 as their Nc5; transferred move h6g5 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Kxg5 (summed QAP objective 139, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



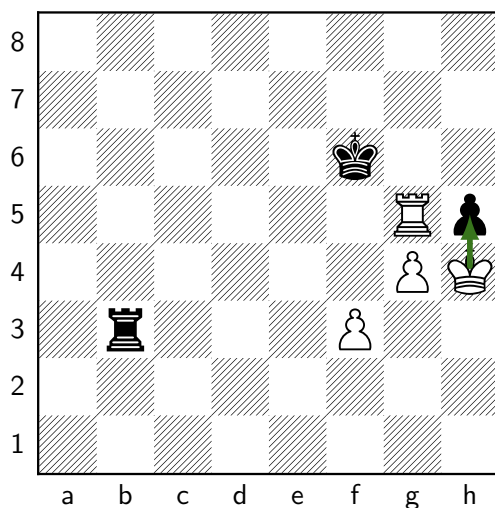
(a)



(b)

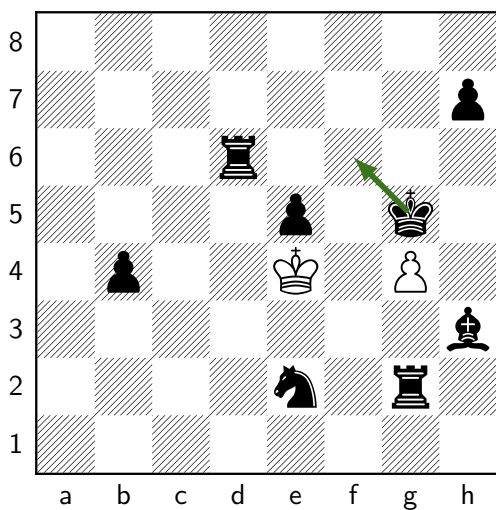


(c)

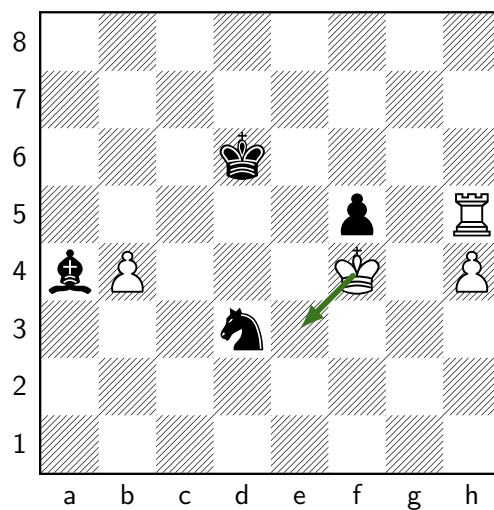


(d)

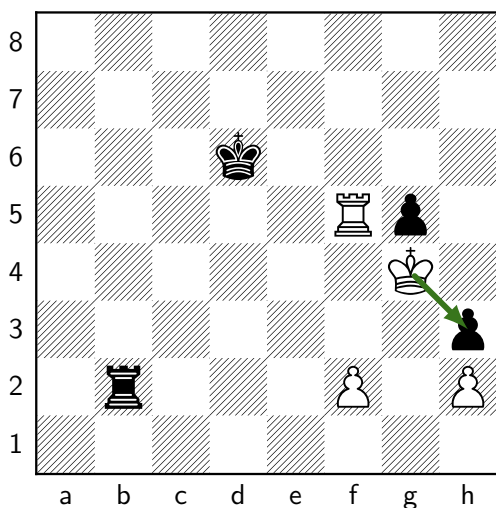
Figure 46: **Move 52, Black: fxe5**. (a) Our position; Melanie plays fxe5 (f6e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1.top_p0002#2239 (ply 95) [structural distance $d=37$, QAP obj=41]: the master played Bxe7; under μ our Pf3 plays the role of that game’s Bd1 (a **cross-type** correspondence); cross-type roles: our Bh3 as their Pd4, our Ne2 as their ph4; transferred move f6e5 (**endorses this move**). (c) Retrieved from ccr1.top_p0002#1486 (ply 119) [structural distance $d=40$, QAP obj=12]: the master played fxg4; under μ our Pf3 plays the role of that game’s Pf4; cross-type roles: our pg4 as their rg8, our Rd6 as their Pg5; transferred move f6e5 (**endorses this move**). (d) Retrieved from ccr1.top_p0000#724 (ply 90) [structural distance $d=42$, QAP obj=18]: the master played Kxh5; under μ our Kg4 plays the role of that game’s Kh4; cross-type roles: our Ph7 as their rb3, our Bh3 as their Pg4; transferred move g5g4. Synthesis: 2 retrieved analogs transferred to fxe5 (summed QAP objective 53, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



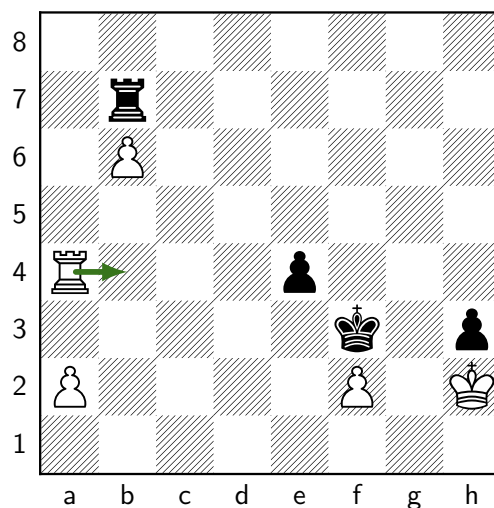
(a)



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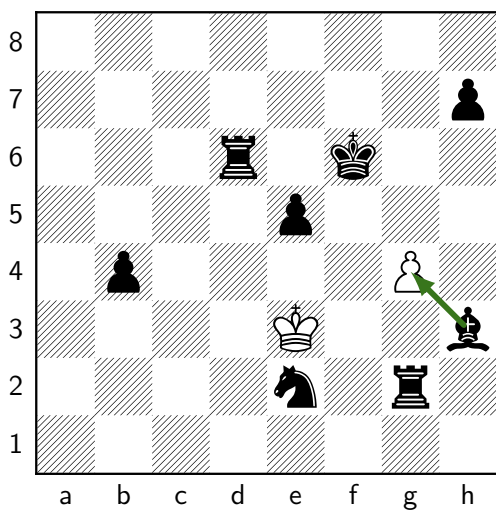


(c)

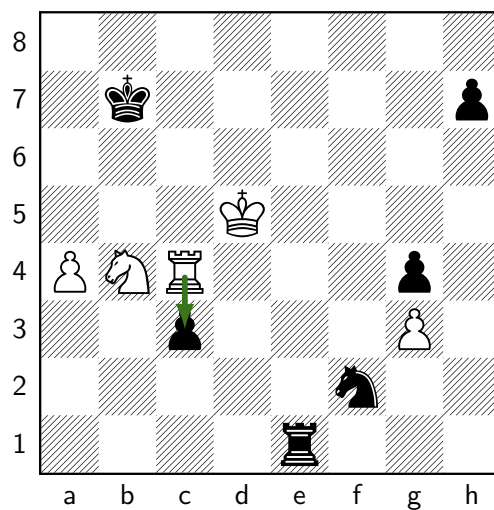


(d)

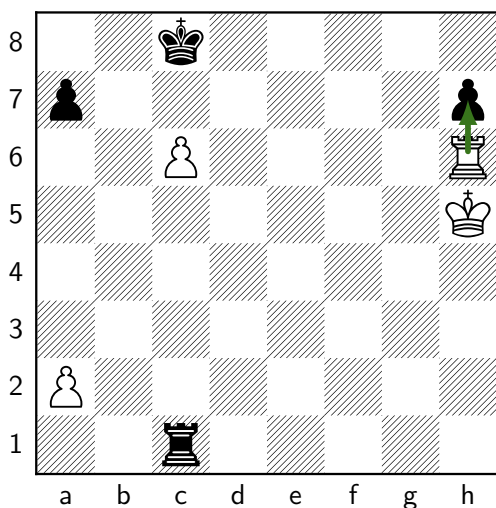
Figure 47: **Move 53, Black: Kf6**. (a) Our position; Melanie plays Kf6 (g5f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1178` (ply 94) [structural distance $d=30$, QAP obj=3]: the master played Ke3; under μ our Kg4 plays the role of that game’s Kf4; cross-type roles: our Ph7 as their nd3, our Rd6 as their ba4; transferred move g5f6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#1540` (ply 112) [structural distance $d=28$, QAP obj=55]: the master played Kxh3; under μ our Kg4 plays the role of that game’s Kg4; cross-type roles: our Rg2 as their Pf2, our Bh3 as their Rf5; transferred move g5h6. (d) Retrieved from `ccrl_top_p0001#2331` (ply 108) [structural distance $d=27$, QAP obj=52]: the master played Rb4; under μ our Ne7 plays the role of that game’s Ra4 (a **cross-type** correspondence); cross-type roles: our Rg2 as their Pa2, our Rd6 as their pe4; transferred move e2g3. Synthesis: 1 retrieved analog transferred to Kf6 (summed QAP objective 3, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



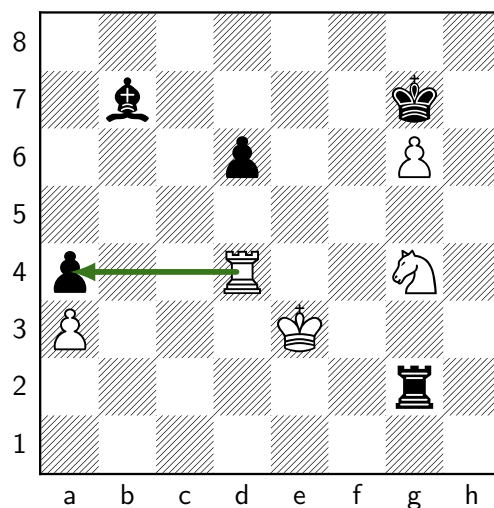
(a)



(b)

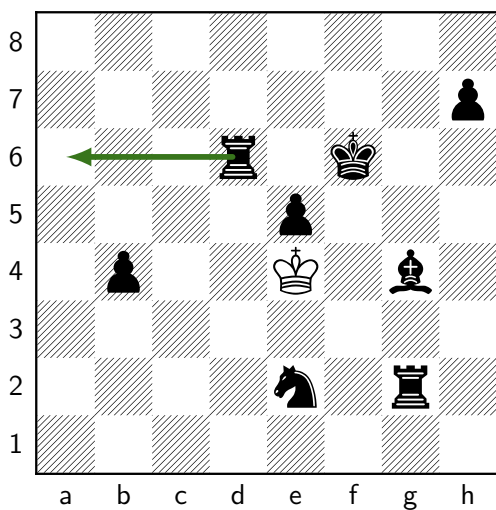


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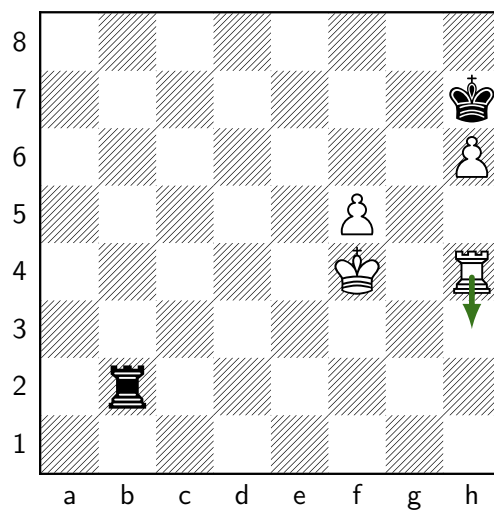


(d)

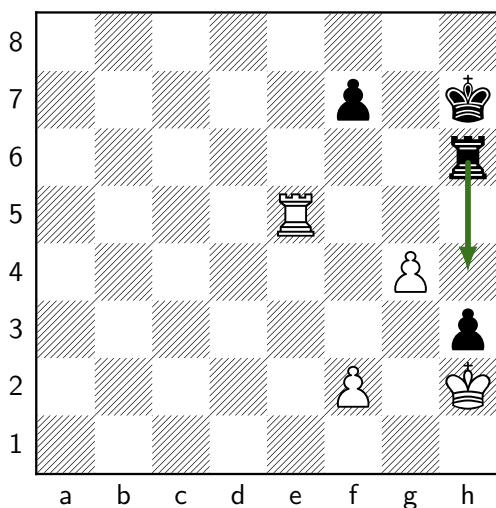
Figure 48: **Move 54, Black: Bxg4**. (a) Our position; Melanie plays Bxg4 (h3g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1133` (ply 118) [structural distance $d=31$, QAP obj=8]: the master played Rxc3; under μ our Bh6 plays the role of that game’s Rc4 (a **cross-type** correspondence); cross-type roles: our Pb4 as their nf2, our Ne2 as their Pg3; transferred move h3g4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2211` (ply 94) [structural distance $d=25$, QAP obj=7]: the master played Rxh7; under μ our Bh6 plays the role of that game’s Rh6 (a **cross-type** correspondence); cross-type roles: our Ph7 as their rc1, our Rd6 as their Pa2; transferred move h3g4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#94` (ply 100) [structural distance $d=27$, QAP obj=12]: the master played Rxa4; under μ our Rg7 plays the role of that game’s Rd4; cross-type roles: our Bh3 as their rg2, our pg4 as their Ng4; transferred move g2h2. Synthesis: 2 retrieved analogs transferred to Bxg4 (summed QAP objective 15, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



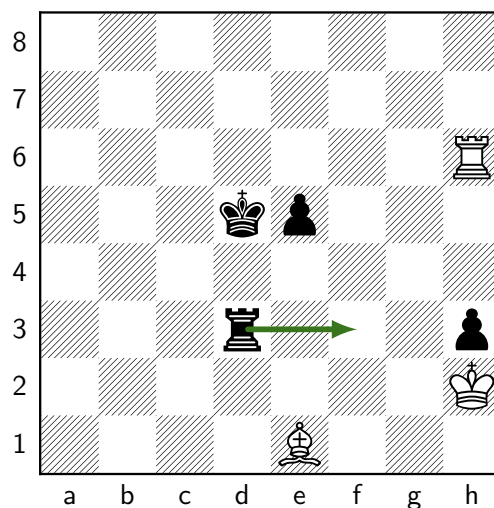
(a)



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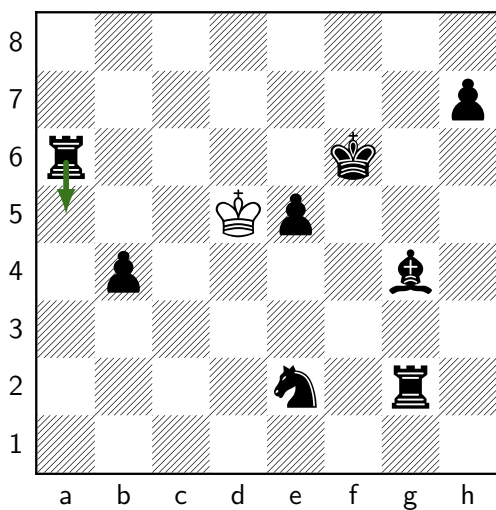


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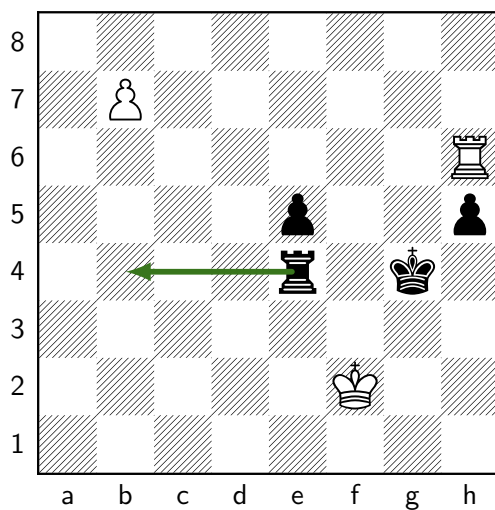


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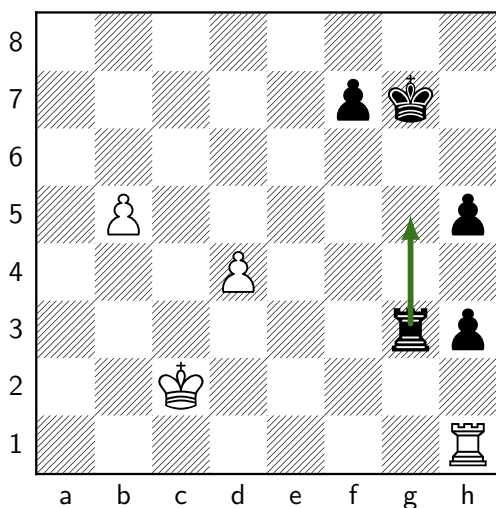
Figure 49: **Move 55, Black: Ra6**. (a) Our position; Melanie plays **Ra6** (**d6a6**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#227` (ply 124) [structural distance $d=36$, QAP obj=21]: the master played **Rh3**; under μ our *Rd3* plays the role of that game’s *Rh4*; cross-type roles: our *Ph7* as their *rb2*, our *Bg4* as their *Pf5*; transferred move **d6a6** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1736` (ply 119) [structural distance $d=31$, QAP obj=20]: the master played **Rh4**; under μ our *Rd3* plays the role of that game’s *Rh3*; cross-type roles: our *Pb4* as their *re5*, our *Bg4* as their *pg4*; transferred move **d6a6** (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2174` (ply 103) [structural distance $d=36$, QAP obj=22]: the master played **Rf3**; under μ our *Rd3* plays the role of that game’s *Rd6*; cross-type roles: our *ke4* as their *rh6*, our *Pb4* as their *be1*; transferred move **d6c6**. Synthesis: 2 retrieved analogs transferred to **Ra6** (summed QAP objective 40, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



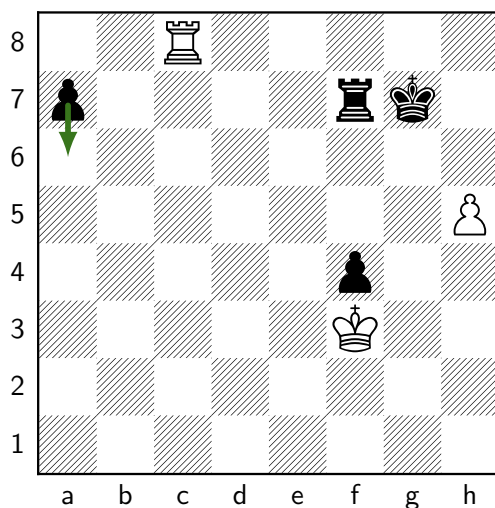
(a)



(b)

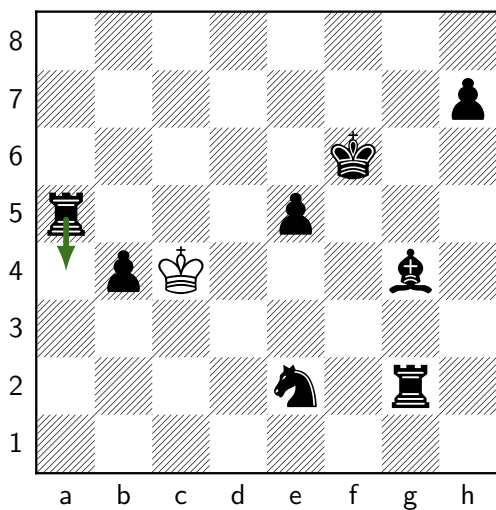


(c)

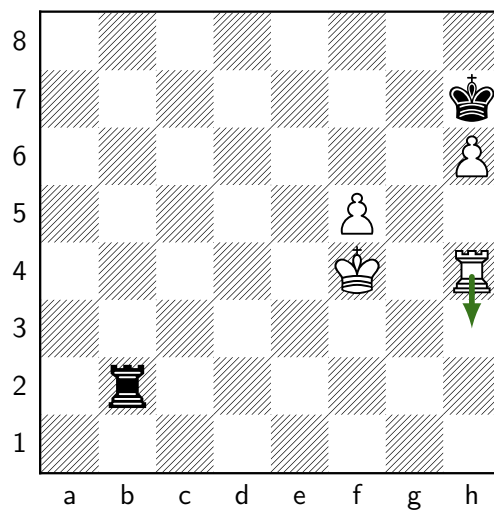


(d)

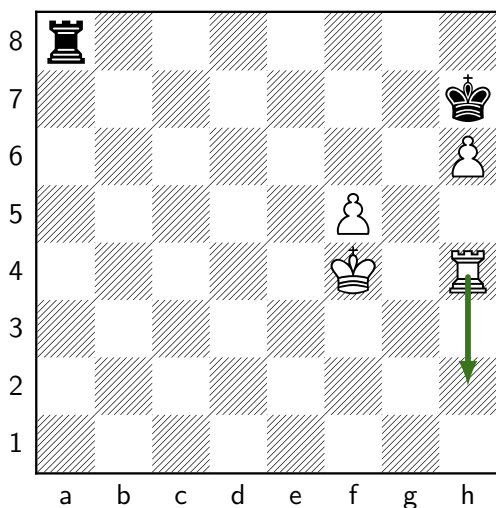
Figure 50: **Move 56, Black: Ra5+.** (a) Our position; Melanie plays Ra5+ (a6a5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3037` (ply 97) [structural distance $d=34$, QAP obj=16]: the master played Rb4; under μ our Ra3 plays the role of that game’s Re5; cross-type roles: our kd5 as their rh6, our Bg4 as their Pe5; transferred move a6a5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1923` (ply 113) [structural distance $d=33$, QAP obj=7]: the master played Rg5; under μ our Ra3 plays the role of that game’s Rg6; cross-type roles: our Bg4 as their pb5, our Ne2 as their pd4; transferred move a6a5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#634` (ply 101) [structural distance $d=33$, QAP obj=18]: the master played a6; under μ our Bg5 plays the role of that game’s Pa2 (a **cross-type** correspondence); cross-type roles: our Ph7 as their rc8; transferred move g4d7. Synthesis: 2 retrieved analogs transferred to Ra5+ (summed QAP objective 23, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



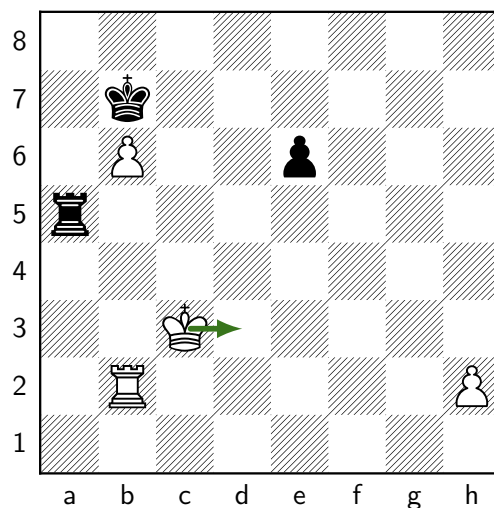
(a)



(b)

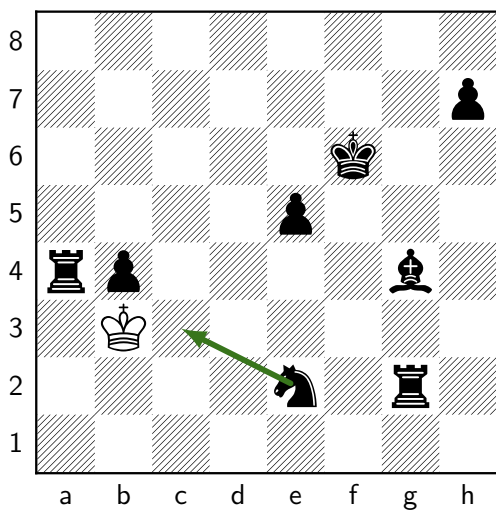


(c)

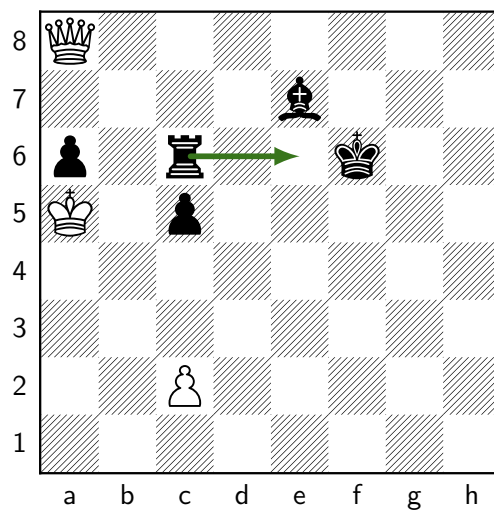


(d)

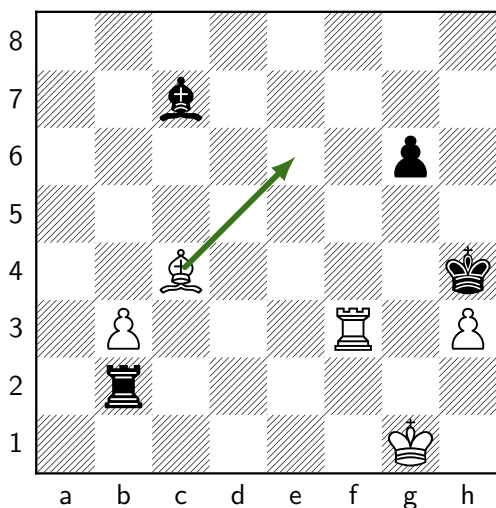
Figure 51: **Move 57, Black: Ra4.** (a) Our position; Melanie plays Ra4 (a5a4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#227` (ply 124) [structural distance $d=31$, QAP obj=19]: the master played Rh3; under μ our Ra4 plays the role of that game’s Rh4; cross-type roles: our Ph7 as their rb2; transferred move a5a6. (c) Retrieved from `ccrl_top_p0001#3184` (ply 102) [structural distance $d=31$, QAP obj=19]: the master played Rh2; under μ our Ra4 plays the role of that game’s Rh4; cross-type roles: our Ph7 as their ra8; transferred move a5a7. (d) Retrieved from `ccrl_top_p0003#1602` (ply 102) [structural distance $d=27$, QAP obj=19]: the master played Kd3; under μ our Kf3 plays the role of that game’s Kc3; cross-type roles: our Pe5 as their ra5, our Bg4 as their pe6; transferred move f6g7. Synthesis: no retrieved analog transferred to Ra4 – the `search` selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



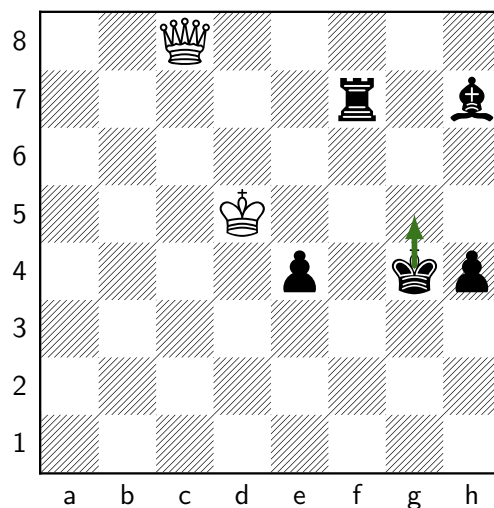
(a)



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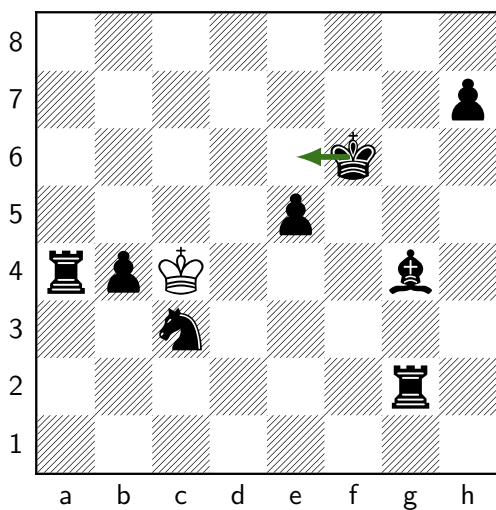


(c)

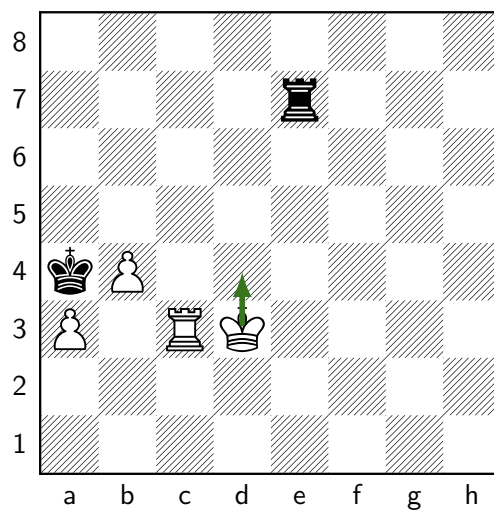


(d)

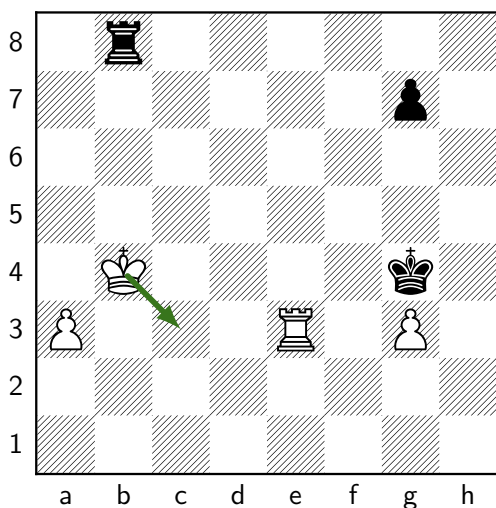
Figure 52: **Move 58, Black: Nc3.** (a) Our position; Melanie plays Nc3 (e2c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2590` (ply 129) [structural distance $d=34$, QAP obj=11]: the master played R6; under μ our R5 plays the role of that game’s R3; cross-type roles: our kb3 as their qa8, our Rg2 as their Be7; transferred move a4a6. (c) Retrieved from `ccr1_top_p0003#1686` (ply 116) [structural distance $d=30$, QAP obj=3]: the master played Be6; under μ our Bg5 plays the role of that game’s Bc4; cross-type roles: our Ne2 as their Pb3, our Pe5 as their bc7; transferred move g4f3. (d) Retrieved from `ccr1_top_p0000#3005` (ply 125) [structural distance $d=34$, QAP obj=3]: the master played Kg5; under μ our Kf3 plays the role of that game’s Kg5; cross-type roles: our Ph7 as their qc8, our Ne2 as their Pe4; transferred move f6g5. Synthesis: no retrieved analog transferred to Nc3 – the `search` selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



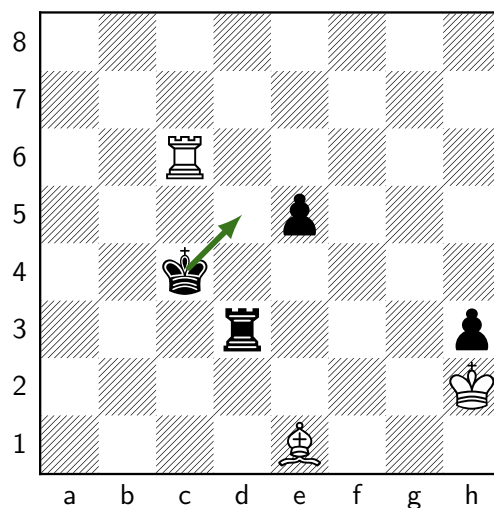
(a)



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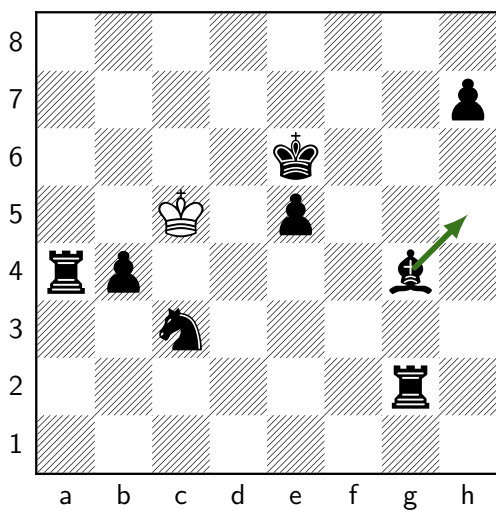


(c)

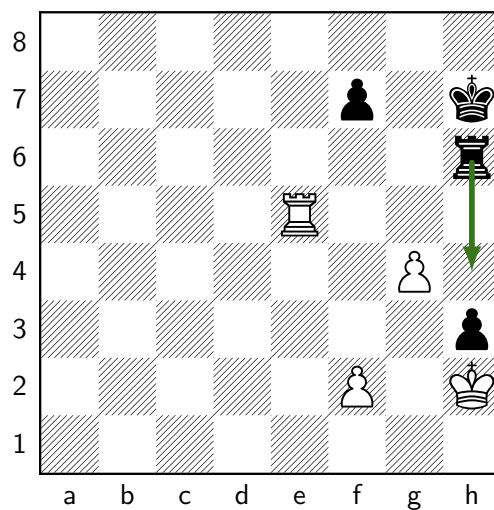


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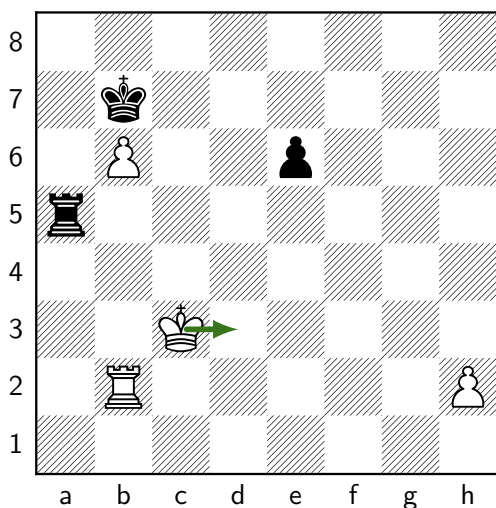
Figure 53: **Move 59, Black: Ke6**. (a) Our position; Melanie plays Ke6 (f6e6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#675` (ply 120) [structural distance $d=27$, QAP obj=21]: the master played Kd4; under μ our Kf3 plays the role of that game’s Kd3; cross-type roles: our Nc3 as their Pb4, our Ph7 as their re7; transferred move f6e6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#3227` (ply 124) [structural distance $d=26$, QAP obj=2]: the master played Kc3; under μ our Kf3 plays the role of that game’s Kb4; cross-type roles: our Bg4 as their rb8; transferred move f6e6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#2174` (ply 101) [structural distance $d=27$, QAP obj=1]: the master played Kd5; under μ our Kf3 plays the role of that game’s Kc5; cross-type roles: our Ph7 as their rc6, our Nc3 as their be1; transferred move f6e6 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Ke6 (summed QAP objective 25, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



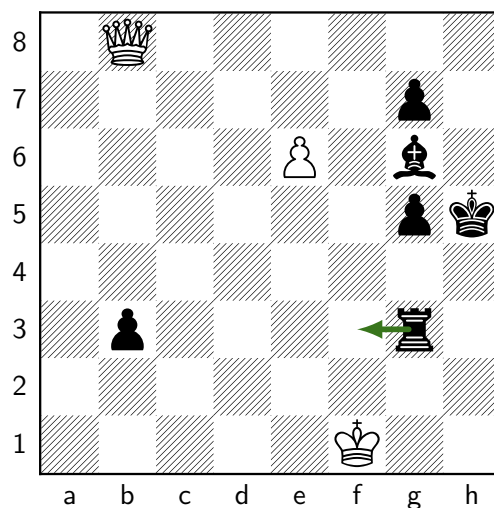
(a)



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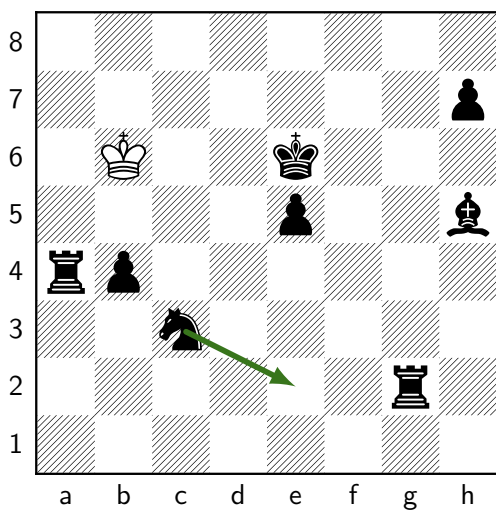


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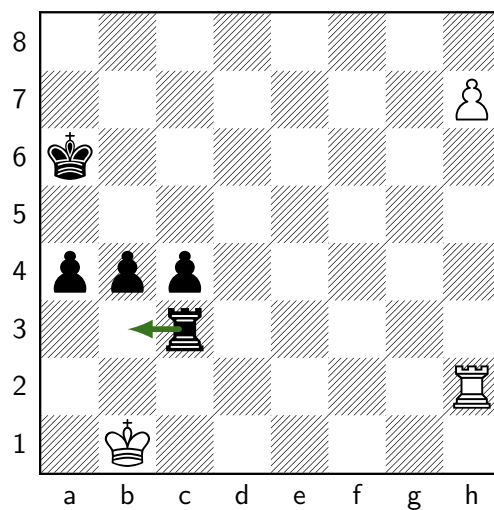


(d)

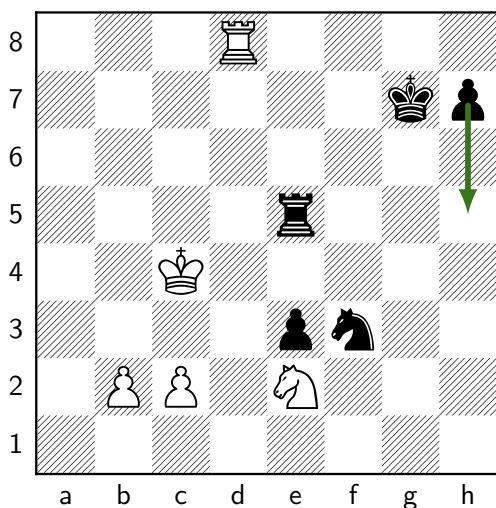
Figure 54: **Move 60, Black: Bh5**. (a) Our position; Melanie plays Bh5 (**g4h5**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1736` (ply 119) [structural distance $d=22$, QAP obj=19]: the master played Rh4; under μ our Bg5 plays the role of that game’s Rh3 (a **cross-type** correspondence); cross-type roles: our Ra4 as their Pf7, our Ph7 as their re5; transferred move g4h5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1602` (ply 102) [structural distance $d=22$, QAP obj=19]: the master played Kd3; under μ our Ke3 plays the role of that game’s Kc3; cross-type roles: our Nc3 as their ra5; transferred move e6f6. (d) Retrieved from `ccr1_top_p0002#2375` (ply 107) [structural distance $d=26$, QAP obj=15]: the master played Rf3+; under μ our Pb5 plays the role of that game’s Rg6 (a **cross-type** correspondence); cross-type roles: our kc5 as their qb8, our Rg2 as their Pg7; transferred move b4b3. Synthesis: 1 retrieved analog transferred to Bh5 (summed QAP objective 19, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



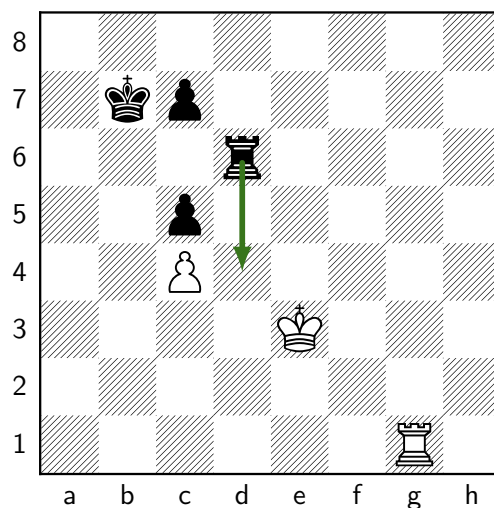
(a)



(b)

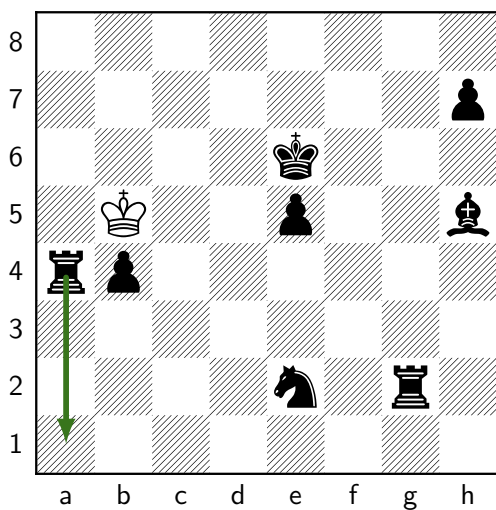


(c)

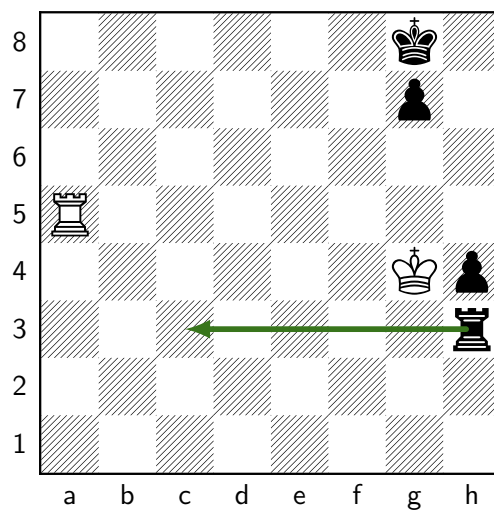


(d)

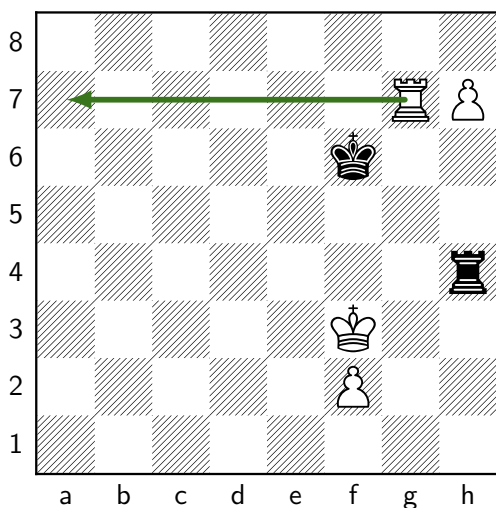
Figure 55: **Move 61, Black: Ne2.** (a) Our position; Melanie plays Ne2 (c3e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1368` (ply 121) [structural distance $d=19$, QAP obj=1]: the master played Rb3+; under μ our Nc6 plays the role of that game’s Rc6 (a **cross-type** correspondence); cross-type roles: our Bh5 as their Pc4; transferred move c3e2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#773` (ply 115) [structural distance $d=19$, QAP obj=3]: the master played h5; under μ our Pe4 plays the role of that game’s Ph2; cross-type roles: our Pb4 as their rd8, our Bh5 as their Pe3; transferred move e5e4. (d) Retrieved from `ccr1_top_p0003#1813` (ply 115) [structural distance $d=20$, QAP obj=2]: the master played Rd4; under μ our Ra5 plays the role of that game’s Rd3; cross-type roles: our Nc3 as their Pc5, our Ph7 as their rg1; transferred move a4a6. Synthesis: 3 retrieved analogs transferred to Ne2 (summed QAP objective 4, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



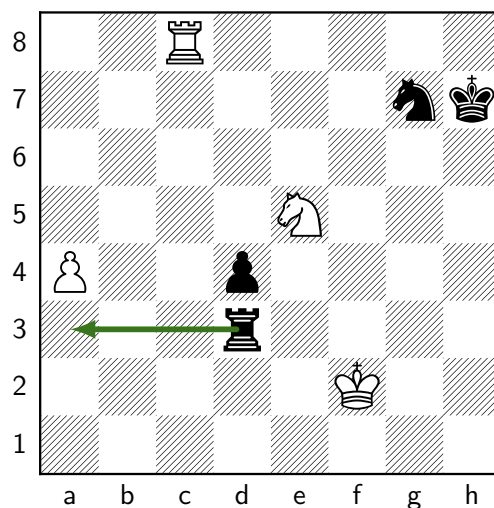
(a)



(b)



(c)



(d)

Figure 56: **Move 62, Black: Ra1**. (a) Our position; Melanie plays **Ra1** (**a4a1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2930` (ply 111) [structural distance $d=32$, QAP obj=30]: the master played **Rc3**; under μ our **Ra5** plays the role of that game's **Rh6**; cross-type roles: our **Ph7** as their **ra5**; transferred move **a4a1** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3059` (ply 112) [structural distance $d=31$, QAP obj=21]: the master played **Ra7**; under μ our **Ra5** plays the role of that game's **Rg7**; cross-type roles: our **Ph7** as their **rh4**; transferred move **a4a1** (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2773` (ply 129) [structural distance $d=32$, QAP obj=8]: the master played **Ra3**; under μ our **Ra5** plays the role of that game's **Rd6**; cross-type roles: our **Bh5** as their **rc8**, our **Pe5** as their **Ng7**; transferred move **a4a1** (**endorses this move**). Synthesis: 5 retrieved analogs transferred to **Ra1** (summed QAP objective 83, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

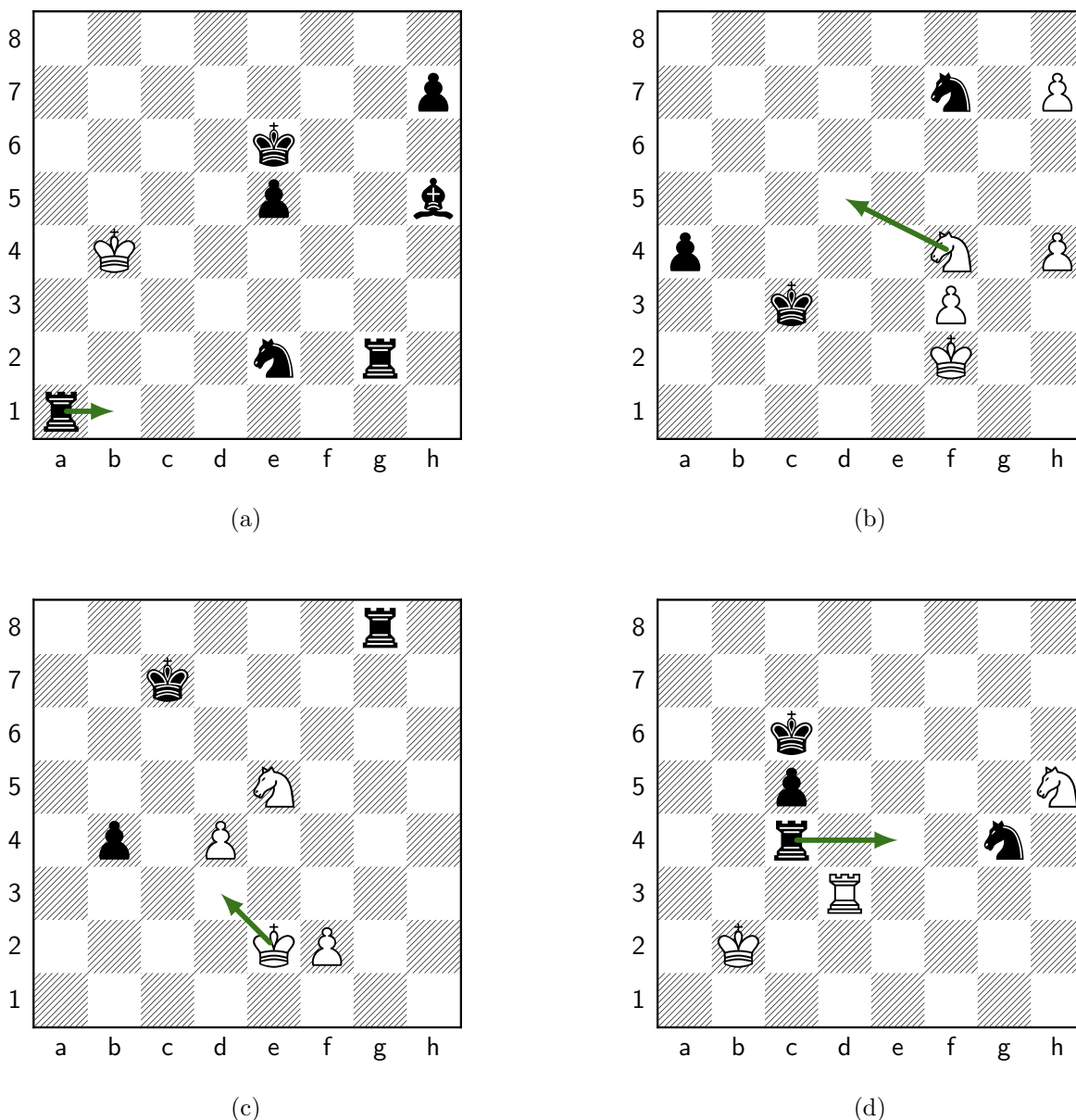
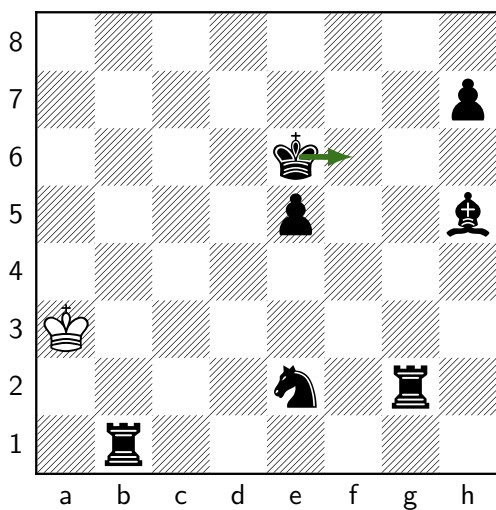
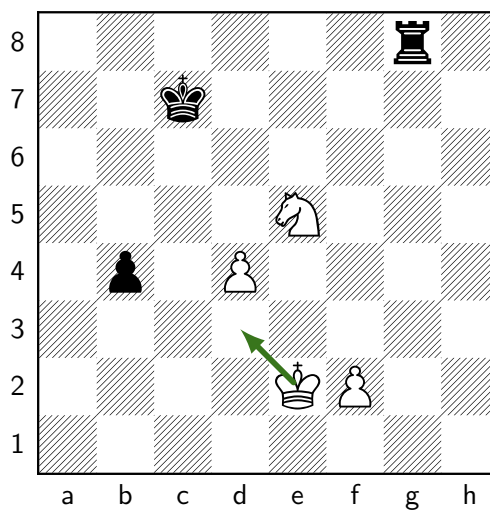


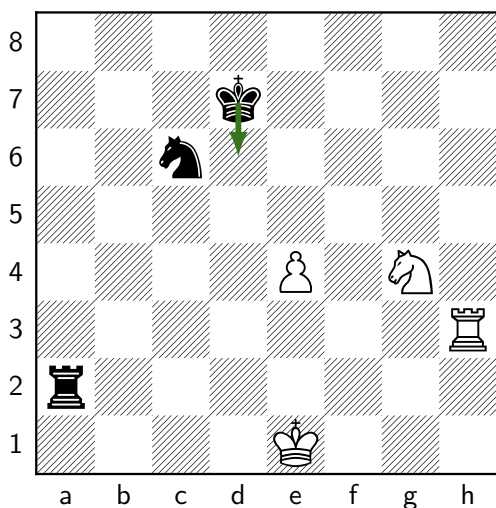
Figure 57: **Move 63, Black: Rb1+.** (a) Our position; Melanie plays Rb1+ (a1b1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1404` (ply 118) [structural distance $d=21$, QAP obj=1]: the master played Nd5+; under μ our *Ra8* plays the role of that game’s *Nf4* (a **cross-type** correspondence); cross-type roles: our *Bh5* as their *Ph4*, our *Rg2* as their *Ph7*; transferred move a1b1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#16` (ply 116) [structural distance $d=21$, QAP obj=2]: the master played Kd3; under μ our *Ke3* plays the role of that game’s *Ke2*; cross-type roles: our *Rg2* as their *Pd4*, our *Bh5* as their *rg8*; transferred move e6f6. (d) Retrieved from `ccr1_top_p0000#3407` (ply 137) [structural distance $d=19$, QAP obj=2]: the master played Re4; under μ our *Ra8* plays the role of that game’s *Rc5*; cross-type roles: our *Ph7* as their *nh5*, our *Bh5* as their *Ng4*; transferred move a1d1. Synthesis: 2 retrieved analogs transferred to Rb1+ (summed QAP objective 3, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



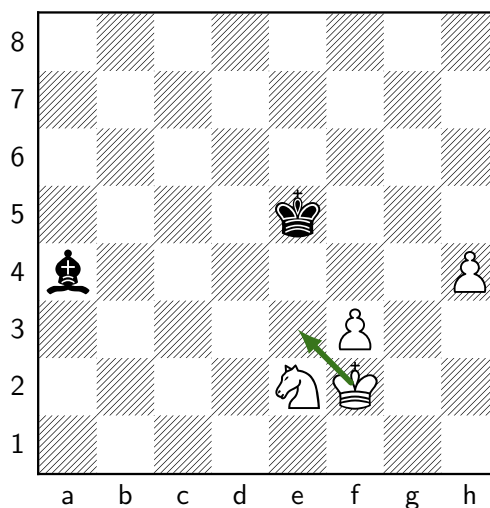
(a)



(b)

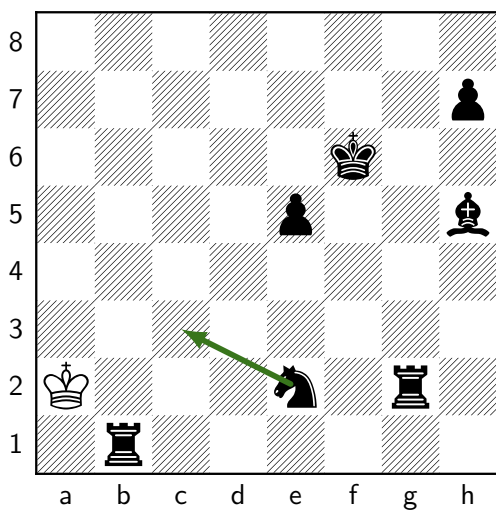


(c)

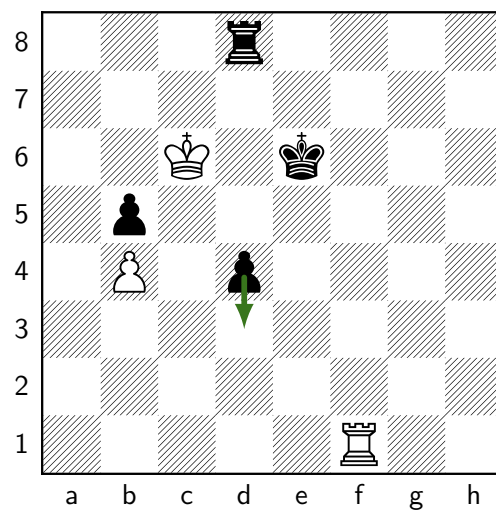


(d)

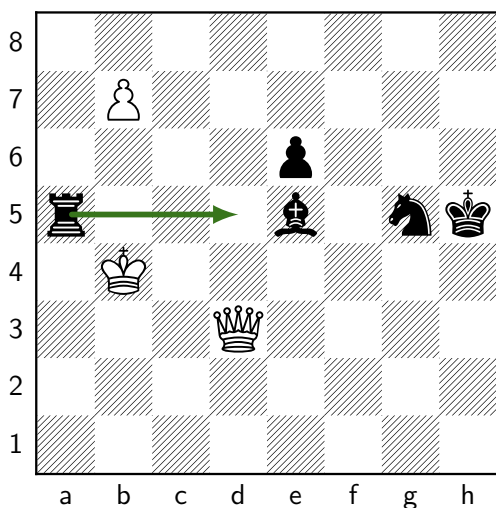
Figure 58: **Move 64, Black: Kf6**. (a) Our position; Melanie plays Kf6 (e6f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0002#16 (ply 116) [structural distance $d=22$, QAP obj=1]: the master played Kd3; under μ our Ke3 plays the role of that game’s Ke2; cross-type roles: our Rg2 as their Pd4, our Bh5 as their rg8; transferred move e6f6 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#3418 (ply 117) [structural distance $d=21$, QAP obj=1]: the master played Kd6; under μ our Ke3 plays the role of that game’s Kd2; cross-type roles: our Pe5 as their Nc6, our Bh5 as their ng4; transferred move e6f6 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#3784 (ply 114) [structural distance $d=22$, QAP obj=1]: the master played Ke3; under μ our Ke3 plays the role of that game’s Kf2; cross-type roles: our Rb1 as their Ne2, our Ph7 as their ba4; transferred move e6f6 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kf6 (summed QAP objective 4, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



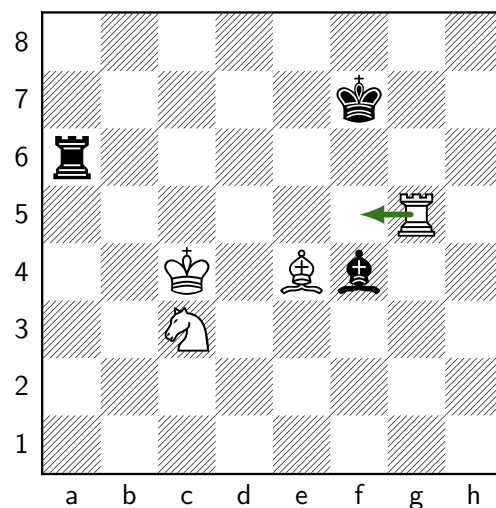
(a)



(b)

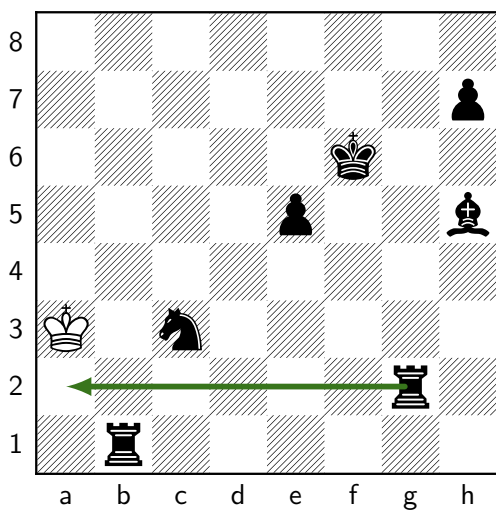


(c)

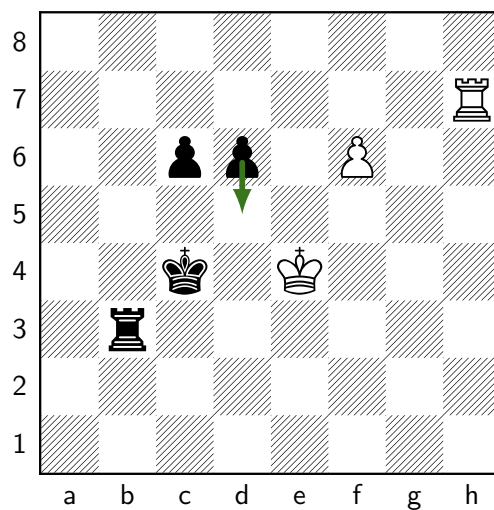


(d)

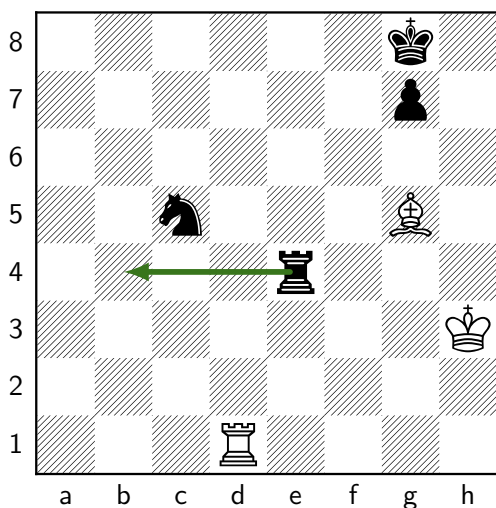
Figure 59: **Move 65, Black: Nc3+.** (a) Our position; Melanie plays Nc3+ (e2c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1302` (ply 119) [structural distance $d=24$, QAP obj=0]: the master played d3; under μ our Ne7 plays the role of that game’s Pd5 (a **cross-type** correspondence); cross-type roles: our Bh5 as their rf1; transferred move e2c3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2560` (ply 139) [structural distance $d=21$, QAP obj=20]: the master played Rd5; under μ our Rb8 plays the role of that game’s Ra4; cross-type roles: our Rg2 as their Pe6, our Ne2 as their Be5; transferred move b1b4. (d) Retrieved from `ccr1_top_p0000#205` (ply 140) [structural distance $d=23$, QAP obj=15]: the master played Rf5+; under μ our Rb8 plays the role of that game’s Rg5; cross-type roles: our Rg2 as their Nc3, our Ne2 as their Be4; transferred move b1c1. Synthesis: 1 retrieved analog transferred to Nc3+ (summed QAP objective 0, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



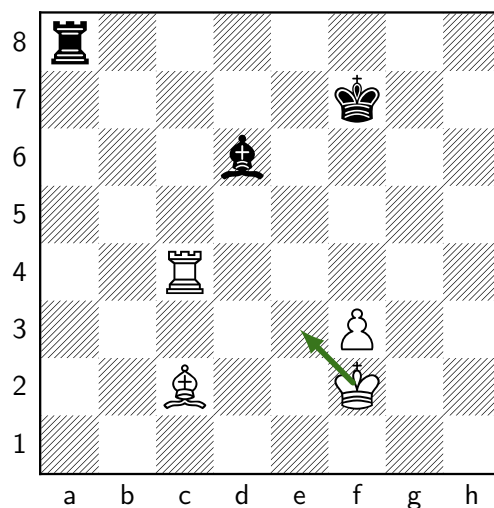
(a)



(b)



(c)



(d)

Figure 60: **Move 66, Black: Ra2#.** (a) Our position; Melanie plays Ra2# (g2a2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2132` (ply 119) [structural distance $d=15$, QAP obj=1]: the master played d5+; under μ our Rg7 plays the role of that game’s Pd3 (a **cross-type** correspondence); cross-type roles: our Ph7 as their rh7, our Nc3 as their Pc6; transferred move g2a2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3575` (ply 127) [structural distance $d=11$, QAP obj=3]: the master played Rb4; under μ our Rb8 plays the role of that game’s Re5; cross-type roles: our Ph7 as their bg5, our Bh5 as their rd1; transferred move b1b6. (d) Retrieved from `ccr1_top_p0000#1799` (ply 142) [structural distance $d=18$, QAP obj=2]: the master played Ke3; under μ our Kf3 plays the role of that game’s Kf2; cross-type roles: our Rb1 as their Bc2, our Nc3 as their Rc4; transferred move f6e6. Synthesis: 1 retrieved analog transferred to Ra2# (summed QAP objective 1, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.